School of Engineering | Lancaster University

INDIVIDUAL MEng PROJECT

‘Investigating the feasibility of energy extraction from the deflection of aerofoils using the aerodynamic phenomenon of vortex shedding’

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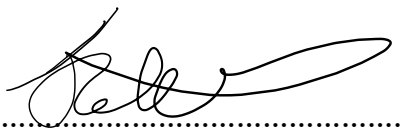
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Signed Declaration on the Submission of a Project

I declare that within this submission the work conducted is my own work and can verify that:

- Experimental procedures followed the wind tunnel risk assessment accordingly.
- Every supervisor meeting was attended to.
- The work conducted is my own work.
- References have been referencing appropriately.

Signed: 

Date: ...27/03/2026.....

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Abstract

Vortex shedding is an aerodynamic phenomenon that's also known as the von Kármán vortex effect that offers a renewable energy opportunity in which hasn't been greatly researched in. With the UK government's strategy to achieve 100% zero-carbon generation by 2030 and net zero by 2050 [1], it's important to prove the feasibility of this technology. The realisation of this method for energy generation and what it provides will help many communities, whether it be rural or urban areas, to increase their energy security whilst providing affordable renewable and reliable energy particularly for rural areas with weaker grid networks.

The project aims to investigate the feasibility of vortex induced vibration (VIV) as a method of energy generation, prove its viability and identify the niche power generation market best suited for a VIV device. During this project the use of CFD within the ANSYS software are used to develop a working scaled prototype of a VIV device. The use of ANSYS aims to verify the mechanisms that the VIV works by and predict the behaviour when it undergoes the periodic oscillations, allowing to determine the displacement of the aerofoil thus the calculating the theoretical energy potential within the device.

One key method utilized within the project to help prove the feasibility of VIV's is the use of software to aid the design and development of the aerofoil. Such software used were SolidWorks, ANSYS Fluent, MATLAB and Model P3 Software. Each software used explores different techniques to develop a device suitable for use. Another method used was the wind tunnel exploring how the aerofoil interacts in a real-world scenario allowing for the comparison between this and the mathematical modelling.

The amplifier device provided significant gains during the testing phase of the project when compared to the differing tip setups excluding it. The overall force produced on the device ended up being lower than predicted from CFD simulations. This was most likely down to the cantilever effect the geometry produced.

In conclusion, the project demonstrates VIV as a feasible method of energy generation, although with limitations. VIV devices are suited to specific low-energy niches and exploring how this technology could be scaled into larger systems by utilising unused infrastructure like powerlines to generate energy presents a valuable research pathway.

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Nomenclature

- *VIV – Vortex Induced Vibration*
- *CFD – Computational Fluid Dynamics*
- *CAD – Computer Aided Design*
- *RMS – Root Mean Squared*
- *FoS – Factor of Safety*
- *2D – Two Dimensional*
- *DC – Direct Current*
- *Cl – Coefficient of Lift*
- $C_{L,max}^{2D}$ *– Coefficient of Lift Peak Value During 2D Simulations*
- $C_{L,rms}^{2D}$ *- Coefficient of Lift RMS Value During 2D Simulations*
- *F – Force (N)*
- F_{allow} *– Allowable Force (N)*
- *Re – Reynolds Number*
- *St – Strouhal Number*
- *f – Frequency (Hz)*
- *U – Fluid Velocity (ms^{-1})*
- *D – Diameter (mm)*
- *v – Kinematic Viscosity (m^{2}s^{-1})*
- σ_y*- Yield Stress (MPa)*
- σ_{allow} *– Allowable Yield Stress (MPa)*
- a *– Point Force Location (mm)*
- I *– Second Moment of Inertia (mm^4)*
- *b – Base (mm)*
- *h – Width (mm)*
- *c – Distance to the Neutral Axis (mm)*
- M_{max} *– Maximum Bending Moment*
- F'_{max} *- Maximum Force per Unit Length (Nm^{-1})*
- F'_{rms} *- RMS Force per Unit Length (Nm^{-1})*
- F_{max} *– Maximum Force (N)*
- F_{rms} *– RMS Force (N)*
- δ *– Deflection (mm)*
- δ_{max} *– Maximum Deflection (mm)*
- L *– Height (mm)*
- E *– Youngs Modulus (MPa)*
- ρ*- Fluid Density (Kgm^{-3})*
- P *– Point Force (N)*
- ω_n*- Natural Angular Frequency (rads^{-1})*
- ω *- Angular Frequency (rads^{-1})*
- f_n *– Natural Frequency (Hz)*
- k_n *– Mode Constant for Cantilever Beam*
- A – Cross-sectional area (mm^2)
- G – Guage Factor
- $\mu\varepsilon$ *– Microstrain*
- V_{out} *– Voltage Out (V)*
- $\theta_{Difflection}$ *– Deflection Angle ($^\circ$)*
- z *– Distance between Force and Pivot (mm)*
- x_{max} *– Maximum Displacement (mm)*
- $x(t)$ *– Displacement at Time t (mm)*
- t *– time (s)*
- $v(t)$ *– Velocity at time t (ms^{-1})*
- $P(t)$ *– Power at time t (W)*
- $F(t)$ *– Force at time t (N)*
- P_{max} *– Maximum Peak Power (W)*
- P_{avg} *– Average Peak Power (W)*

Chapter 1 - Introduction

As the global demand for energy continues to rise along with increasingly rigorous carbon reducing polices, the ever-growing necessity for pioneering new renewable energy mechanisms in which supplement the current infrastructure of wave, wind and solar is pivotal to a net zero future. Dams and conventional wind turbines subjugate large-scale generation while requiring even larger scale development to implement. Offshore wind turbine farms and the construction of dams require a large amount of capital and prior planning to construct which provides limitations into how swiftly and effectively they can be implemented into the grid. Furthermore, the low wind speed and the mechanical complexity limitations of conventional wind turbines have driven interest into differing aerodynamic devices that are able to effectively capture the energy in lower fluid flow conditions whilst being mechanically efficient.

Vortex shedding has been conceptually known for over a century with the studies of Theodore von Kármán, which expanded upon the work conducted by Vincenc Strouhal and his experiments involving fluid flow around a wire. With the global influence and growing necessity for renewable energy, using these fundamental principles it's important to convincingly prove the feasibility of energy generation using VIV devices, like that of which Strouhal initially used during his experiments.

VIV devices are well documented within civil and mechanical engineering disciplines, where typically its seen to be an engineering challenge to mitigate the oscillatory effects that it perpetuates. Existing research primarily focuses on suppressing the periodicity of the phenomenon in effort to prevent structural failures of buildings subject to wind loading. This provides a significant gap in research, specifically, the small-scale power generation by amplifying the aerodynamic oscillations to a controllable effect as to where energy extraction is possible using a VIV device.

The focus of this project is to verify the feasibility of the VIV device by investigating methods to amplify oscillations by iterating designs based upon mathematical reasoning to determine a more efficient and transient geometry best suited for the small-scale development of a bladeless aerofoil.

1.1 – Project Aims and Objectives

This project aims to analyse and produce a vortex shedding based scale aerofoil capable of proving VIV devices are capable option for renewable energy generation, as a cheaper, simpler and a more economical alternative to traditional wind turbines. This involves determining whether the forces produced from vortex shedding can produce significant and repetitive enough forces in which functional energy could be extracted. The project also seeks to elaborate upon the gap between the detrimental effects VIV devices can cause to the theoretical exploitation of the behavioural characteristics' vortex shedding provides.

Furthermore, the aim to develop a fully computationally driven design methodology and validate its accuracy to the physical iteration of the VIV device is another aim of the project. This is to determine whether the use of CFD is a pivotal tool in the development of a VIV device, concluding if the addition of computationally driven design benefits predicting the physical behaviour to an accurate degree. This approach allows the project to iterate swifter and predict behaviour without wastefully prototyping. Additionally, the comparison between the mathematical predictions and the physical device will follow.

Finally, the project aims to identify a suitable application for the device to address should it be deemed a viable source of renewable energy generation when the project concludes.

The objectives for the project are as follows:

- Produce a mathematical model to determine the theoretical energy potential within the design of the VIV device.
- Produce and design a CAD model within SolidWorks that's modular and capable of vortex shedding. Iterate this design to produce the most efficient design that's achievable within this project's constraints.
- Test it within CFD software (ANSYS Fluent), determine optimal mesh and turbulent settings when undergoing CFD runs to produce reliable 2D planar simulations.

- Test the physical model within the wind tunnel and measure its displacement and strain. Cover a range of wind speeds to determine a wide range of behaviours dependent on fluid flow.
- Finally, analyse the results and determine the feasibility of the design regarding energy generation, scalability and adverse weather effects it could be subject to.

1.2 – Project Methodology

The VIV device will be designed and developed to operate within the Lancaster University wind tunnel boundaries, to produce a consistent and predictable oscillatory movement in which energy could be theoretically harvested from. To achieve this several steps within the project are due to be undertaken to permeate this project's aims. Demonstrating this the methodology is divided into stages elaborating on each key phase to progressively develop, verify and analyse the progression of the aerofoil and thus, the project as a whole via mathematical modelling, computational design, simulation and physical experimentation.

With the functionality of the aerofoil being paramount to the success of the project, succinct and thorough computational analysis of the structure will be essential. Therefore, the geometrical structure will be designed within SolidWorks, the CAD software of choice, as this allows for rapid and cohesive iterations to be constructed based upon mathematical reasoning with seamless integration with ANSYS Fluent, improving workflow. During the design phase, the aerofoil is developed such that its properties are modular and scalable, allowing for effective revisions to characteristics such as cross-sectional profile, span length, type of central rod/support and characteristic diameter along the length of the structure. These parameters directly influence the Reynolds and Strouhal number effecting the efficiency and behaviour of the aerofoil. Therefore, it's mandatory that a cohesive design is produced with reasoning behind each iteration in the pursuit of greater amplification of vortex induced vibrations in which the VIV device can benefit from. Considerations within the design that are significant to its functionality will consist of: the CAM capabilities of Lancaster Universities Engineering departments workshops, mounting dimensions and capabilities within the wind tunnel, the structural strength of the structure under cyclic

loading conditions and the potential future scalability of the project and VIV device as a whole.

To quantify the behaviours of the aerofoil once experiencing VIV, simulation will be mandatory with the comprehensive use of ANSYS Fluent. Due to Fluent's industry leading accuracy, integration with other ANSYS tools and its robust post processing capabilities that allow for data extraction which is able to be exported analysed in 3rd party software such as MATLAB.

Multiple CFD runs will be conducted to optimize mesh settings and determine the correct turbulence model suited to the project's objectives, most notably being speed and accuracy of the simulations. Mesh optimisation will commence, initially with ensuring mesh independence is established by monitoring the parameters of: C_L, C_D, shedding frequency, pressure coefficient and residuals ensuring they converge. From this the mesh will be analysed and made coarse to save the computational load making solutions solve faster with an acceptable degree of accuracy. Key mesh properties in which will be modified will be infiltration layers around aerofoil wall, face splits with denser mesh to capture vortex shedding wake with greater resolution and the transient time step resolution. Boundary layer refinement won't commence as the wind tunnel boundaries will be the boundaries within the simulations as 600mm x 600mm.

Following, 2D planar simulations will be conducted to explore a range of flow velocities and object geometries, computationally determining the optimal conditions for the VIV devices successful operation. Areas of particular interest whilst the simulation of the planar geometries is running are the identification of significant events such as, the onset of vortex shedding, shedding frequencies that are dominant over others, regions of the flow separation and sensitivity of Reynolds number.

To validate the computational results of CFD, the physical manufactured scale model will be run within the wind tunnel. Wind tunnel testing allows for the obtaining of vital performance metrics, such as the magnitude of oscillation, shedding frequency and the structural response. This provides real-world validation of the VIV device principle, aiding the project's goal of proving the feasibility of this method of energy extraction. A key method during this phase of the project to obtain results, is the use a strain gauge

circuit. The strain gauge circuit will allow for precise readings of the oscillatory motion of the structure, allowing for the determination of the periodicity of the motion of the device along with the force that its experiencing. To do this, the strain gauge will be connected to the central rod of the device in which the aerofoil oscillates around allowing for a concentrated high-resolution area in which that strain gauge can be located. Furthermore, as the wind tunnel boundaries are transparent, cameras can be used along with visual measuring methods to determine the displacement of the device and verify the functionality of the strain gauge circuit by ensuring that both the periodicity of the oscillatory movement match each other.

Using the wind tunnel allows for a wider scope of testing such as pursuing the use of fluid flow conditions larger than which would be within the operational expectations of the designed device. This allows for an in-depth behavioural analysis into how the device negotiates operating in extreme conditions in which the wind tunnel can produce. This would determine the theoretical max fluid flow conditions where the device can function in without structurally failing. This is important because if proven feasible, during energy generation it's vital to know the limitations of the device to prevent structural failure causing potential harm and damage.

Concluding the methodological framework previously implemented, a final analysis into the performance of the device will be conducted. Such analysis will assess the proximity of the results obtained during the testing of the physical device and the simulations ran.

1.3 – Literature Review

It's a necessary step while undertaking this project to carry out a literature review, doing so by providing a necessary amount of intellectual material to comprehend this field of aerodynamics. This is to specify the direction of present research and to contrast as to where this project is pursuing, in which is a lesser covered field, specifically the amplification of the vortex induced oscillations and the small-scale adaptation of the bladeless turbine design. This is opposed to the typical researched direction in the suppression induced vortices in the civil engineering aspect of this phenomenon. Vortex shedding can be an important aerodynamic trait as each shedding of the vortices generates a force; this force can be utilised in clean energy harvesting further pursuing the global renewable energy agenda. Following this the literature review will further discuss the known aerodynamic concepts in greater detail, typical use case with context of the problems it can cause, known mitigation methods, the chosen alternate use case, critical review of the researched literature, further dive into the gap in research and conclude the findings in that literature.

1.3.1 – Known Concepts

Vortex shedding is the oscillating flow of a fluid that passes past a non-streamlined (bluff) body such as a cylinder whilst generating lift forces doing so and occurs at specific velocities dependent on the size and shape of the structure in the fluid flow. For vortex shedding to occur initially a Reynolds number roughly between the range of 40-150 will induce the laminar vortex shedding regime [3]. This induces initially a consistent form of vortex shedding providing efficiency within the design as these values for the Reynolds number will produce a predictable and consistent vertex shedding effect. However, to fully realise the effects of vortex shedding we require the subcritical turbulent flow regime over our geometry [4]. This generates the strongest and most periodic version of vortex shedding which is a highly sought after property within this project as we look into increasing the strength and periodicity of the aerodynamic phenomenon for the small-scale device constructed. Furthermore, another dimensionless constant is required to be understood to fully characterize the oscillating flow mechanisms called the Strouhal number. This constant relates the 3 variables of: frequency oscillation, diameter of cylinder and fluid velocity with the equation: $St = \frac{f \cdot D}{U}$

The Strouhal and Reynolds number have a complex relationship with an ideal Strouhal number of ~ 0.2 being optimal for vortex shedding to occur, without the laminar boundary of the flow developing turbulent transition thus disrupting the periodicity of the aerodynamic phenomenon [2].

In the figure above we can see that the greater majority of Reynolds numbers are within the 0.2 figure as previously stated. Thus, the aim for the project will be to achieve a

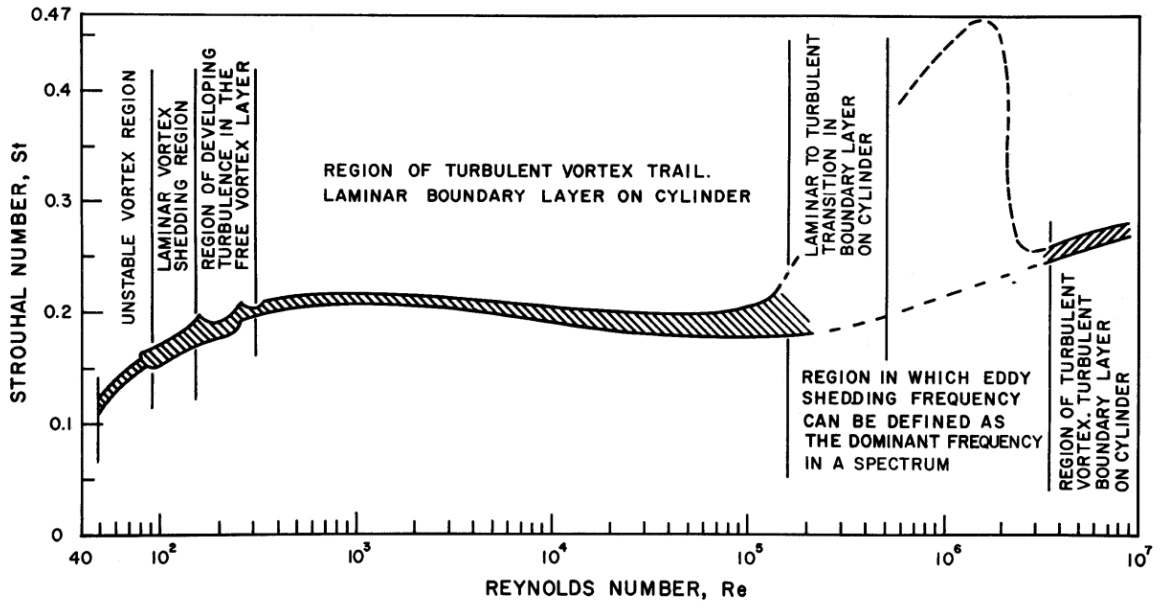


***Figure 4.1 The Strouhal-Reynolds Number Relationship for Circular Cylinders as Defined by Existing Data** [2].*

Reynolds number between the values of 10^3 and 10^5 to ensure that we are in the 0.2 window ensuring we're maximising the effect. The equation for Reynolds number of a cylinder is: $Re = \frac{U \cdot D}{v}$, and the re arranged Strouhal equation for vortex shedding frequency is: $f = \frac{St \cdot U}{D}$, this directly shows us that fluid velocity directly impacts the shedding frequency and provides validation within the graph above for a predicted behavioural pattern.

With this we would want to achieve resonance within the design of the aerofoil and relate that to a similar shedding frequency to optimise the design for certain wind speeds. This is as when we match the natural frequency with the shedding frequency the amplitude of the oscillations will increase due to resonance, this is another highly important aspect of this project as it increases the wanted amplification factor and

general increase in the aerodynamic principle allowing for more energy to be generated from the structure. The equation for the natural frequency of an object oscillating is as follows: $f = \frac{1}{2\pi}\sqrt{\frac{k}{m} - \left(\frac{c}{2m}\right)^2}$ [5]. Using this aforementioned equation the intention will be to achieve a suitable frequency for the structure to oscillate at with the aid of the design of the central rod within the structure which will later be expanded upon.

1.3.2 – Typical Use Case and Problems

The aerodynamic phenomenon of vortex shedding has typically been an obstacle in civil engineering requiring solutions to minimise its oscillatory effect. Typically, solutions are found on large cylindrical chimneys and car antennas by creating a screw-like spiral around the surface of the cylindrical structure which are called helical strakes [6]. Without mitigation methods this can become a very significant issue for structures that aren't equipped with a tuned mass damper or helical strakes, as without substantial failures can occur. Thus, potentially causing harm to people in the vicinity of the failed structure and could cause more structural damage to the buildings in the surrounding area specifically for larger structures subject to wind conditions.

There have been historical failures where vortex shedding hadn't been considered within the design of the structures affected. Most notably the Tacoma Narrows bridge is a great example of this as the 1940 accident showcases the aerodynamic forces and resonance to a great effect portraying the energy potential in which could theoretically be harvested. These oscillations generated from the periodic aerodynamic forces partly caused by vortex shedding, demonstrated how this sustained wind load could affect the bridges movement exiting the structure due to the lack of sufficient damping and tortional stiffness within the design of the bridge. While the disaster ultimately was found to have failed due to aeroelastic flutter, the role in which vortex shedding contributed mustn't be forgotten as the initial periodic forces that caused the bridge to move in the first place were caused by the phenomenon.

The dynamic instability of aeroelastic flutter is caused by the interaction between the aerodynamic forces applied and the corresponding elastic response of the structure. Unlike vortex shedding, aeroelastic flutter is a self-excited phenomenon where with a constant fluid flow alone causes the effect rather than external forces. The forces that

are generated along the structure are in phase with the natural frequency of the structure that typically happens when bending and torsion modes couple transferring energy between them. This happens at the flutter speed which generates the positive feedback loop where the altered angle of attack generates lift on the structure allowing for the torsion to rotate it back once at its maximum elastic potential. Once the flutter speed is exceeded the structure becomes unpredictable as the energy input exceeds that of the damping making the structure increasingly unstable. This principle and vortex shedding in the Tacoma Narrows bridge disaster emphasised the necessity to consider these principles when designing a structure for the public by implementing mitigation methods.

1.3.3 – Existing Mitigation Methods

As previously discussed, the use of mitigation methods is vital in the safety of built structures and must be considered during the design phase of the project at hand. There are several types of mitigation in which could be used depending on the situation in which the design is subject to. The first being helical strakes in which is typically found on cylindrical objects that experience a high value of fluid flow around its surface. Helical strakes work by disrupting the coherent fluid flow over the body of the cylinder by creating destructive interference with the induced vortices, resulting in a significant decrease in the vortex induced vibration. However, due to the increased surface area that the strakes provide a larger coefficient of drag is generated. This means more structural load is applied on the surface of the structure resulting in the need for more supports and structural strengthening for the strakes to work effectively. Car antennas, however, use helical strakes to remove broadband noise the vortex shedding creates with its periodic oscillations as well as its oscillatory movement when the car is up to speed.



Figure 1.2 Example of Helical Strakes on a Car Antenna (Photo Taken of a 2018 VW Polo)

Tuned mass dampers are another example of ways to mitigate the oscillatory motion caused by vortex shedding. The mitigation is caused by the counteracting movement of the mass damper to the structure's natural frequency. This out of phase methodology in counteracting the oscillations is a reliable and effective method and can be done with multiple techniques. A few are as follows: spring dampers, pneumatic dampers using fluid and pendulum systems. Each provide simple solutions by opposing the induced movement of the structure, these devices have been used on many structures with and without vortex induced vibrations such as skyscrapers and F1 cars proving their versatility.

Another simple counter to vortex induced vibrations is the aerodynamic shaping of the affected geometry. By shaping the structure to avoid the shear layer from separating and increasing the size of the boundary layer making it have a greater reluctance to separation. Examples of this are: spoilers and guide plates which prevent the roll up of vortices and vortex generators and fins which reduce the drag coefficient and increase the boundary layer to a fuller state aiding the structural stability of the geometry.

However, during this project I am researching as to whether increasing this aerodynamic principle's oscillatory effect can produce a large enough displacement

from its oscillations where energy can be extracted during it periodic movement rather than minimising it. It should be noted that while conducting this researched reduction in vortex induced vibration, valuable information regarding what aspects cause and amplify the oscillations such as aeroelastic flutter will be key to implement within constraints to this project. Furthermore, the aerodynamic principles of how to reduce the oscillations allow the design process to avoid these structural elements, allowing enhanced propagation of the desired oscillations for energy take off thus providing a more efficient device.

1.3.4 – Alternative Applications

Energy generation from bladeless wind turbines are an under investigated area in the renewable energy sector due to the lack of research in the field. Despite this there is interest in this area as the European Commission of the EU have invested €1,328,687.50 into a vortex shedding based wind turbine to the company "Vortex Bladeless" [7]. If proven feasible the vortex shedding wind turbine would provide a clean and cheap energy source that could benefit the medium-small sized enterprises (SMEs) power generation market.

The principle of vortex induced vibration using a bladeless turbine is largely less researched due to the lack of span in which the turbine is generating energy. Whereas a traditional wind turbine spans 360° the bladeless turbine will span approximately 45°. This is an 8x reduction in the area under which the device captures energy from, resulting in research becoming less appealing. However, I determine it to be further expanded upon as the development of these turbines are cheap and can be scaled. Furthermore, the lack of any moving parts within the "Vortex Bladeless" design shows that in the long run the reduced need for maintenance saves the consumer finances which otherwise could be invested into a greater volume of these devices, thus potentially generating a greater amount of energy for a similar capital.

Vortex shedding is also used in another instance in measuring fluid flow called vortex flowmeters. Using the principles of the rate of oscillations it's able to determine the rate of fluid flow within pipes.

Chapter 2 - CFD Configuration for the 2D Simulations

To perform a preliminary simulation, initial mesh and simulation settings must be implemented to estimate the forces expected on a uniform diameter and thus the onset of vortex shedding. During this section of the report, the transient response with respect to the variables: diameter and fluid velocity, will be recorded and iterated upon to set the primary design parameters. Specifically, the determination of C_L, force and shedding frequency are the numerical values which provide the information to progress and inform the development direction of this project.

2.1 - Geometry Initialization

The geometry constructed for the following 2D simulations were developed within ANSYS' Design Modeller, as it provided a simple interface to construct 2D geometry on and aided seamless integration into the meshing stage with iterative functionality. Firstly, the boundary conditions needed to be set up with accordance to the wind tunnel dimensions, therefore the CFD boundary conditions where modelled as a square with 600mm x 600mm being the dimensions.

Dimensions: 5	
D1	50 mm
H2	300 mm
H3	600 mm
V4	600 mm
V5	300 mm

Figure 5.1 Design Modeller Dimensions Used for Boundary Conditions

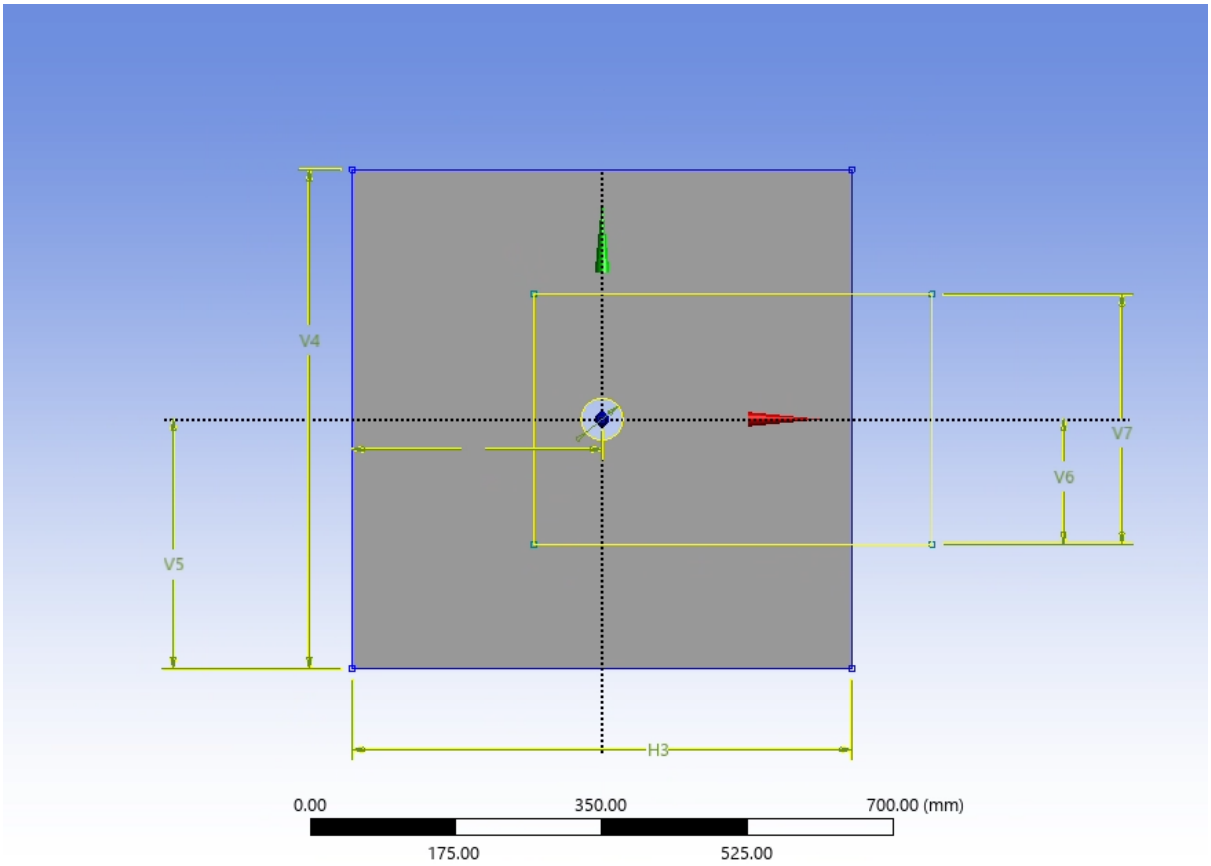


Figure 2.2 Design Modeller Reference Geometry

With the vortex shedding regime permeating after the fluid flow interacts with the bluff body, in this case being the diameter of the cylinder. The implementation of a face split was necessary to improve the accuracy of the simulation where a finer mesh can be incorporated, which can be seen in the figure above defined by V6 and V7. To provide ample space for the development of the vortices avoiding traveling through varying mesh refinement levels, V7 is half the size of V4 at 300mm wide providing the space necessary for a reliable and accurate simulation of the development of the Von Kármán vortex sheet. The cylinder was cut through the 2D geometry providing the necessary boundary for the bluff body to interact with allowing for vortex shedding to permeate.

2.2 - Mesh Setup

The mesh is configured in 3 major sections, the overall coarse mesh, the fine mesh using face sizing and the infiltration layers surrounding the cylinder where each layer has its purpose and reason for its implementation. Initially the coarse mesh was generated around the entire geometry providing a baseline that can be iterated upon

until a suitable mesh materializes. Furthermore, this baseline coarse mesh is used around the boundaries of the simulation where little aerodynamic interest is located. This is to minimise computational workload required for the simulation to conclude, reduce size of the CDF files produced and decrease the resolving time for the simulation, allowing for a greater iterative approach and thus minimizing time spent on computation allowing for the prioritization of other tasks. Reducing the file size was a particular concern for this project, as ANSYS' verification of appropriate licencing restricts the use of cloud storage and must be stored locally. Therefore, the use of locally stored simulations is mandatory and with only 10GB of allocated memory per student at Lancaster University, every reduction in unnecessary computational increase in file size that doesn't decrease the accuracy of the simulation was sought after. The coarse mesh thus used has an element size of 0.005m and a growth rate of 1.1.

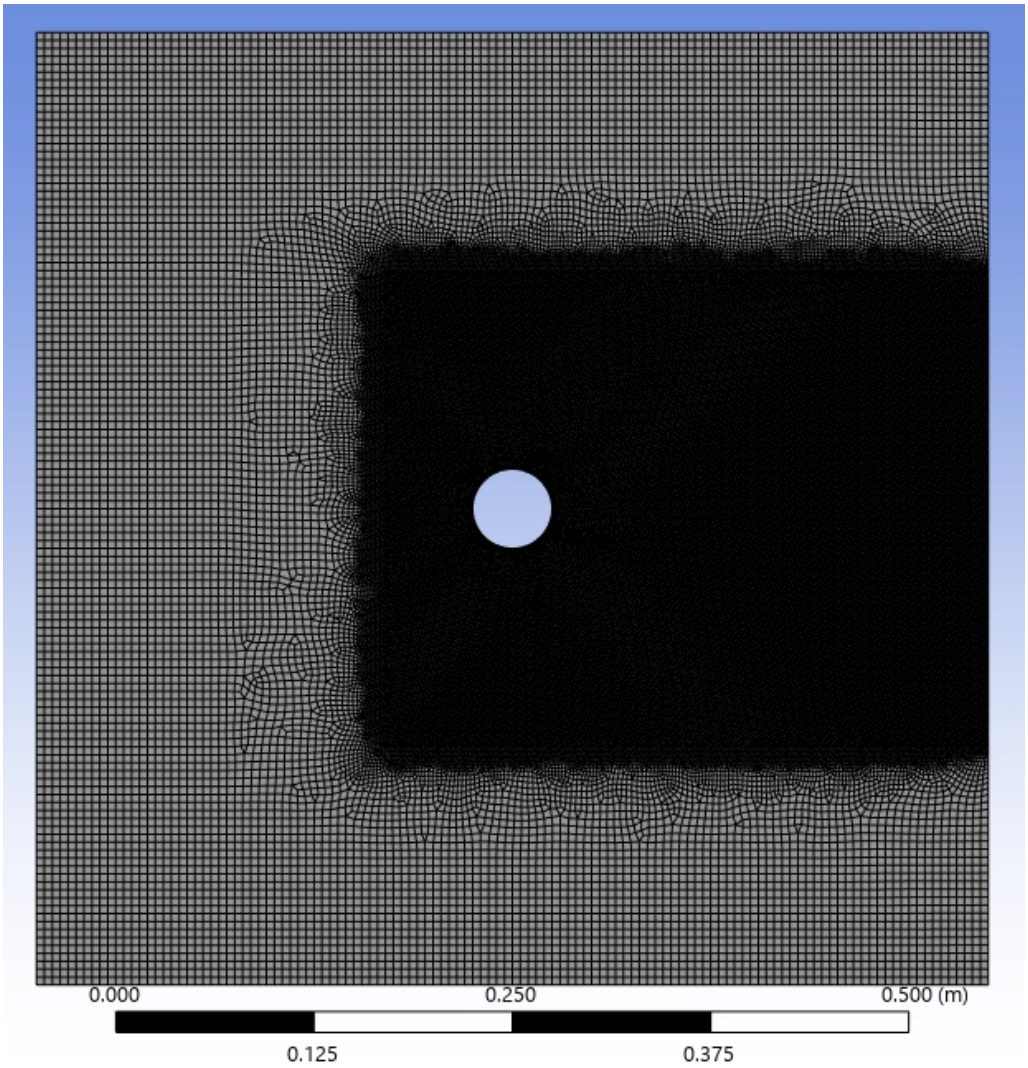


Figure 2.3 ANSYS Mesh Generated Showcasing the Finer Mesh in the Wake Field

The fine mesh generated within the face sizing function on the face split created in Design Modeller has a substantially smaller element size of 0.0012m and the same growth rate of 1.1. This 4.17x finer mesh provides the essential increased resolution for the wake field behind the bluff body. Thus, increases the accuracy of the simulation as shed vortices are simulated with greater precision due to the smaller cell sizes within this region. This finer meshed region starts in front of the bluff body due to a high-pressure region that develops in front of the cylinder that needs to be simulated with a greater resolution as its pivotal to the accurate onset of vortex shedding and the further flow field behind the bluff body.

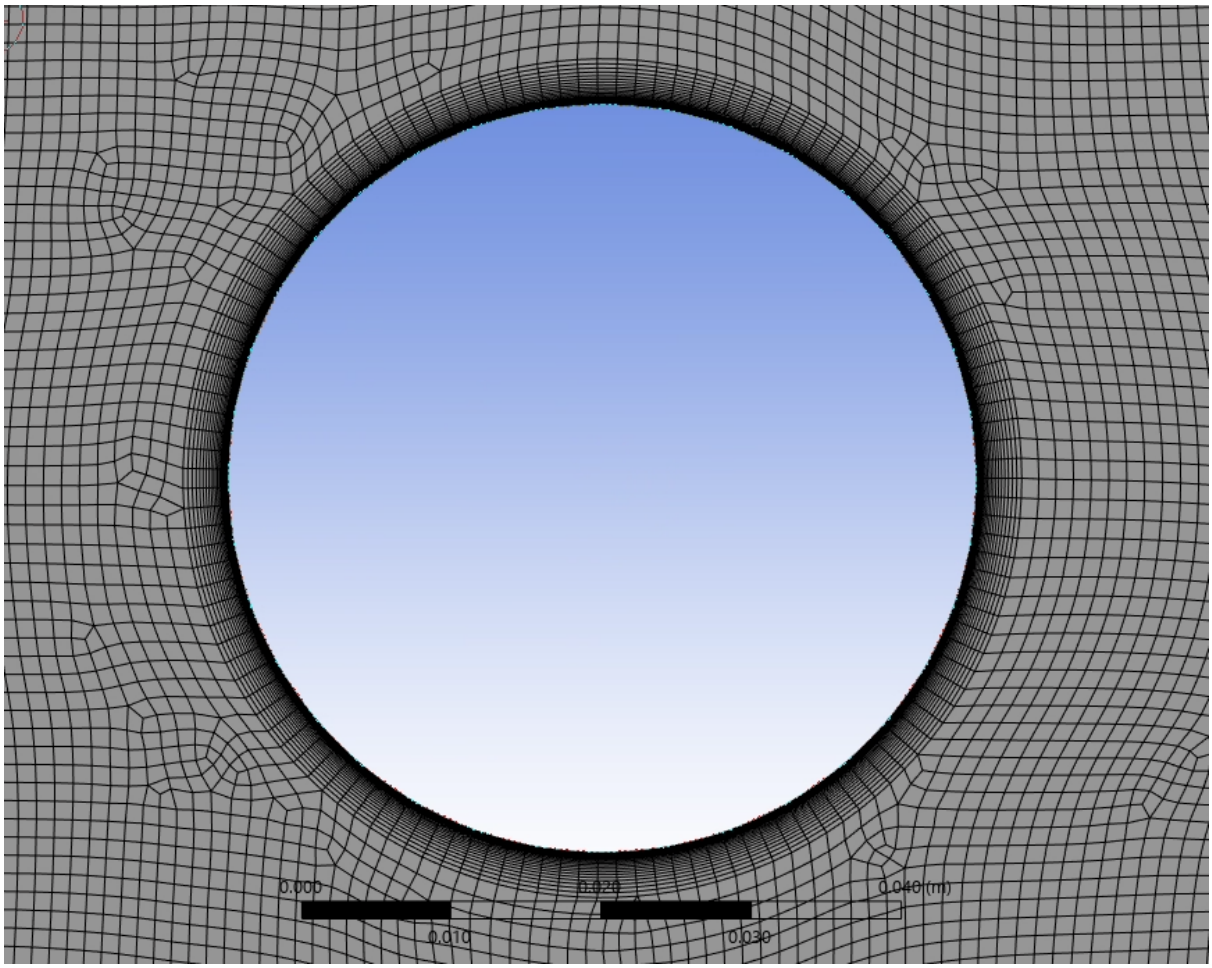


Figure 2.4 Depiction of the Added Inflation Layer to Better Capture the Viscous Layer

An inflation layer was added to the surface of the cylinder within Meshing to improve the accuracy of simulation around the surface of the cylinder. As the bluff body being the only geometry within the simulation, the boundary layer around the surface of the

cylinder requires particular attention. With vortexes being periodically shed from both sides of the cylinder and the transient behaviour of the boundary layer, detailed and precise measurements must be taken within this region of the simulation. With this in consideration, the inflation layer has a maximum thickness of 0.003m with the same fine growth rate of 1.1. Due to the particular interest in this boundary layer around the cylinder, 20 inflation layers were added to fully incorporate the geometry.

Within Meshing, Named Selections were added for Fluent to reference. These were necessary additions as the direction of fluid flow had to be established and therefore, the inlet and outlet needed to be defined. The inlet was defined as the left edge and the outlet the right edge of the square boundary creating linear flow. To finalize the boundary conditions, wall sections must be added to the simulation, these wall sections were the top and bottom edges of the square boundary creating a fully isolated system. In addition, the cylinder was also a wall section providing the physical geometry for the fluid flow to interact with. However, a different named section called cylinder was created, so that its aerodynamical properties could be later extracted into numerical form.

2.3 - Mesh Analysis

The mesh analysis determines whether the mesh is suitable or not, to achieve an accurate simulation, an accurate and consistent mesh is required. Within the simulations conducted purposeful metrics were obtained to indicate the quality of the mesh. These are: Average Y^+ value, Minimum Orthogonal Quality, Maximum Aspect Ratio and Cell Count.

The average Y^+ value represents the quality of the resolved mesh adjacent to boundaries in near wall flow regions, whether that be the wind tunnel boundaries or the cylindrical surface of the VIV device. The resolved mesh had an average Y^+ value of 1.1348253, as this value is ~1, indicates that the mesh is well resolved around the walls. This means that the initial mesh cell sits firmly within the viscus sublayer providing a very accurate boundary layer resolution. With the use of k-ω SST as the turbulence model having a Y^+ of ~1 is mandatory for the simulation to resolve accurately. With this low value, the drag coefficient and the skin friction coefficient will be simulated to a significant level of

accuracy as wall shear effects are captured with greater resolution. The Y^+ value holds significant importance to this project due to the transient behaviour of the boundary layer during vortex shedding. Therefore, the Y^+ value being ~1, vastly improves the degree of accuracy of the simulation in which informed choices can be taken from [8].

```
Calculation complete.

            Area-Weighted Average
                        Wall Yplus
--------------------------------- --------------------
                          cylinder            1.1348253
```

Figure 2.5 Calculation of the Average Wall Yplus Within Fluent's Console

Minimum orthogonal quality states how perpendicular the mesh cells are in comparison to the adjacent line of cells preceding the minimum cell. Therefore, if the mesh was perfectly square the value would be 1, providing a solution with a great stability and accuracy. Within the simulation, the minimum orthogonal quality was 0.371349, this falls within the accepted range for suitable simulation unlikely creating numerical instability within the results. This is acceptable within the scope of this specific simulation, due to the inclusion of circular geometry which can decrease the orthogonal quality.

The maximum aspect ratio informs the most stretched cell in the domain, in this case being 22.9878, which is slightly out of the acceptable values for this metric with the limit at 20. However, the cell in question, is within the coarse section of the mesh where greater orthogonal quality is situated due to its square like geometry. Therefore, the coarse section would approximately have a 4.17x larger ratio than that within the fine section where the detailed resolving is occurring in the wake field. This aids the desired adoption of the coarse mesh's suitability, as it overall remained within accepted limits where little to no interaction occur around the domain's boundary conditions.

```
Mesh Quality:

Minimum Orthogonal Quality =  3.71349e-01 cell


Maximum Aspect Ratio =  2.29878e+01 cell 5402 o
```

Figure 2.6 Calculation of the Minimum Orthogonal & Maximum Aspect Ratio Within Fluent's Console

The mesh has a total of 96508 cells spanning the entire computational domain. A predominantly quadrilateral mesh was utilized to encapsulate the cylinder to improve numerical accuracy and minimize the diffusion caused by triangular cells. This is as triangular cells have irregular orientations causing high skewness producing a weaker solution that appears smoother, particularly affecting the wake simulation and shear layer resolving. Approximately 60 cells span the diameter of the 50mm diameter cylinder providing excellent resolution to capture the shear layers aiding the production of a numerical result with a high degree of accuracy.

```
Mesh Size

Level    Cells    Faces    Nodes   Partitions
    0    96508   193334    96826            4

 1 cell zone, 6 face zones.
```

Figure 2.7 Calculation of the Mesh Size Within Fluent's Console

2.4 - Turbulence and Fluid Flow Configuration

To configure turbulence and thus fluid flow, it's important to analyse the expected behaviour of vortex shedding. Initially, as fluid flows around the surface of the bluff body the boundary layer develops along the surface inducing steady laminar flow. As the pressure gradient builds up along each side of the cylinder, the boundary layer undergoes separation inducing a vortex. These separated boundary layers become shear layers due to them periodically adversely shedding off the surface of the bluff body in an intended aerodynamically unstable manor. It's the cyclic shedding of vortices that generate the periodic forces from the adverse pressure gradient along spanning the bluff body in which the cylinder experiences.

The turbulence flow within the simulation was modelled using the k-ω SST turbulence model. This is as it provides both the advantages of k-ω and k-ϵ models [9]. Specifically, the combination of the precise near wall behaviour simulation and improved free stream simulation was ideal for this application. This model improves the numerical accuracy when calculating C_L and force on the cylinder as the boundary layer separation is simulated to a greater degree and then the vortex sheet behind the cylinder is simulated with increased precision to better predict wake flows and shedding frequency[10].

Precise turbulent conditions were defined using turbulence viscosity and turbulent viscosity ratio inside fluent. The turbulence intensity metric defines the magnitude of velocity fluctuations; with this we can compare it to the measured wind tunnel velocity variations. Practical measurements taken from the wind tunnel approximately varies ~±0.25ms^{-1} across all the wind velocities being tested. However, this variation could be caused by instrument uncertainty in the measurement process as fluid velocity updated roughly every 0.5s or fan speed variation due to the variable frequency drive system the wind tunnel operates on. Therefore, based off several trial simulation runs a 1% turbulence intensity was chosen as it represents the expected low turbulence flow within the wind tunnel and only minor changes occurred during testing different values of intensity. The turbulent viscosity ratio describes the inlets interaction of momentum transfer due to generated turbulent eddies with respect to the molecular viscosity of the fluid flow. A value of 10 was chosen due to its commonly used nature with external flows around a bluff body, for a moderate degree of turbulence, which represents the wind tunnel conditions.

2.5 - Solver and Transient Settings

Vortex shedding requires a transient solver to simulate correctly. This is as the shedding of vortices is a cyclic process that involves fluctuating forces, periodic flow behaviour and generates unsteady wake structures that all depend upon change over time to numerically simulate. As the fluid flow doesn't experience constant force and pressure differential that create developed flows without oscillations, a steady solver cannot be used in this application.

When optimising the fluid flow simulation, the time step size had to be determined to a degree that'll incorporate all the shed vortices with appropriate resolution to simulate the minimum and maximum points of each transient response. Initially, the use of the largest fluid velocity was used to identify the time step, this is as using the shedding frequency equation: $f = \frac{St \cdot U}{D}$, the larger with fluid velocity for a constant diameter the greater the rate of vortex shedding. Consequently, using 20ms^{-1} as the maximum fluid velocity and setting a fine and course time step of 0.005s and 0.0005s respectively results in the graph below:

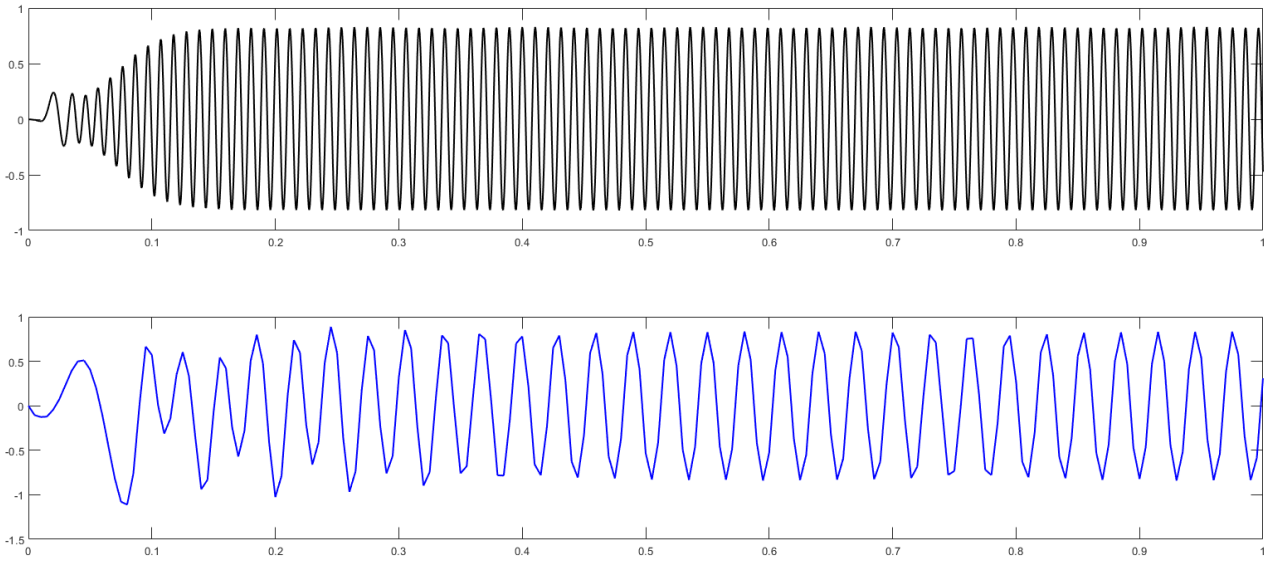


Figure 2.8 Comparison Between 0.0005s Time Step and 0.005s Time Step

With a velocity of 20ms^{-1} an expected shedding frequency of 80Hz is anticipated when taking Strouhal's number to be 0.2. The period of 80Hz is 0.0125s, therefore, to fully simulate each vortex being shed, a step size less than 0.0125s is required. It is recommended by ANSYS that 20 – 25 time steps per shedding cycle are used [11]. This results in 0.0005s being the recommended time step which provides a significant increase to the resolution surrounding the peaks of each vortex shed and captures all the oscillations neglecting data loss through unsimulated time steps. This can be seen within the figure above as the 0.005s time step showcases sharper peaks and has less oscillations than the mathematical model suggests. The trade-off for using this finer time step size is the significant increase in computational demand, as 10x more data points are being taken than that of the 0.005s time step interval results in 10x longer computational time required. To achieve this, the autosave feature had to be reduced from saving every time step to every 10 reducing the file size to a workable size.

It was decided that the number of time steps taken would be 2000 to achieve a full second of simulation. Allowing for ample time for the simulation to stabilize and produce a consistent wake field. This would also allow for a range of stabilized data to be used to produce an RMS value based upon a larger more consistent data set.

Maximum number of iterations per time step iteratively updates the flow variables and equations to approach a convergence on a result before passing onto the next time step. To ensure sufficient convergence at the end of each time step, the maximum number of iterations per time step is configured to a value of 25. This provides a reasonably fast and accurate convergence solution to the simulation.

Chapter 3 - Design of the VIV Device

The overall design aims to measure the strain and thus force that the VIV device experiences during vortex shedding. To achieve this, the VIV Device was developed to be assembled with 6 key parts, 2 of them being the steel rod and baseplate assembly and the remaining 4 parts being the made from PLA for the surface of the aerofoil. A key decision that's part of the design, was to make the aerofoil modular with parts able to be switched in and out, making use of rapid prototyping to test, iterate and improve the design. The ability to substitute parts allows for comparisons to be measured between development directions and creates the opportunity to compare them in the same conditions.

3.1 - Overview of the Aerofoil Design

As stated before, the aerofoil is comprised of 4 segments each one having its own purpose and ability to be swapped. Measuring how each part effects the overall behaviour of the oscillatory movement provides multiple data points to compare allowing the determination of which one operates with greater efficiency and thus captures a greater share of theoretical energy from the fluid flow. These 4 parts are as follows: the bottom of the aerofoil, centre of the aerofoil, central clips and tip designs.

The VIV device is based upon varying cylindrical geometries, due to the well documented and studied Structural behaviour, with the Strouhal number ranging from 0.18 – 0.22 when experiencing subcritical flow. This allows for comparisons to be drawn to scientifically accepted values and how the device behaves physically when under load to the theoretical behavioural traits. Furthermore, cylindrical bluff bodies provide strong, consistent and predictable shedding patterns which are able to be compared between physical, simulation and mathematical interpretations. With a cylindrical

cross section also being symmetrical there is no preferential angle of attack for the vortices to shed from, unlike standard aerofoils or other asymmetric geometry. This could account for any ununiform flow within the wind tunnel and mitigate the effects of it. With the predicable shedding on the cylinder, lock in frequencies can be calculated and studied to increase the amplitude of oscillations and therefore increase the theoretical energy output from the VIV device.

An optimum diameter must be chosen to operate suitably within the wind tunnel, that provides a suitable amount of force with the compromise of size. This is to ensure that it fits within the wind tunnel and firmly fits within the plate when 3D printing. This reduces the mass of PLA used and time printing meeting a suitable compromise creating geometry that's fast and easy to iterate whilst providing a great enough force for the functionality of the VIV device to be studied.

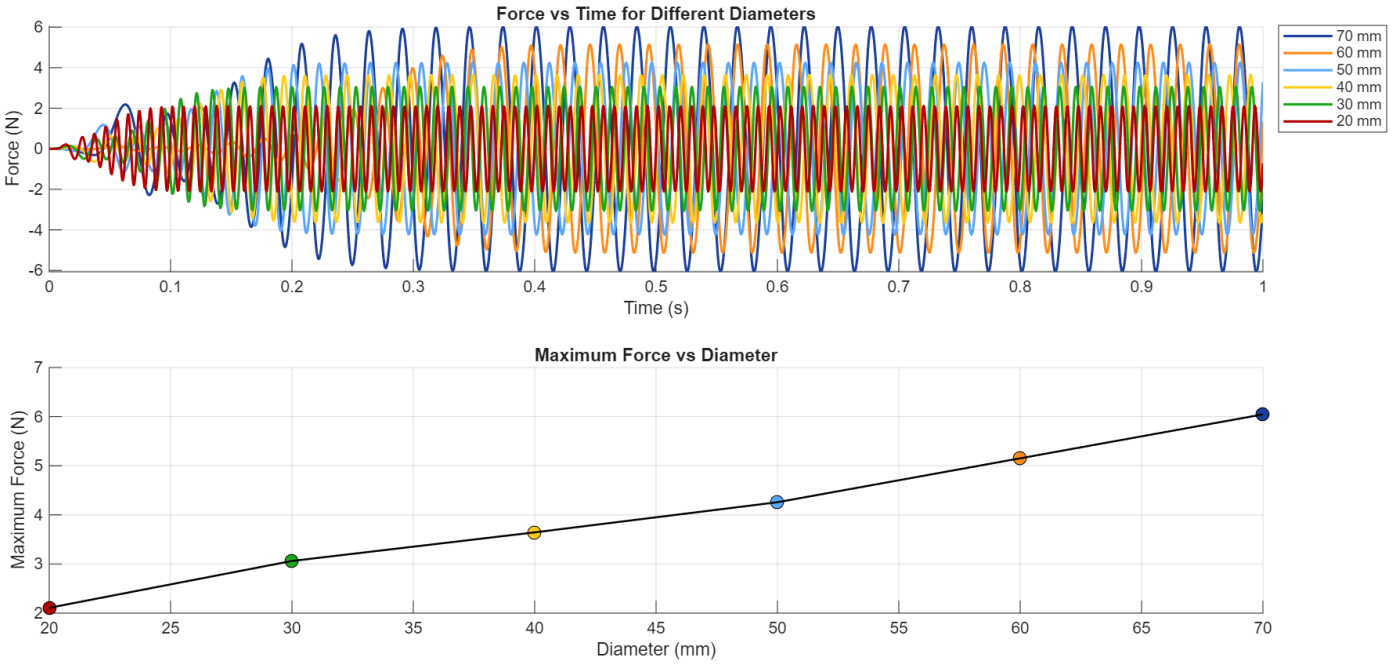


Figure 6.1 CFD Results for Force vs Diameters at 10ms^{-1} Fluid Velocity

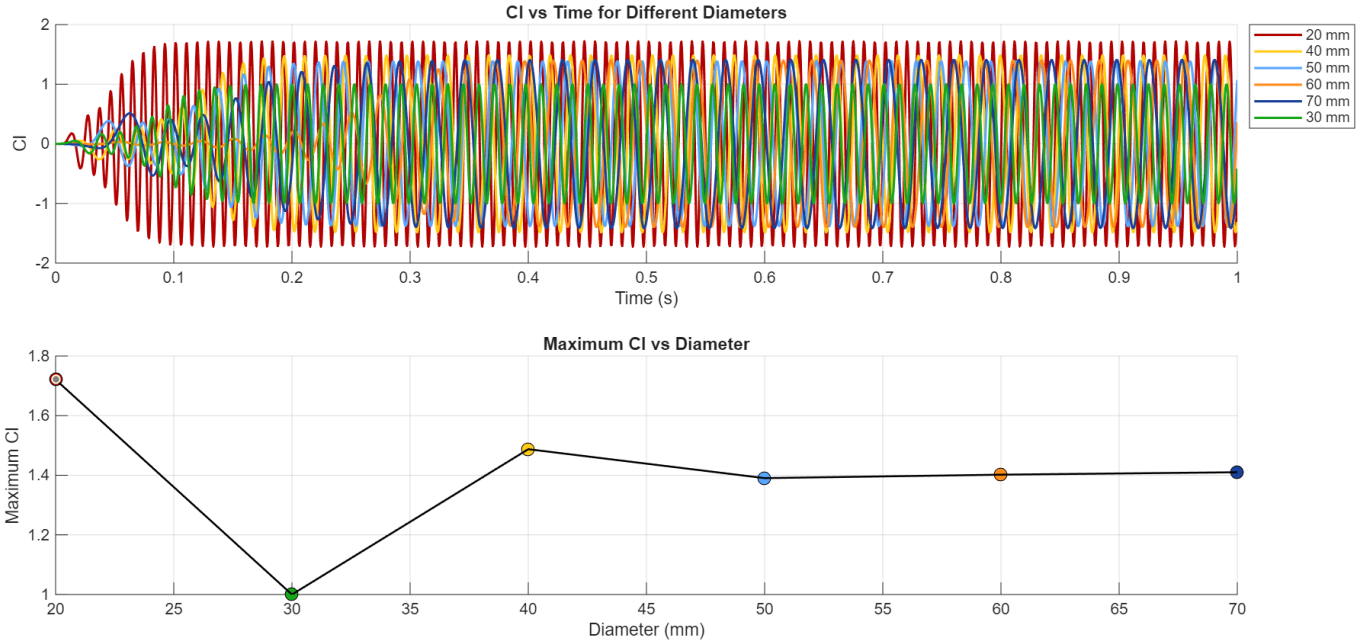


Figure 3.2 CFD Results for Cl vs Diameters at 10ms^{-1} Fluid Velocity

From the figures above there are clear trends which can be deciphered and analysed. Initially, the decrease in shedding frequency with the increase in diameter is evident, which is an expected outcome thus, increasing the validity of the CFD runs as the intended behaviour has been observed as the equation: $f = \frac{St \cdot U}{D}$ has held true. Secondly, there appears to be a linear increase in force as diameter increases whilst the Cl values start high then typically decrease to the plateau at ~1.4. Notably, the Cl value for 30mm diameter is sitting in a local minimum, dipping below the other diameters in a noticeable manor. With all of these diameters being tested at 10ms^{-1}, using the Reynolds equation: $Re = \frac{UD}{v}$, the minimum and maximum Reynolds numbers are: 1.3x10^4 and 4.7x10^4. This range of Reynolds numbers fully resides within the turbulent transitional vortex shedding with a laminar boundary layer on the cylinder [2], therefore, the local dip is most likely due to the shedding of vortices having a greater symmetrical shedding pattern when compared to the surrounding diameters reducing the Cl value.

Considering these factors, to select an optimum diameter, the decision was made to prioritise the onset of stable and predictable oscillatory vortex shedding rather than the largest. This compromise provides the best of both worlds as 50mm was selected as the diameter for the aerofoil. The 50mm choice provides a large enough diameter to produce meaningful forces with a maximum of 4.26Nm^{-1} whilst having a suitable shedding frequency to produce this force more consistently. In addition, the Cl value for

50mm is this first in the Cl plateau at ~1.4, this results in having a greater Cl value without having to increase the diameter to one larger than necessary. Whilst the 20mm option looks appealing to choose in the Cl graph the small diameter presents different challenges with the force it produces as the vortices shed are of a smaller size and thus produce less force despite the higher Cl.

3.2 - Control Design

In aid for the control test to be performed, the geometry must be as similar to that of the simulated one. As 2D simulation only performs a calculation based on an infinitesimally small slice of the cylinder, the diameter must remain constant for this comparison to occur as all constants and force values are calculated per meter of a constant diameter. Therefore, the central aerofoil remains at 50mm diameter throughout its span accompanied with a constant sized tip to achieve its full size. Using this model within the wind tunnel will allow for direct comparison to the simulated version, providing a significant and useful reference point for the project to analyse from.

For all tests the lower segment of the aerofoil will remain constant. The design contains the singular constant diameter of 50mm that spans 210mm down the length of the metal rod and provides the central supportive mechanism for the entire aerofoil to connect to, which operates as a push through and twist into place system. To prevent the wrong direction of installation of the lower segment and incompatible geometry that's larger than the cutouts situated on the steel rod. Block improper connections from happening, therefore, only the intended direction can occur, preventing incorrect assembly.

In addition, the clips used across assembly point of the aerofoil will remain constant throughout with their 50mm diameter. This is to ensure that the flow around the non-streamlined features that the mounting hardware exhibits doesn't interfere with the periodic vortex shedding surrounding the surface of the VIV device. This allows for the instillation of the central aerofoil piece to be performed without negatively impacting the flow around the centre of the aerofoil and preserving the flow regime without adding unnecessary turbulence as a variable in the design.

3.3 - Design Choices

3.3.1 - Tip Designs

Several Tip designs were developed to determine whether altering the tip geometry effects the magnitude of the oscillations, the rate of shedding or both metrics. The tip of the VIV device provided a significant area for iteration due to the tip contributing to the greatest bending moment throughout the design, thus, having the potential increase the magnitude of oscillations with unique geometry. Due to this a total of three designs were drafted based upon differing design approaches and principles. These are: a dome top design, expanding tip design and a tapered tip design.

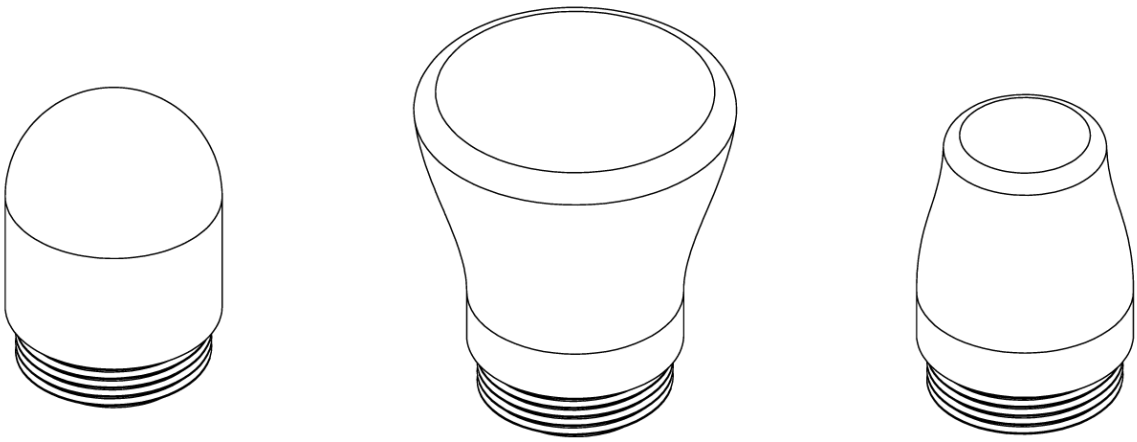


Figure 3.3 Differing Tip Designs of Dome, Expanding and Tapered Exported from SolidWorks

The dome tip design was created to examine how minimizing the geometry to a single point changes the behaviour of the oscillations, with smaller diameters providing an increased rate of shedding with a smaller magnitude, it's predicted that this tip will increase the rate of shedding potentially increasing the amount of theoretical energy that the VIV device could harvest. The dome also seeks to see if that tip increases the magnitude of the oscillations in the interaction with the wake field and shed vortices. Each tip design features a screw system to enable the connection and interchangeability of each of the tip designs with it being safe and reliable to operate with.

The expanding top follows the opposite development path of using a larger diameter to produce vortices with a greater magnitude. However, as mentioned before the trade-off of this is the decreased shedding frequency at the tip. This has the potential to generate

larger forces at a smaller frequency which permits the comparison to be made of the expanding top's average force to the others. This determines whether the focus on increasing the rate at the tip or magnitude is the correct development path for the VIV device.

Similar to the dome tip, the tapered tip seeks to understand whether ending the tip with a constant diameter impacts the rate compared to the dome on which doesn't. It shares the development path of increasing the rate of oscillations but with a stop. With these three tips benefits and negatives will be able to be produced when looking at the physical results determining whether they impact the design or not and if so, which ones operate better than the others will be a key part in increasing the feasibility of using vortex shedding as a method of generating electricity.

3.3.2 - Central Aerofoil Concept

The central aerofoil exhibits a taper along its length to seek and increase in shedding frequency along its span. This is in an effort to produce a more consistent onset of shedding further towards the top of the VIV device where the larger bending moment is. With the choice to taper the central aerofoil, the intention is to increase the RMS force with the increased shedding frequency, to envelop around the surface of the device hence, increasing the efficiency. The top of this component exhibits the thread for the tips to screw into and form their secure bond enabling the tip analysis to occur.

3.3.3 - Amplifier Design and Theory

The amplifier design consists of a basket like attachment that surrounds the top section of the VIV device. It aims to increase the magnitude of oscillations by producing larger more influential vortices. It does this by producing a region of high pressure that's encapsulated by the basket. The high-pressure region is a transient process as the central cylindrical shape provides the needed vortex emission which allows the release of the high-pressure air encapsulated within the basket. It should be noted that the added semicircular geometry itself doesn't increase the amplitude of oscillations as no added force is applied to that structure, it's rather the feedback loop it provides enables the vortices to generate more consistently and with a greater more energetic magnitude. During testing this was proven true as a cupped hand placed behind the circular

geometry in the wake field generated the same effect with the residual low fluid momentum after a test run had been carried out.

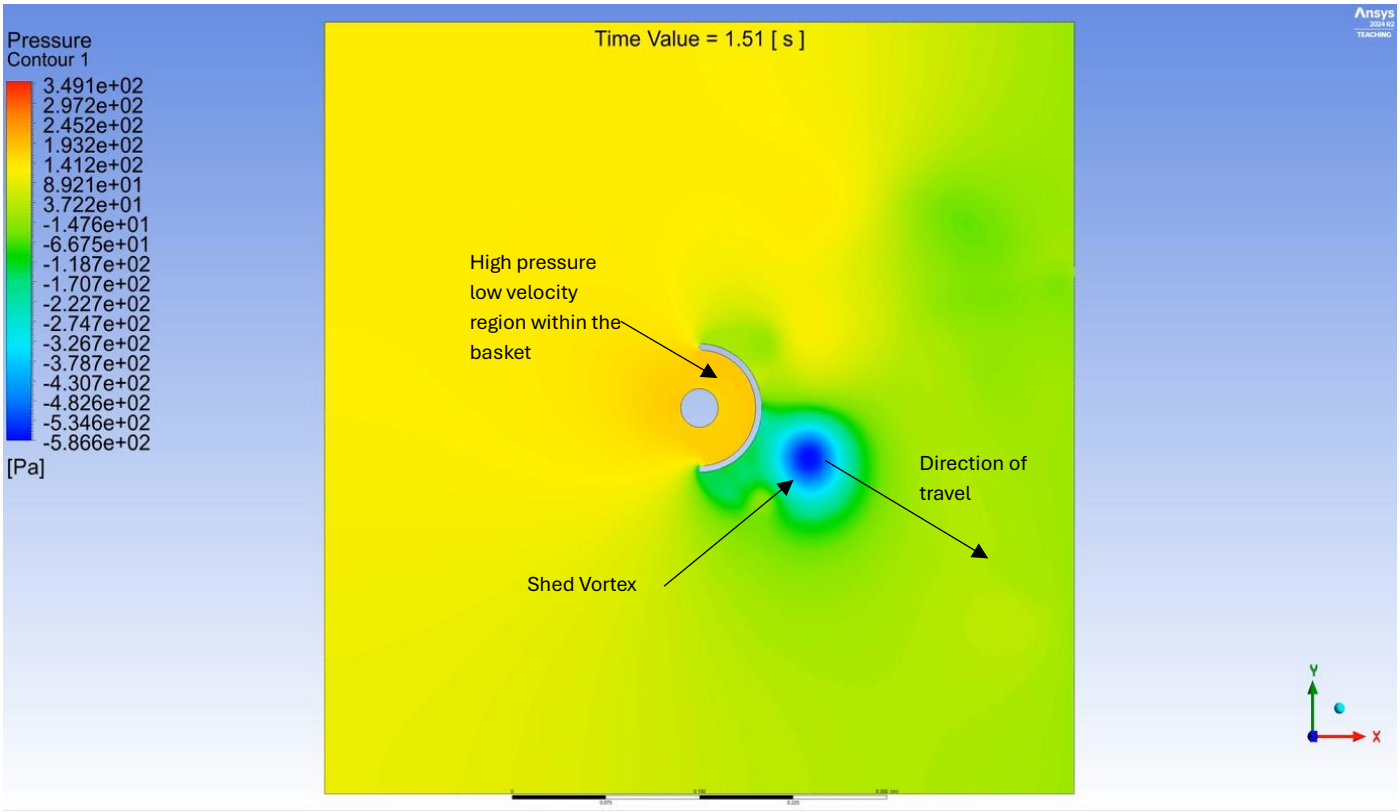


Figure 3.4 Contour Plot of Vortex Behaviour with the Amplifier Devices Feedback Loop

The contour above portrays the process of ejecting a vortex from the amplifier design in a 2D simulation. The vortex shed travels at a greater angle than regular vortex shedding therefore, a greater component of the force that the low-pressure region applies to the y axis. This is beneficial as the direction of desired travel is in this dimension, thus producing a larger displacement for the device to benefit from.

The use of the amplifier will increase the efficiency of capturing the theoretical energy potential within the fluid flow past the cylindrical geometry. Consequently, the low velocity performance will be increased to a level more in line with the UK average wind velocity, providing the necessary real-world context that this device aims to achieve. With the basket being a semicircular shape the geometry seeks to auto centre, resulting in its steady state to be always facing into the fluid flow direction making it useful for further evaluation with a baring system and an omnidirectional setup beyond this projects scope.

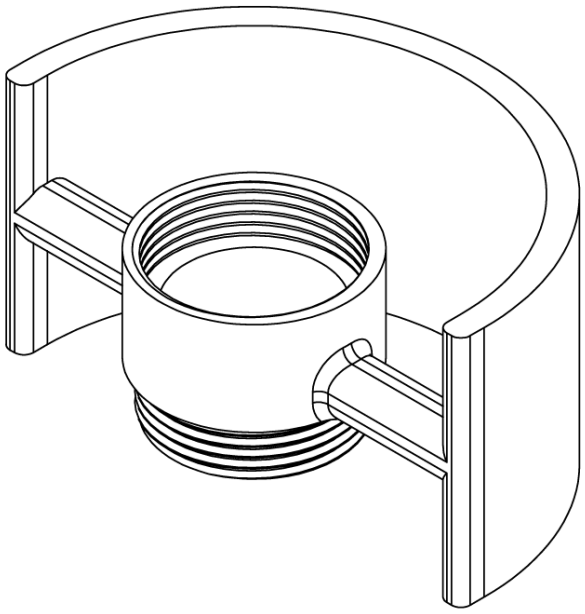


Figure 3.5 Amplifier Design Exported from SolidWorks

The design for the amplifier allows for the installation of the tips into the top of the device as well as directly connecting to the central aerofoil itself. The supporting arms are made symmetrical to avoid any adverse lift conditions produced from their surface resulting in the amplification factor being solely down to the added basket geometry. With this device the greater majority of the force produced is applied on the basket rather than the central circular geometry. The role of the central cylinder within the basket enables the emission and transient behaviour that the amplifier experiences.

3.4 - Material Selection for the Central Rod

Several materials were considered for the central rod which are accessible and reasonably priced. These are as follows: mild carbon steel, stainless steel, aluminium and PLA. The initial material to be ruled out for use of the rod is PLA. Namely, the problematic implementation of a strain gauge to the surface of the material due to the structural layout of the polymer chain without free electrons to measure the strain within the structure. For this purpose, specialized strain gauges would be required for the accurate measurement of forces acted upon the rod [12]. Furthermore, the weak layer adhesion within the 3D print creates weak and promote fractures underload potentially breaking the rod.

Aluminium has benefits like the ability to withstand corrosion. However, the young's modulus the material is approximately 70GPa [13], this means when in comparison to steel at 200GPa the difference roughly 3x softer. Hence the rod will deflect 3x more potentially surpassing the elastic limit of the material, doing so would render the rods' elastic ability to be nulled. This is important as the rod is required to spring back into the neutral position in order to produce the needed conditions for resonance lock in to occur.

Stainless steel wasn't chosen due the higher expense than regular steel. Furthermore, stainless steel has a slightly smaller young's modulus which increase the deflection as stated before. The main advantage that stainless steel has over regular steel is the corrosion resistance the metal exhibits.

Steel was chosen as the material of choice due to its high young's modulus which permeates reasonable deflection, without developing stress fractures when under consistent cyclic loading. Furthermore, the steel is available in varying sheet widths to tailor it to an appropriate depth for a suitable deflection magnitude. In addition, the use of the water cutter to manufacture the geometry provides a reliable and accurate method for producing the physical models for the rod and the base plate for the rod to stay on. The higher young's modulus also helps prevent warping when being water cut which prevents geometry misalignment for assembly purposes.

3.5 - Rod Thickness Calculations and Selection

Within the engineering department several widths of steel sheet are available to be used. To determine which one is best suited to the operational conditions in which the VIV device experiences, calculations of the width must be completed with the options provided of: 1mm, 1.2mm. 1.5mm and 2mm. These calculations will consider the amount of force that will be needed to surpass the elastic limit of the steel with an in-depth beam analysis. The maximum fluid velocity that will be tested in the wind tunnel is $20ms^{-1}$ and thus the forces generated from that will be simulated within Fluent with a uniform diameter of 50mm. This is to be used in this section to aid the selection of which thickness is best suited for the rod, as the forces that expected from CFD can be used on the beam analysis.

To determine the optimal depth, we need to know the elastic limit of the rod, as if it's surpassed the functionality fails and at what magnitude of force that it surpasses the accepted safe limit. To initially perform this calculation the yield stress of steel is required, $\sigma_y \approx 250MPa$ [14] and apply the accepted level of FoS (factor of safety) of 1.5. This provides an allowable yield stress of $\sigma_{allow} \approx 166.67$.

These calculations use the cantilever deflection equation with a point load at location *a*. Location *a* is due to the design having the aerofoil mounting hardware 7.5mm from the top of the structure where all the force generated from the aerofoil is focused into.

$$a = 250 - 7.5 = 242.5mm$$

For this calculation to progress the calculation of the maximum bending moment must be pursued, using the maximum bending moment formula for all depths, followed by determining the force that's applied to the mounting location to achieve the structural limit. For the 2mm steel sheet depth using mm within the calculations:

$$I = \frac{bh^3}{12}$$

$$M_{max} = \frac{\sigma_{allow} I}{c}$$

$$c = distance\ from\ the\ neutral\ axis = \frac{h}{2} = \frac{2}{2} = 1mm$$

$$I = \frac{30 \cdot 2^3}{12} = 20mm^4$$

$$M_{max} = \frac{166.67 \cdot 20}{1} = 3333.4Nmm$$

$$F_{allow} = \frac{M_{max}}{a} = \frac{3333.4}{242.5} = 13.75N\ to\ 2.d.p$$

Using the deflection equation to find the max deflection caused by the calculated force:

$$\delta_{max} = \frac{F_{allow} a^2 (3L - a)}{6EI}$$

$$\delta_{max} = \frac{13.75 \cdot 242.5^2 (3 \cdot 250 - 242.5)}{6 \cdot 200{,}000 \cdot 20} = 17.09mm\ to\ 2.d.p$$

Using the method above the results for the following depths are as follows:

Table 3.1 Showcasing the Allowable Limits for Each Rod Thickness

Depths (mm)	*c (mm)*	*M_{max} (Nmm)*	*F_{allow} (N)*	*δ_{max} (mm)*
2	1	3333.40	13.75	17.09
1.5	0.75	1875.04	7.73	22.79
1.2	0.6	1200.02	4.95	28.49
1	0.5	833.35	3.44	34.19

With these values, to initiate a decision to determine rod thickness, the maximum force must be calculated as to what the VIV device expects to experience when inside the wind tunnel. To obtain this value the highest wind velocity that's intended to be tested will be simulated.

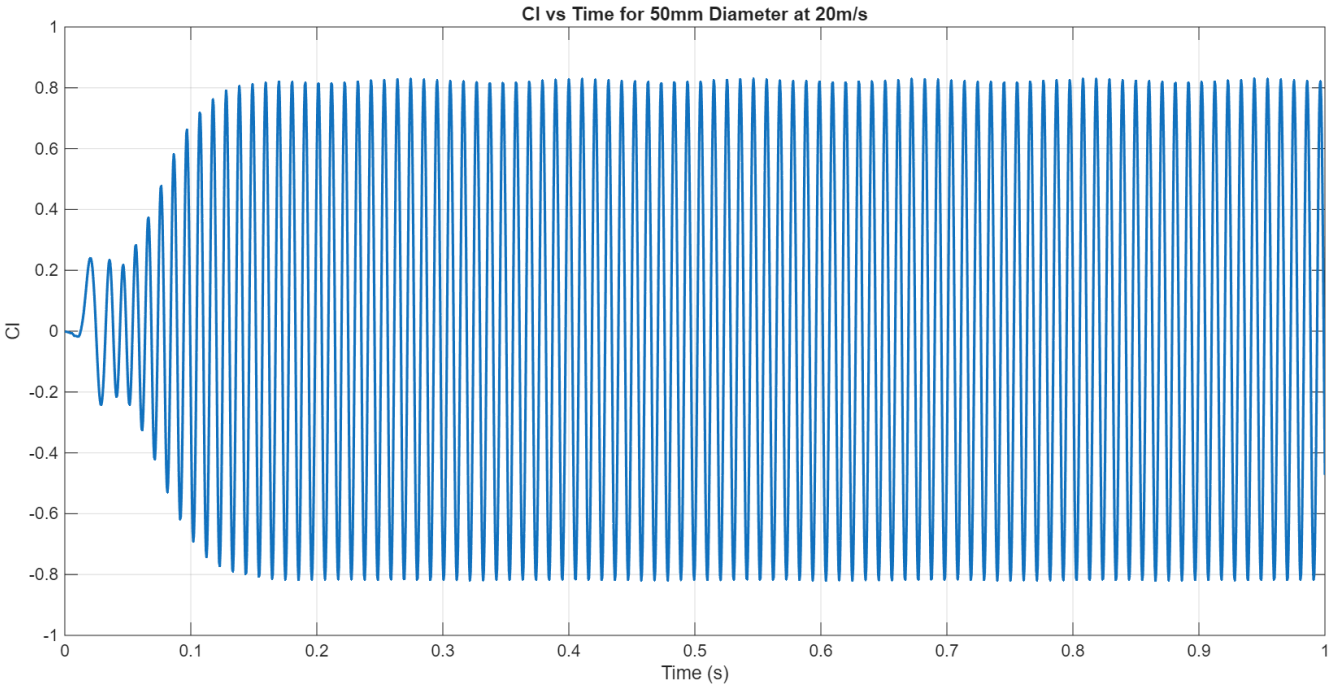


Figure 3.6 CFD Plot of Cl vs Time for 50mm Diameter at 20ms^{-1}

The maximum value for $C_{L,max}^{2D}$ within the CFD simulation is 0.8303m^{-1}, which is within the expected range for a cylinder this size within this magnitude of fluid flow which will be used to determine the maximum force. The $C_{L,rms}^{2D}$ value is 0.58672m^{-1} (calculated in MATLAB), which represents the average force that the VIV device is succumbed to, was taken after the 300th iteration as that's where the simulation converges on the oscillatory behaviour which is at 0.15s for the step size of 0.0005s. As the CFD preformed was a 2D simulation the $C_{L,rms}^{2D}$ number is per meter, as the Fluent assumes the depth to extend

infinitely in and out of the plane when calculating. Therefore, that must be considered when performing the calculations.

Assuming that the entire aerofoil is a 50mm diameter x 500mm height cylinder, we can use the lift equation of:

$$F'_{rms} = \frac{1}{2}\rho U^2 D C_{L,rms}^{2D}$$

Within the above equation the area of the aerofoil is just D (diameter), this is as during 2D simulation as stated above, the fluid flow is assumed to extend infinitely in and out of the plane direction. Consequently, there is no finite area in the planes 3rd dimension resulting in the reference area of the cylinder being simply just D. This means when determining the full force, it assumes that the diameter is constant throughout the aerofoil. Doing this allowed for a fast way to approximate the behaviour of the full-scale aerofoil without preforming a full 3D simulation.

The RMS force per unit length is as follows for the 2D slice:

$$F'_{rms} = \frac{1}{2} \cdot 1.225 \cdot 20^2 \cdot 50 \times 10^{-3} \cdot 0.58672 = 7.19 Nm^{-1} \; to \; 2.d.p$$

For 500mm being the height of the uniform cylinder the force is as follows:

$$F_{rms} = 7.19 \cdot 0.5 = 3.59N \; to \; 2.d.p$$

To calculate the maximum force that a simple cylinder would experience we simply take the peak of the graph above. The highest iteration from within the 2D simulation was $C_{L,max}^{2D} = 0.8303 m^{-1}$. Using the same equation as before the maximum force per unit length is:

$$F'_{max} = \frac{1}{2}\rho U^2 D C_{L,max}^{2D} = \frac{1}{2} \cdot 1.225 \cdot 20^2 \cdot 50 \times 10^{-3} \cdot 0.8303 = 10.17 Nm^{-1} \; to \; 2.d.p$$

$$F_{max} = 10.17 \cdot 0.5 = 5.06N \; to \; 2.d.p$$

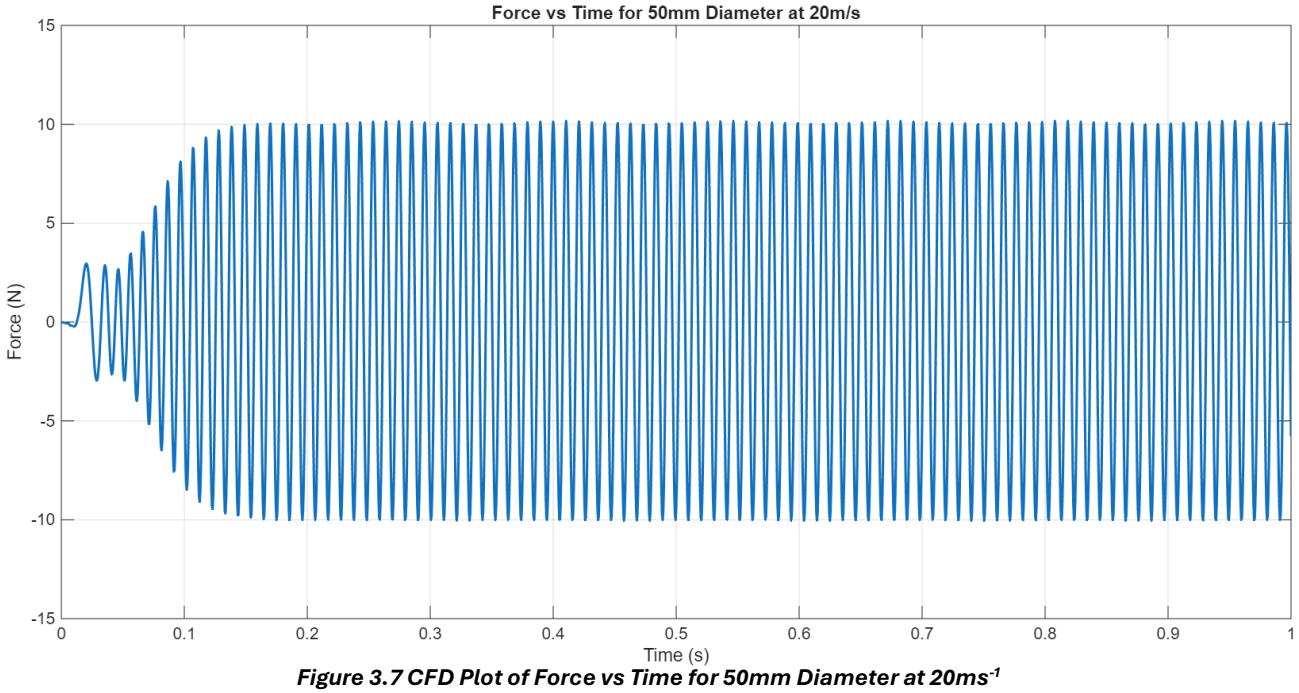


Figure 3.7 CFD Plot of Force vs Time for 50mm Diameter at 20ms^{-1}

The comparison can now be made between the calculated values and the values produced through simulation and whether they align will further increase the validity of the preformed CFD simulations. With the use of MATLAB, the maximum value of F'_{max} produced within the simulation displayed above has a value of 10.1713Nm^{-1} with the F'_{rms} having a value of 7.1873Nm^{-1}. Both of these values when extracted from CFD and analysed within MATLAB share the exact same values as the calculated ones using the Cl graph and thus holds firm.

Using this data, we can now firmly rule out the use of 1mm and 1.2mm as the force produced using a 500mm long device in theory should produce a maximum force of 5.06N, which is larger than the allowable forces of 3.44N and 4.95N for 1mm and 1.2mm diameters respectively. To specify the choice between 1.5mm and 2mm, an optimal deflection of 1cm was chosen as a metric that would suit the operation of the VIV device when under load.

The equations used for this calculation use mm as the length and are below:

$$\delta = \frac{Pa^2(3L - a)}{6EI}$$

$$P = \frac{\delta 6EI}{a^2(3L - a)}$$

Calculation for 2mm sheet depth:

$$P = \frac{10 \cdot 6 \cdot 200{,}000 \cdot 20}{(242.5)^2 \cdot (3 \cdot 250 - 242.5)} = 8.04N \; to \; 2.d.p$$

For 1.5mm depth:

$$P = \frac{10 \cdot 6 \cdot 200{,}000 \cdot 8.4375}{(242.5)^2 \cdot (3 \cdot 250 - 242.5)} = 3.39N \; to \; 2.d.p$$

From these calculations, to achieve the desired 1cm deflection to achieve a steady and consistent oscillatory cycle, 2mm is the preferred option as 3.39N is lower than the F_{max} of 5.06N. Furthermore, the addition of the amplifier geometry increases the magnitude of force applied to the structure and having that ~2N buffer to the 1cm deflection and the overall safely limit of 13.75N, the device is shielded from surpassing the elastic limit and will typically oscillate within the desired range of motion. In addition, the larger the depth of the structure the less it will tend to warp under construction, mounting and operation. This is particularly important as the rod is subject to water cutting and welding processes in order to achieve the final assembly.

3.6 - Natural Frequency Calculations

For identification purposes, it's important to know what frequencies the metal rod and shedding frequency undergo resonant lock in behaviour. To identify, the resonant frequencies the formula below must be used:

$$\frac{f_n}{2\pi} = \omega_n = k_n^2 \sqrt{\frac{EI}{\rho A L^4}}$$

Where $k_1 = 1.875$ and $k_2 = 4.694$ [15].

$$f_1 = \frac{1.875^2}{2\pi} \sqrt{\frac{200 \times 10^9 \cdot 2 \times 10^{-11}}{7850 \cdot 0.002 \cdot 0.03 \cdot 0.25^4}} = 26.09Hz$$

$$f_2 = f_1 \left(\frac{k_2}{k_1}\right)^2$$

$$f_2 = 26.09 \left(\frac{4.694}{1.875}\right)^2 = 163.51Hz$$

Using the shedding frequency calculation with a 50mm diameter, to obtain the first natural frequency at 26.09Hz, the onset of resonant lock occurs at ~ 6.5ms^{-1}. The second natural frequency has a lock in velocity of ~40.88ms^{-1}. With 6.5ms^{-1} being the velocity of the first resonant lock in for this rod, the effects won't be as evident as the wind tunnel won't be able to hold that exact velocity as it varies with roughly 0.5ms^{-1}. Secondly, the wind tunnel testing will not be testing at the excessive velocity of 40.88ms^{-1} or higher so the second and beyond harmonics won't be reached.

3.7 - Rod Assembly

There are key features with the rod assembly that require mentioning, these are the notch and assembly mechanism. Within the rod there is a notch 2cm from the bottom edge of the structure, this allows for an amplification of the strain within that region. Thus, providing the ideal point for the installation of the strain gauges and reducing the magnitude of noise within the electrical circuit helping to identify the actual force from the physical data.

The base plate is to be made from the same sheet metal as the rod. This is to reduce material waste and provide a strong stable platform for the rod to be welded into place. The decision to weld the rod to the centre of the base plate was taken due to base plate requiring a stable flat connection to the bottom of the wind tunnel. Consequently, a tab bending method of rod with a bolted connection wasn't a feasible option, thus, the welding option provided the necessary strength and preserved the natural frequency of the rod.

The strain gauges used are the KFG-1-120-D16-11L1M2S by Kyowa as they are specialised for the use on steel with temperature compensation. They were installed between the notches in the centreline of the rod to achieve a greater defined measurement. They have a gauge factor of 2.08 ± 1.0% and a gauge resistance of 120.4 ± 0.4 Ω. The adhesion onto the steel rod was conducted through a precise and methodical process with the use of sand and glass paper, prepping and cleaning chemicals and adhesion chemicals. The process of applying the strain gauges is delicate as the gauge has to have a perfectly clean and have an unobstructed

connection to the metal, the gauge itself mustn't get dirty from dust or other air particulates and the connections to the wires from the gauge must be preserved without detaching them.

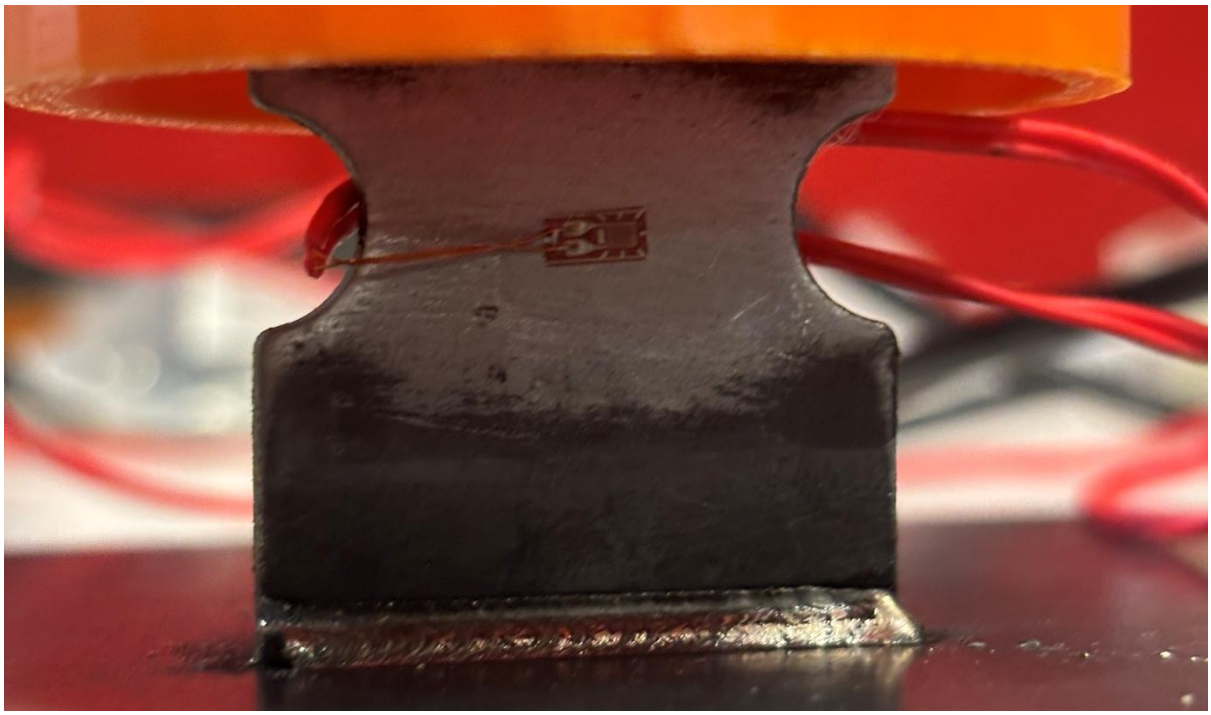


Figure 3.8 Strain Guage Location Situated Between the Notches

Chapter 4 - Manufacturing Methods

Several manufacturing methods were used within this project. This section seeks to elaborate on them and provide the context as to why and how they aided the progression of the project.

4.1 - Additive Manufacturing for Aerofoil Surface

The use of additive manufacturing within the project, specifically, the use of the Bambu Lab A1's 3D printers with the use of Bambu Studio slicer produced the physical models for all of the geometry created within SolidWorks. This provided fast and reliable iterations to be made from early mock-up designs to the large prints of the bottom and centre aerofoil parts without failing.

Key features used within the slicer was the ability to change the quality of print. This was specifically used when developing the clips that surround the mounting hardware, as the scale of the small-scale geometry produced required the use of the finer options.

Therefore, 0.08mm extra fine was used in this case to provide the needed strength and accuracy within the print.

As the geometry submitted to CAM typically contained cylindrical geometric parts, this enforced the use of high definition 3MF files to produce the parts with more polygons, thus, increasing the overall quality of the print and surface for the vortices to shed from. 3MF files allow the use of multiple meshing types when constructing the 3D shape for manufacturing producing a finer more refined 3-dimensional mesh in which the circular geometry benefits from [16].

4.2 - Subtractive Manufacturing for Central Steel Rod

Water cutting was used as the method to produce the steel rod and base plate. Using water cutting on steel promotes the onset of corrosion within the cuts produced. However, the tolerance of 0.25mm when water cutting provided the needed accuracy for the rod and base plate. When developing the models for water cutting, DXF files are required to transfer the data between SolidWorks and the water cutter. For this reason, the development within CAD had to be considered as a 2D shape excluding geometry with depth.

4.3 - Final Assembly Configuration of the VIV Device

Assembling the VIV Device, requires the use of 2 M5 x L16 mm bolts to interlock the rod with the bottom and central aerofoil section. To spread and encourage even loading during oscillations, the tabs where the bolts pierce through are offset by the rods width and each tab is mirrored onto the other side of the rod, therefore, one tab is always pushing the other pulling. This enables the secure setup of the device when the bolts are tightened sufficiently. The base plate contains 8 holes for mounting options within the wind tunnel. These holes are designed to fit M6 bolts through, enabling a strong connection to the bottom of the wind tunnel disabling unwanted movement. Tape was used to cover the small joining seams within the design, specifically, the clips utilized this as the added support was necessary and the added aerodynamic shaping added performance during the testing phase.

Chapter 5 - Physical Experimentation

Physical experimentation was conducted using several methods to determine the forces applied to the VIV device during operation. These were the use of strain gauges that measure the direct strain within the structure at their mounting point and visual analysis of the deflection that allow for the cross comparison of oscillation frequency to the strain gauge readings.

5.1 - Experimental Methods

Several methods were used during testing. Notably, the use of a half bridge with amplifier circuit in conjunction with a Nucleoboard for data extraction was trailed. However, even with the inclusion of bypass and shunt capacitors, the electrical noise produced from the circuit resulted in an unreadable output. This setup used a differential op amp circuit with a gain of 1000 to produce the needed resolution for the forces that the rod experiences. Minius the electrical noise, the circuit worked as intended as larger forces visibly displayed on the live MATLAB graph, however, it's not applicable for the VIV application. Therefore, the use of specialized equipment was needed. Resulting in the P3 Strain Indicator and Recorder by Micro Measurements being used as the primary data extraction tool used in this project. This is due to the build in low pass filter, automatic circuit matching and excitation, analogue and digital data readings and the ability to use the half bridge configuration.

5.1.1 - Strain Guage Configuration

As mentioned previously, a half bridge configuration is used as the strain gauge setup. This is due to the temperature compensation this setup provides as well as the increased resolution within the readings as one gauge is in compression and the other in tension. With one gauge being in compression and the other in tension the potential difference between the output of the Wheatstone bridge is greater which provides that increased resolution.

In order to implement the strain gauges to the P3, the correct setup is needed. Following the instructions inside the lid of the P3, pins P+, S- and P-, were used to set up the half bridge configuration. Once the wiring is implemented, the P3 has to know the channel and type of bridge being used. Channel 3 was used so channel 3 was set to "HB shear",

this completes the half bridge circuit with internal resistors matching the 120Ω resistance of the gauges exactly, compensating for any tolerance error with the gauges. The shear option allows for the P3 to understand the direction of the deflection being used and the shear strain is the metric required with the intended bending direction.

5.1.2 - Strain Guage Calibration

To produce a meaningful output from the P3, factors such as the gauge factor, full scale value, mV/V and balancing function, have to be configured. When the P3 is first turned on the reading of με is just electrical noise, to achieve the expected neutral reading of 0με, the reading must be auto balanced. Once the auto balance procedure is completed the strain reading will be 0 and deflecting the rod will produce a strain that's positive or negative depending on the direction of the bend. For the reading of strain to be accurate the gauge factor (G) has to be set within the settings of the channel output, for this case the G of 2.08 was used to match the gauges values.

When measuring force from the P3, the full-scale value and mV/V settings have to be calibrated. As the strain expected shouldn't surpass more than 1000με, 1000 was set as the full-scale value, providing the correct order of magnitude within the P3. To calibrate the scaled force the mV/V had to be set in order to sync the digital values to the real magnitude of force applied [17]. Using a newton meter and incrementally increasing the value of mV/V the value of 15.000 provided the best synchronisation to that of the measured force to that on the digital reading.

5.1.3 - Data Extraction

When using the P3 in digital mode for data extraction, the use of a USB-B cable was required with the use of the P3 software to enable data extraction. Using the record function within the P3 software and selecting channel 3 as the active channel, the minimum sampling rate available is 1Hz. This creates a large uncertainty within the values as lots of data is lost due to the slow sample rate. However, the forces and strain that it does capture are accurate at that instantaneous moment, this then allows for the calculation of the average force provided that the data is collected over a wide time range which should be comparable to the RMS force simulated within Fluent.

The P3 also provides analogue data output which can be used to transfer the data to an oscilloscope which eliminates the sample rate errors taken with digital data readings. The analogue data extraction uses the low pass filter which eliminates noise substantially when compared to the initial differential amplifier circuit produced, this was signified by the 60Hz rejection setting. Furthering noise reduction a coaxial cable was used when transferring the analogue data from the P3 to oscilloscope enabling all wind speeds tested to be displayed and thus analysed.

5.1.4 - Camera Mounting and Visual Aids

A wide-angle lens was used to capture the motion of the VIV device within the wind tunnel directly above the centreline of the model through the Perspex glass. A recording of the motion allowed for the cross comparison between rate of oscillations between visual and processed data results, as well as the comparison between the magnitude of deflection between visual and processed data. The magnitude of oscillations will be calculated using the known diameter at the top of the VIV device and comparing it to the maximum displacement measured with a scaled ruler to the true scale of the image.

5.2 - Wind Tunnel Testing

Wind tunnel testing consisted of testing and measuring all 3 tips, control model and the amplifier device. Each configuration of the VIV device had strain, force, oscilloscope readings and visual recordings taken. This allows the data to be verified and cross examined between methods of recording, increasing the validity of the results taken. All recordings of force and strain with the digital setup were taken over a time period greater than ~3 minutes to produce a large data set. For all devices wind speeds of: $10ms^{-1}$, $15ms^{-1}$ and $20ms^{-1}$ were taken as they produced a force great enough for accurate data extraction. The amplifier device has $4ms^{-1}$ and $7ms^{-1}$ wind speeds tested in addition the previous three as the increased force produced has a great enough magnitude to be measured. These lower wind speeds provide the necessary real-world context for everyday operation within the UK climate with an average wind speed of $3.3ms^{-1}$.

5.2.1 - Control Results

The results provided within this test will allow for significant and direct comparisons between the predicted CFD results and the physically tested ones. The following results were taken using the P3 with digital readings for both force and micro strain.

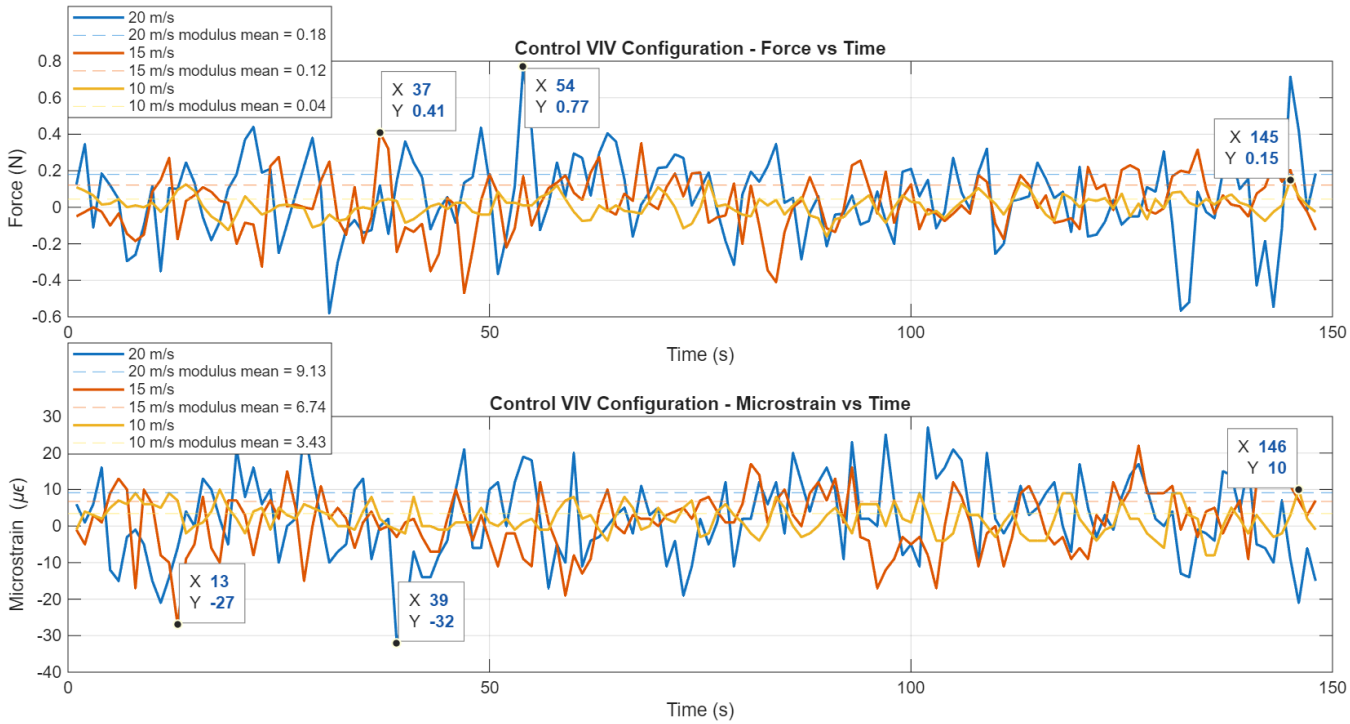


Figure 5.1 Control VIV Configuration Digital Results from the P3 for Differing Velocities Displaying Force and Microstrain

Most notably, the 1Hz sampling rate can clearly be seen with this graph as each peak is pronounced via 1 point. This signifies extreme data loss, specifically, when determining the shedding frequency as there is no coherent oscillatory behaviour evident within these results. Secondly, the absence of wind speeds below that of $10ms^{-1}$ is due to the forces and strain being produced at these lower wind speeds become increasingly harder to interpret from each other as the values generated are harder to extrapolate from the noise produced by the measuring device at these lower magnitudes.

The peaks of this graph most likely represent the value when the sampling rate coincided with the peak deflection point in that instantaneous moment, however, the exact peak is most likely higher due to the data leakage with the sampling rate. The data produced from this P3 in digital mode can however produce a value for the mean force that the VIV device experiences during operation as the values captured are perfect for

this role. The mean had to be calculated taking the modulus of the values to avoid it tending towards 0 due to the negative values taken when deflecting in the negative direction, this is so that the comparison to the RMS force simulated within Fluent can be made.

To verify the results taken with the P3 the comparison between the visual max deflection and maximum peaks observed within the digital readings can be made. To display the maximum deflection 3 images are overlayed 1 of the neutral position (marked with the black arrow), 1 of the maximum left displacement (marked with the red arrow) and the last with the maximum right displacement (marked with the blue arrow). A scaled ruler is set to measure the displacement in the maximum displacement conditions, due to the known diameter of the tip of the control test is 50mm, which is shown with the black arrows being 5cm apart.

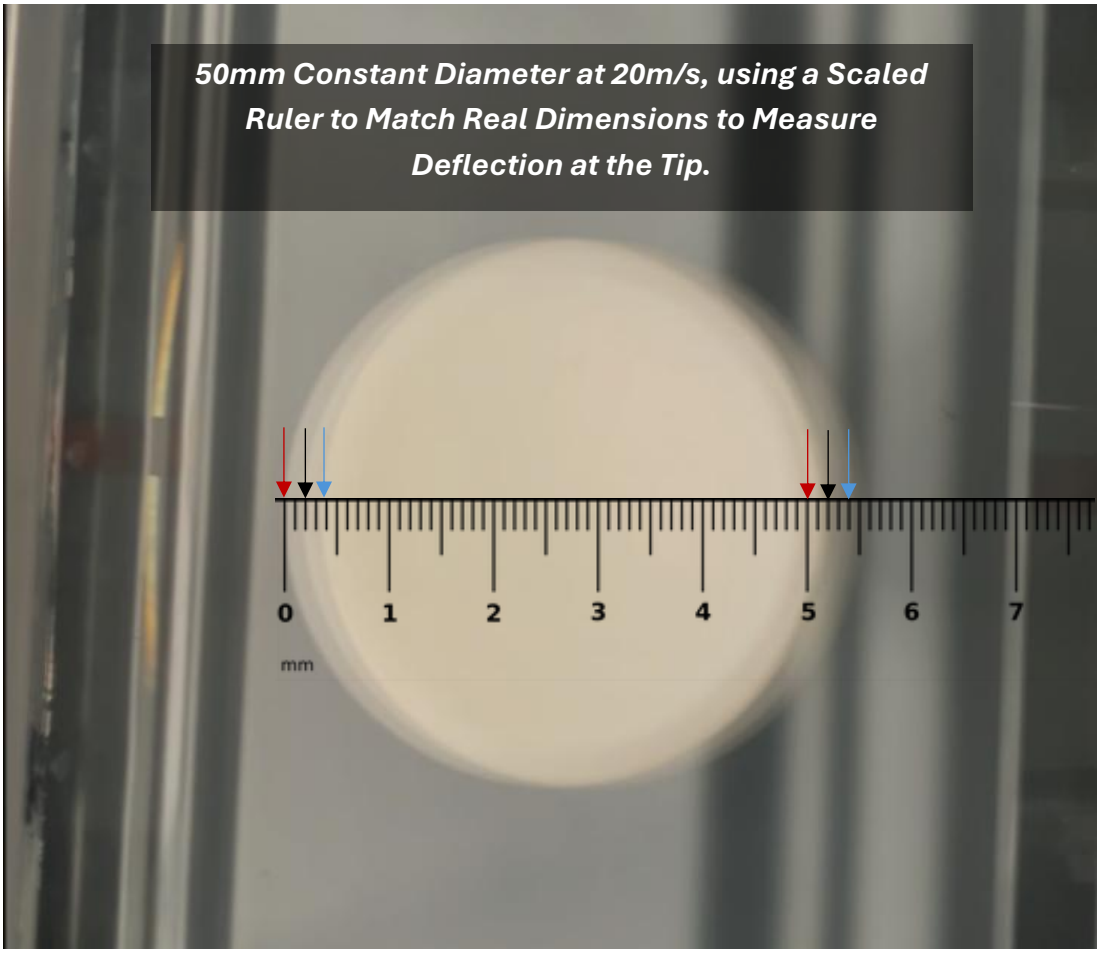


Figure 5.2 Visual Deflection of Overlayed Images of the Control Setup at 20ms^{-1}

Within the image created the maximum displacement is displayed to be 2mm in each direction respect to the neutral position. To calculate the force required to initiate this displacement, the determination of the deflection angle with the use of Pythagoras has to be performed to quantify this displacement at the location at mounting point situated on the metal rod.

$$\theta_{Difflection} = \sin^{-1}\left(\frac{2}{500}\right) = 0.229° \; to \; 3.d.p$$

Therefore, the deflection at the rod is $250\sin(0.229) = 1mm$. The deflection can be used with the beam deflection equation to calculate the force that the rod requires to achieve that displacement. The calculation used mm as the length unit and the values used are previously calculated in the rod thickness section:

$$F_{Max\ Load}^{20m/s} = \frac{3EI\delta}{a^3}$$

$$F_{Max\ Load}^{20m/s} = \frac{3 \cdot 200000 \cdot 20 \cdot 1}{242.5^3} = 0.84N \; to \; 2.d.p$$

The maximum force measured using the P3 in digital mode for the 20ms^{-1} wind tunnel test was 0.77N. Compared to the maximum force deciphered from calculation based upon visual deflection of the device, 0.84N was predicted is an 8.33% difference in values. The peak of 0.77N is a singular large peak on the P3 measurement tool indicating the need to run the simulation for a large enough period of time to achieve a peak value were the shedding and sampling frequencies align.

To further refine the value and shedding frequency of that the device operates with, the oscilloscope readings can be analysed. The oscilloscope is set to a scale of 50mV per division and has 250ms roll. This allows for the oscillations to be clearly displayed within the display window where data analysis can be performed. The P3 provides and excitation voltage of 2.5V into the half bridge strain gauge setup when using the low range.

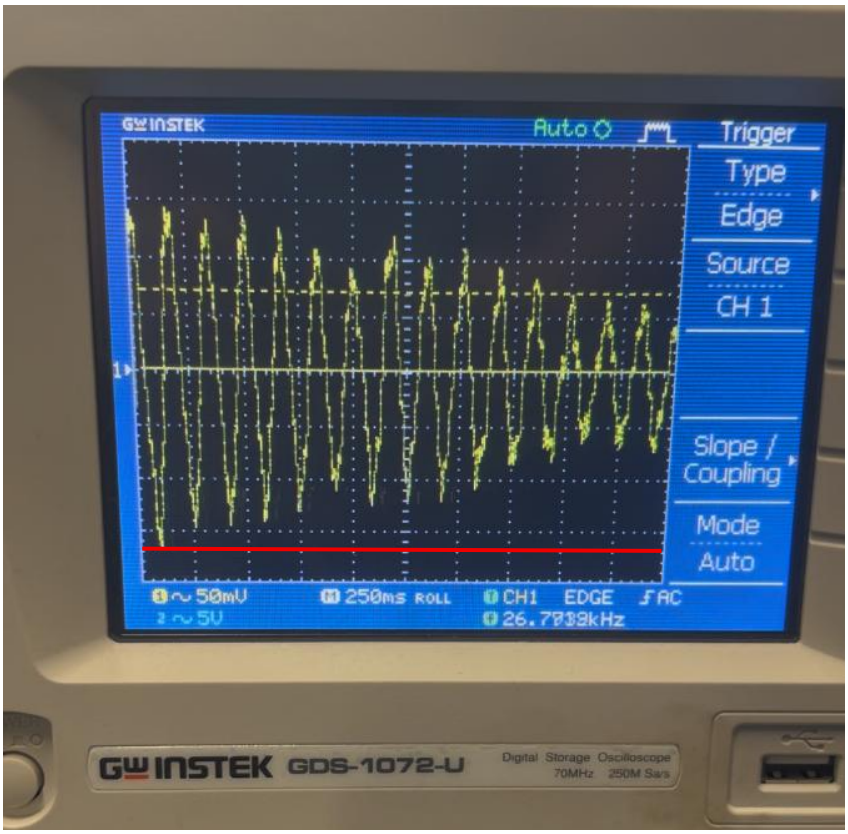


Figure 5.3 Peak Oscilloscope Reading for Control Setup at 20ms^{-1}

The image of the oscilloscope shows the moment in which the oscillations where at its absolute maximum during the wind tunnel run as shown by the red line. This peak has a magnitude of 170mV. In order to determine the force at that point, first the strain must be calculated from this peak value. Its key to note that the full-scale value is 1000με at 2.5V excitation. With this information the equation to determine strain can be setup with the sensitivity equation:

$$\mu\varepsilon = \frac{1000}{2.5} \cdot V_{out} = 400V_{out}$$

With 170mV being the output voltage V_{out} = 0.17V:

$$\mu\varepsilon = 400 \cdot 0.17 = 68\mu\varepsilon$$

This can also be compared to greatest magnitude strain reading taken from the digital output from the P3 for 20ms^{-1} at 32με which is less than half of the absolute maximum strain that the rod experiences from the peak reading from the oscilloscope, further idealising the digital readings to be used as averages.

With the strain value now computed to be 68με, the bending moment can now be quantified using the equation:

$$M = \varepsilon \frac{EI}{y}$$

Using values previously calculated in mm again:

$$M = 68 \times 10^{-6} \left(\frac{200000 \cdot 20}{1} \right) = 272Nmm$$

In order to use the bending moment equation, the distance between the location of the bending moment calculation at the strain gauge location 2cm from the bottom of the rod to the applied force location 7.5mm from the top results in the distance for the calculation being: $z = 250 - 20 - 7.5 = 222.5mm$.

This can now be used in the equation:

$$F = \frac{M}{z} = \frac{272}{222.5} = 1.22N$$

For future calculations this entire process can be compacted down to the simplified equation of $F \approx 7.192V_{out}$ and the deflection caused by this force with the constant rod geometry has the compacted equation of $\delta_{mm} \approx F \cdot 1.188$.

The force of 1.22N is 31.15% increase from 0.84N obtained from the visual analysis. However, this could be down to a number of reasons. Firstly, the visible deflection follows an arc and therefore part of its travel in in the y-axis rather than the x-axis, thus increasing the distance travelled as the deflection caused by 1.22N is 1.45mm, in which it's highly possible that the arc travels 0.45mm downwards in its trajectory. Furthermore, this maximum value was obtained after recording the oscilloscope for a longer time period than the visual analysis and in conjunction with the higher resolution with the oscilloscope minor increases in oscillations can be identified to a greater extent. Moving forward, the visual analysis will provide a supportive case to the calculated force and displacement from the oscilloscope readings acting as a verification method for these readings.

The CDF simulations provided a value of 5.06N as the maximum force value for the control 500mm VIV device however the real device tested received a force of 1.22N. This is a 75.9% difference, which is a substantial margin of error within these two results. This could be down to several factors but most notably the shedding frequency and the metal rod's natural frequency not working in harmony thus, destructively interfering with each other. For example, the rod is elastically moving right as a vortex is shed on the left numbing the effect the shed vortex has. Within the simulation and calculations for this velocity and diameter the scientifically accepted values the shedding frequency should be 80Hz, whereas the actual device oscillates at 5Hz when counting the oscillations in the oscilloscope and the video recording of the device in operation. These further aid this hypothesis onto why this device behaves this way when under load. Furthermore, after physical analysis it could be determined that the lower quarter of the device activity seeks to dampen these oscillations due to the cantilever affect the rod induces therefore the bottom of the rod requires opposite edge vortices to be shed when compared to the top, this is a limitation in the design when seeking to obtaining a consistent wake field.

5.2.2 - Different Tip Results

To compare the performance of each tip, the average was calculated with each parameter of fluid velocity and tip design for both force and strain. The digital output for each of these data points is also plotted against each other providing visual representation of the performance differential.

Table 5.1 Average Force Values with Differing Tip Designs at Multiple Velocities

	AVERAGE FORCE VALUES (N)		
TIP TYPE:	***10m/s***	***15m/s***	***20m/s***
Control	0.04763	0.11494	0.17402
Dome	0.04613	0.07803	0.15900
Taper	0.04886	0.05247	0.14510
Expanding	0.04109	0.04600	0.08120

Table 5.2 Average Microstrain Values with Differing Tip Designs at Multiple Velocities

	AVERAGE MICROSTRAIN VALUES (με)		
TIP TYPE:	***10m/s***	***15m/s***	***20m/s***
Control	3.32937	6.73034	9.12838
Dome	2.95985	4.19477	9.69132
Taper	3.09506	3.13293	7.88593
Expanding	2.84270	2.75278	4.19106

On average, the tips preformed worse than the control setup. Specifically, the expanding top noticeably underperforms in all of the calculated averages with each value consistently being the lowest of all of the tested metrics. In addition, the expanding tip doesn't noticeably increase its value in either the force or the micro strain wind tunnel runs in the step from 10ms^{-1} to 15ms^{-1}. In fact, the average micro strain actually decreases from 2.84270 to 2.75278 in this step. This could be down to the 1Hz sampling rate neglecting the peaks of the run during testing or the fact that the expanding top could disrupt the development of the onset of a consistent wake field. Nether the less, the expanding top clearly ranks as the worst tip tested, as its peak force and strain 50% worse than the control counterpart. This disproves the assumption that an expanding tip could create a larger force due to the increased size of the diameter at the location of the largest bending moment throughout the VIV device.

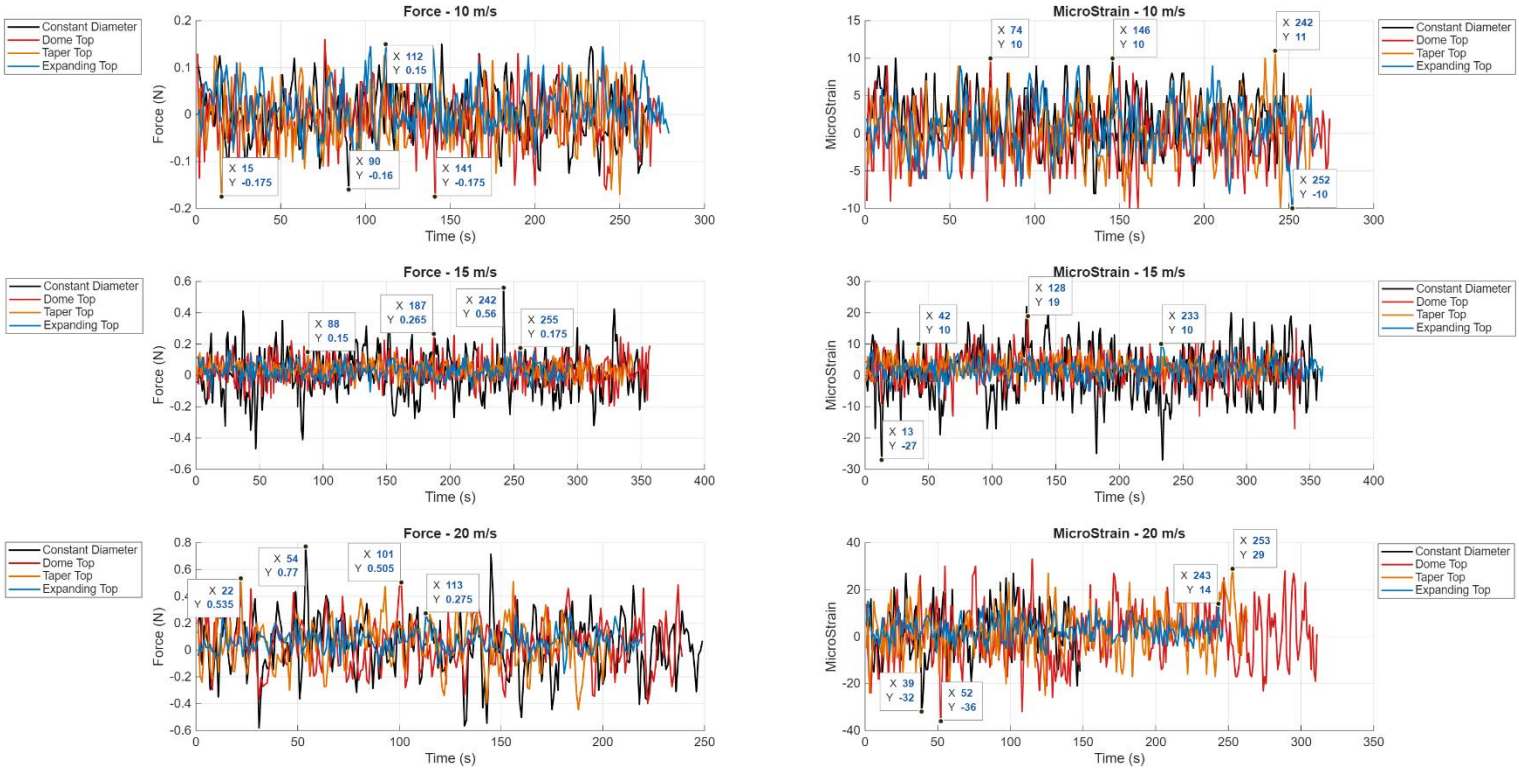


Figure 5.4 Force and Microstrain Digital Values from P3 for all Tested Tip Designs and Velocities

Both the dome and taper tip options outperformed the control diameter in certain wind speed tests, particularly, the values of dome micro strain at 20ms^{-1} and taper force at 10ms^{-1}. This could signify that these tips have a preferred operational window and thus perform better at certain wind speeds. This is also evident in the maximum points provided throughout the wind tunnel tests as only within half on the tests did the control test have peaks larger than that of the tips. The control tests performed better across the 15ms^{-1} tests for both strain and force and the 20ms^{-1} force test.

Disparity between force and strain is apparent due to the inability to record both simultaneously on the P3. This creates these scenarios where in the force graph, one tip looks preferable than in the other strain graph a different tip looks preferable. Nether the less, the averages provide an insight into which tips operate with greater efficiency, in this case the tips that taper in are better as it appears that increasing the shedding frequency increases the average force applied to the device. To further the analysis the oscilloscope readings will determine the true peaks of the oscillations as all recordings of the oscilloscope are for a minute.

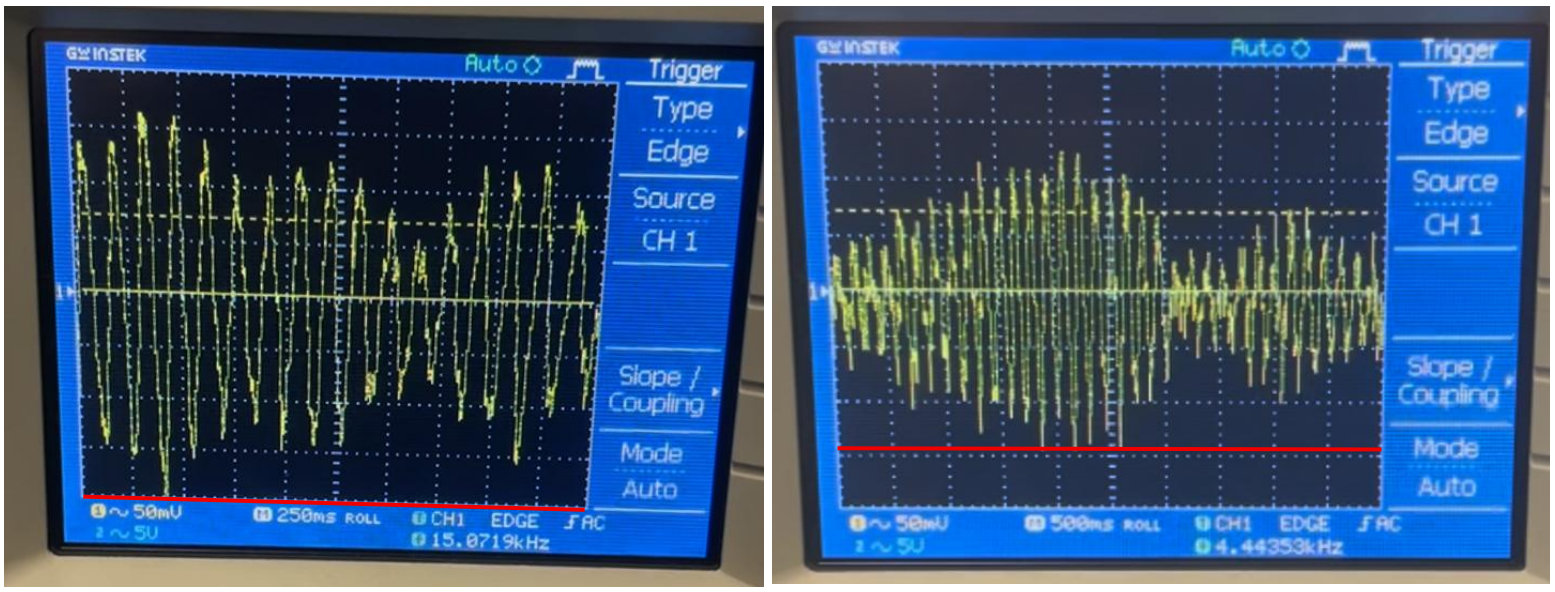


Figure 5.5 Peak Oscilloscope Readings for Dome and Taper Tips at 20ms^{-1}

The oscilloscope images show the dome and tapered tips maximum magnitude voltage reading during the 20ms^{-1} test with the red lines. Importantly, the tapered tips oscilloscope reading has a time scale of 500ms per division rather than the 250ms one shown in the dome top. The dome top exhibits a maximum voltage of 200mV whereas the tapered top has a peak voltage of 149.5mV. These equate to 1.4384N and 1.0752N respectively using the previously resolved equation with a deflection of 1.608mm and 1.202mm at rods mounting location. In comparison to that of the control test, the dome top out preforms it however, the tapered top doesn't. It's important to note that the control diameters length is 2cm larger than that of the tip tests. This is due to the aerofoil being specifically made of the span of the length which interacts with the fluid flow to be 500mm, whereas the tip the hight is 500mm which could slightly alter the results obtained.

Furthermore, the dome tip exhibited less natural cyclic damping when operating compared to the control test, reducing aerodynamic losses and thus increasing the potential energy harvesting the device is theoretically capturing as the RMS force is greater. Both of these tips exhibit an oscillation frequency of 8Hz which is greater than the 5Hz the control test operates at, this is a significant increase as the 3 extra oscillations of 1.4383N over large time periods substantially increases the power output from the device. Furthermore, the dome tip has a 15.18% larger force per oscillation creating more power.

To calculate the instantaneous power that this device produces the formula: $P(t) = F(t) \cdot v(t)$ can be used. However, the signal produced is oscillatory, this means that maximum power produced is when there is 0 displacement due to the rod having the greatest velocity traveling through the neutral position. In order to calculate the maximum velocity, the modelling of the displacement must be sinusoidal.

$$x(t) = x_{max} \sin(\omega t)$$

Using the modelled displacement formula, to determine the velocity at a specific time during oscillations the formula can be differentiated.

$$v(t) = \frac{dx(t)}{dt} = \omega x_{max} \cos(\omega t)$$

Similar to that of the displacement formula, the force produce must also be modelled as oscillatory function which is in phase with that of the velocity formula. This results in the equation:

$$F(t) = F_{max}\cos(\omega t)$$

With the use of the equations formulated, the substitution into the instantaneous power equation can be made.

$$P(t) = (F_{max}\cos(\omega t)) \cdot (\omega x_{max}\cos(\omega t)) = F_{max}\omega x_{max}\cos^2(\omega t)$$

This equation can then simplify down due to the maximum and average values of the cos term being 1 and ½ and with ω = 2πf.

$$P_{max} = 2\pi F_{max} f x_{max}$$

$$P_{avg} = \pi F_{max} f x_{max}$$

Using these equations the power generated during the cycles are as follows to 3.s.f: P_{max} = 0.116W = 2.79Wh and P_{avg} = 0.0581W = 1.39Wh for the dome tip and P_{max} = 0.0650W = 1.56Wh and P_{avg} = 0.00325W = 0.780Wh. Assuming that the power produced can be extracted fully, these power values are significantly smaller than that of a conventional hobby wind turbine. The power produced during the 20ms^{-1} is more comparable to a micro turbine.

5.2.3 - Amplifier Results

Notably, the amplifier allowed for wind speed tests down to 3-4ms^{-1}, this is as the force produced at these lower wind speeds was great enough to produce meaningful results. Importantly, the distinction of the lowest wind speed being tested at 3-4ms^{-1} was used due to the relatively poor ability of the wind tunnel to hold a steady velocity at the lower wind speeds therefore, the range of this wind speed is displayed as 3 to 4ms^{-1}. This is significant for this project as this lower wind speed range allows for the direct comparison to real world wind speeds. With the device now being able to be tested at these lower wind speeds the validity of this method of power generation becomes greater.

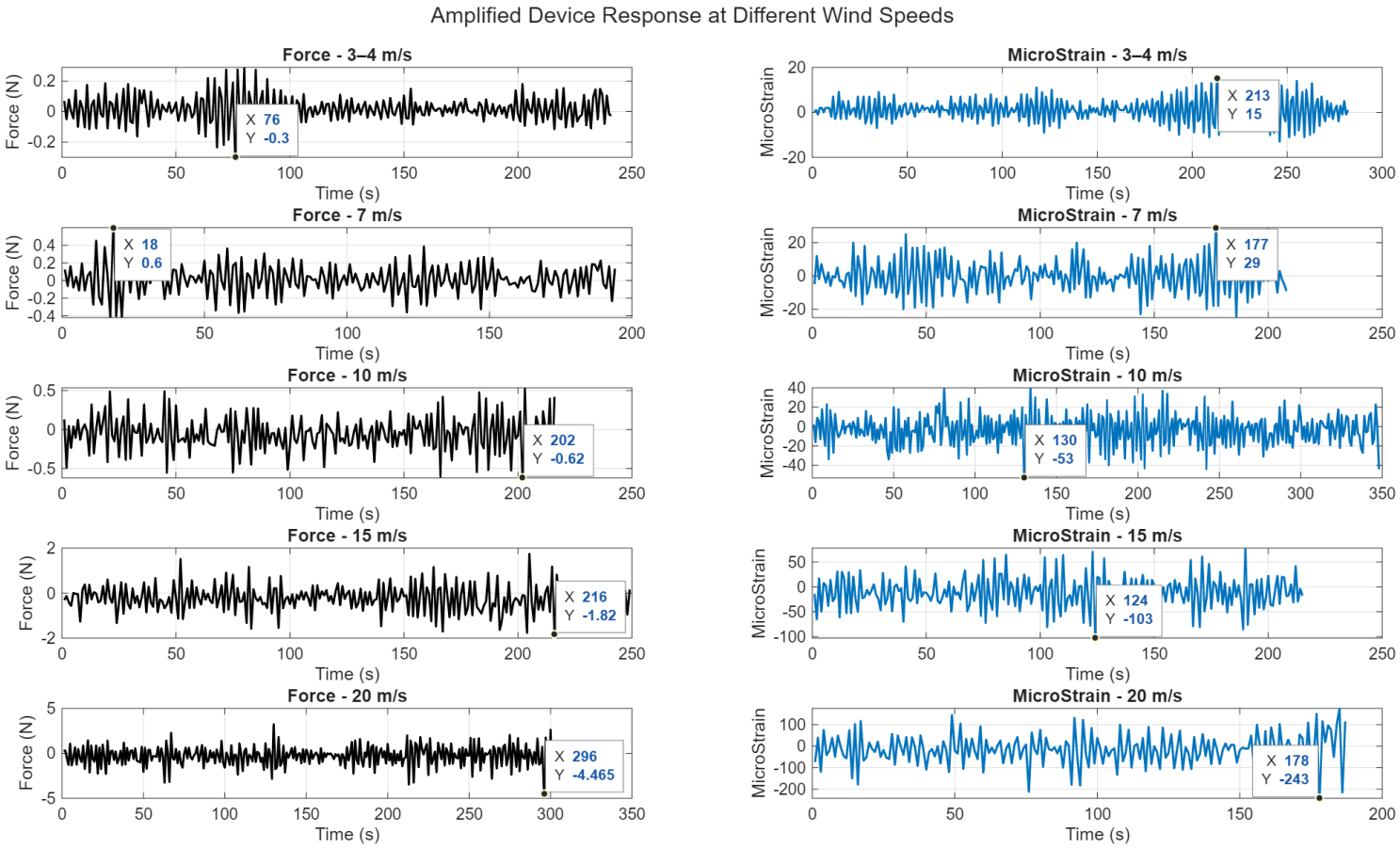


Figure 5.6 Amplified Device Response for Force and Microstrain at Differing Wind Speeds

Impressively, VIV resonance lock in can be identified within the 7ms^{-1} test. This is due to the predicted resonance lock in velocity to be at 6.5ms^{-1} and with the wind speed variation during the experimental runs caused by the wind tunnel, the correct conditions are met for this phenomenon to occur. The maximum point during the force 7ms^{-1} test, produced a value of 0.6N whereas the 10ms^{-1} test only produced 0.62N, clearly displaying the amplification factor caused by the resonance lock in. Notably, 6.5ms^{-1} is an achievable velocity to target during real world operation and therefore, can be targeted for peak efficiency for a greater developed model using this projects rod setup.

The cyclic damping pattern experienced across all tests performed can be clearly seen within the 3-4ms^{-1} test. This is due to the reduced frequency that the device oscillated at allowing the P3 to capture most of this behaviour and displaying it accordingly.

Table 5.3 Average Amplifier Results of Force and Microstrain for Differing Velocities

	AVERAGE AMPLIFER RESULTS				
Unit:	**3-4m/s**	**7m/s**	**10m/s**	**15m/s**	**20m/s**
Force (N)	0.07407	0.13930	0.19100	0.50888	0.92628
Microstrain (με)	4.26241	8.17788	13.36782	28.45116	59.80214

With the use of the amplifier device, the average force and strain values appear to exponentially increase with wind speed. To determine the actual peak force and power produced form the maximum and minimum fluid velocities, the largest oscilloscope readings must be taken. The smallest and largest tests will be analysed to produce the greatest contrast between results with wind speed being the only variable.

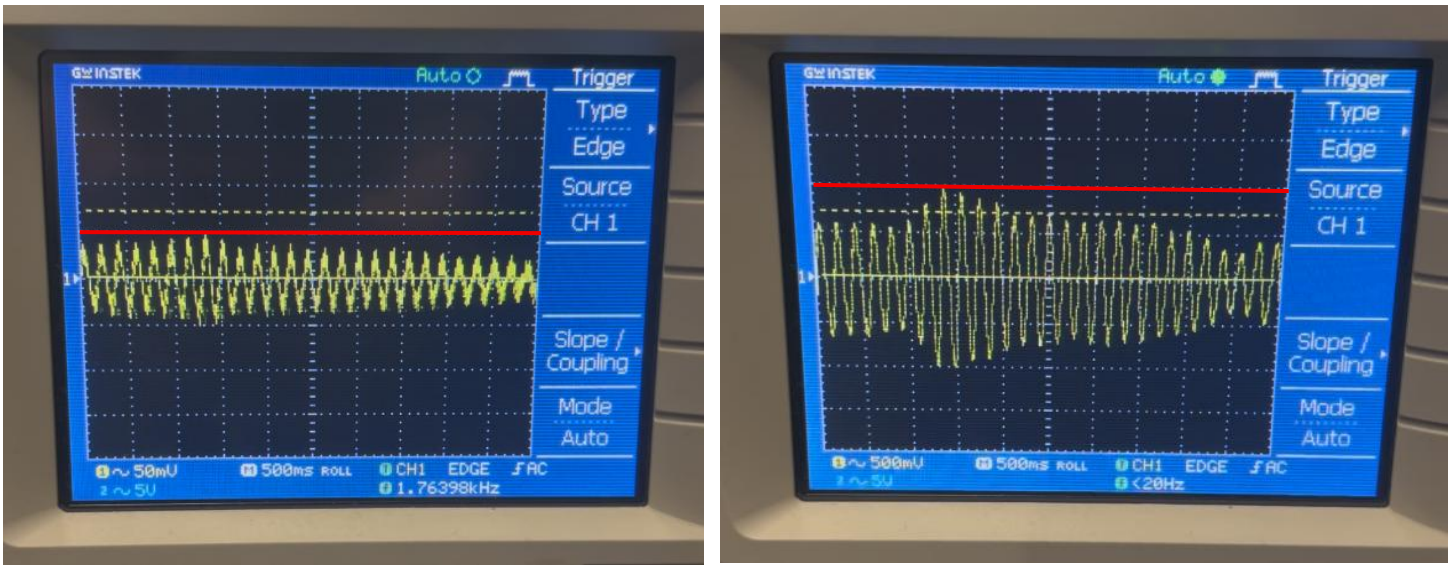


Figure 5.7 Peak Oscilloscope Readings for 3-4ms^{-1} test and 20ms^{-1} test for the Amplifier Device

The left oscilloscope reading displays the 3-4ms^{-1} reading and the right image shows the 20ms^{-1} reading. Notably the 3-4ms^{-1} reading uses 50mV per division compared to that of the 20ms^{-1} reading with 500mV per division. This was due to the fact that the response generated during the 20ms^{-1} was much greater than the domain displayed within the 3-4ms^{-1} test. With this the maximum voltage response during the physical experimentation was 50mV and 1V respectively. The difference within the electrical noise can be seen as the smaller forces produced have increased widths and fuzzy response with wider peaks when compared to the clearly defined response within the 20ms^{-1} test.

The oscillation frequency differed between these two experiments however, the difference in frequency is minimal. This is as the 3-4ms^{-1} test has a frequency of 5Hz compared to the 20ms^{-1} having 6.5Hz as its operating frequency. This could be down to the mass and thus momentum of the metal rod counteracting the desired shedding frequency as the rod physically cannot move faster than it is with the forces applied.

The peak force and displacement produced during these experimental runs are as follows. The 3-4ms^{-1} run with the maximum voltage reading of 50mV using the force equation previously formulated experiences 0.3596N under peak load. This equates to a displacement of 0.427mm to 3.d.p. The 20ms^{-1} run generated a force of 7.192N with 1V of generated oscillation, resulting in an 8.54mm deflection. Comparing this to the maximum force permissible of 13.75N solidifies the decision of using a 2mm thick metal rod as the elastic limit isn't breached whereas in the other options would have failed during this test.

To determine the power during these runs the use of the maximum and average power equations for the peak force moment will again be used. This results in for the 3-4ms^{-1} run being P_{max} = 0.006W = 0.15Wh and P_{avg} = 0.003W = 0.075Wh. The 20ms^{-1} test produced P_{max} = 1.909W = 45.82Wh and P_{avg} = 0.954W = 22.91Wh.

The power produced is very small and inefficient, for example to power a 5W led light, assuming 100% efficiency ran for 24 hours at 20ms^{-1} would only run for 9 hours. Therefore, the average power produced is comparable to that of the POFET 1W mini vertical wind turbine. The size of this turbine is comparable to that of the tested VIV device as typically turbines need to be elevated to work effectively. The lower wind speeds although detectable produce not enough force to become a suitable method of generation, this is particularly significant as the lower forces are produced at wind speed that real world wind velocities occur, reducing the ability of this device to capture and generate a suitable amount of power during operation.

5.2.4 - Comparison of Results

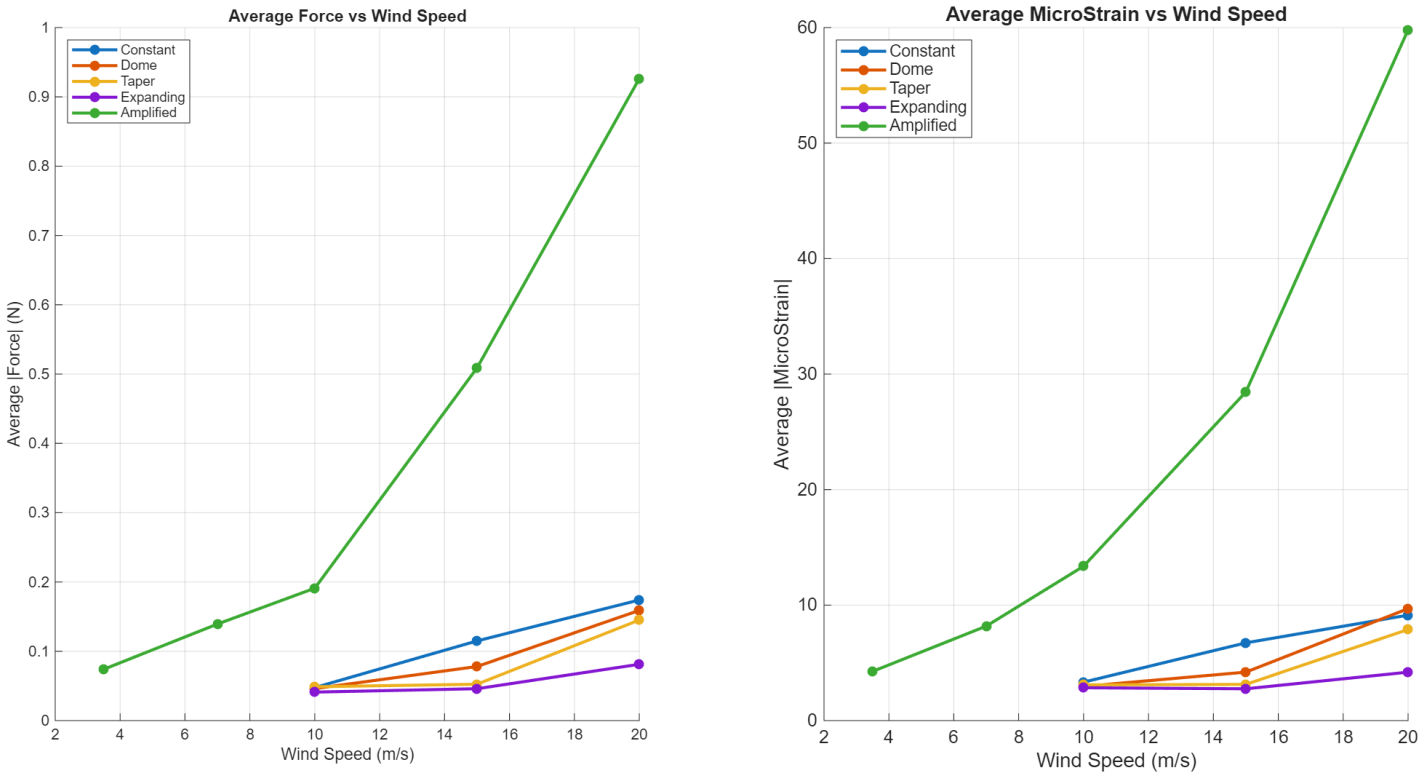


Figure 5.8 All Tested Devices Digital Average Force and Microstrain Against Wind Speed

The graph above shows how each configuration compared to each other. Clearly, the amplifier drastically increases the amount of force captured, creating a distinction when compared to that of the tip devices. There is a 5.32x increase in efficiency when compared to the best performing force value without the amplifier device. This is a significant gain in the amplification factor that this project sought after, with this device alterations to known limitations further improvements can be made to increase this further allowing the geometry to reach its theoretical maximum efficiency according to CFD.

The exponential nature of the amplification graph can be clearly seen in the figure above whereas the graphs representing the tips appear disjointed. With the control and expanding tip graphs there appears to be a linear pattern when increasing the wind speed, particularly, the expected outcome for the control diameter is a linear pattern due to the relationship between the diameter, Strouhal number and wind velocity but the expanding tip also appears to follow this trend but to a less efficient extent. Both the taper and dome tip appear to increase exponentially, however, due to the minimal data

points taken during the tip tests due to the inability to decipher the response from noise produced in the slow speed tests created this limitation. It is in hindsight possible to take data readings from fluid velocities of 12.5ms^{-1} and 17.5ms^{-1} to bridge this gap and produce a graph with greater coherence. However, with the P3 in digital mode actually deciphering the differences and the 1Hz sample rate perfectly aligning with a peak in an oscillation could skew the results inappropriately. To counter this all peak oscilloscope readings from all of the tests are displayed below.

	Peak Voltage (V_out)	Peak Force (F_max)	δ_max (mm)	P_max (W)	P_avg (W)	OSCILLATION FREQUENCY (Hz)
			DOME			
10ms	0.0750	0.5394	0.6408	0.0163	0.0081	7.5000
15ms	0.1000	0.7192	0.8544	0.0290	0.0145	7.5000
20ms	0.2000	1.4384	1.7088	0.1081	0.0541	7.0000
			TAPER			
10ms	0.0750	0.5394	0.6408	0.0163	0.0081	7.5000
15ms	0.1250	0.8990	1.0680	0.0452	0.0226	7.5000
20ms	0.1495	1.0752	1.2773	0.0647	0.0324	7.5000
			EXPANDING			
10ms	0.0600	0.4315	0.5126	0.0097	0.0049	7.0000
15ms	0.0650	0.4675	0.5554	0.0114	0.0057	7.0000
20ms	0.1250	0.8990	1.0680	0.0422	0.0211	7.0000
			CONTROL			
10ms	0.0450	0.3236	0.3845	0.0047	0.0023	6.0000
15ms	0.0500	0.3596	0.4272	0.0068	0.0034	7.0000
20ms	0.1700	1.2226	1.4525	0.0669	0.0335	6.0000
			AMPLIFIED			
3-4ms	0.0500	0.3596	0.4272	0.0048	0.0024	5.0000
7ms	0.1150	0.8271	0.9826	0.0306	0.0153	6.0000
10ms	0.1600	1.1507	1.3671	0.0494	0.0247	5.0000
15ms	0.3000	2.1576	2.5632	0.1911	0.0956	5.5000
20ms	1.0000	7.1920	8.5441	2.5096	1.2548	6.5000

Figure 5.9 All Oscilloscope Values and Calculations

It's evident that the amplifier device oscillates at a slower rate than the other devices however, the force it produces is significantly larger. All devices seem to experience the same inconsistent oscillation frequencies throughout increasing wind speeds, in contrast to the anticipation is that they should increase. However, during some velocity variations it decreases and some it increases, in particular the control diameter going from 6Hz to 7Hz back to 6Hz. The surface of the aerofoil could play a part in such decrease in frequency due to the slight polygonization of the surface. This could result in the vortices not shedding correctly during certain wind speeds reducing the operational frequency the VIV device oscillates at. The dome tip performed better than the control test throughout all tested metrics during the oscilloscope readings, this clearly displays the limitations of the 1Hz sampling rate as used previously. Therefore, for future developments in this field a domed top provides the best efficiency.

Chapter 6 - Conclusions

In conclusion, the VIV device operated successfully within the wind tunnel validating the computationally driven design methodology. However, the forces and power produced ended up lower than anticipated but with the significant gain provided by the amplifier device suggests that VIV devices remain a highly promising, low-maintenance solution to powering small scale remote applications where a traditional wind turbine is unsuited to. While the aerodynamic principle was proven, further refinement of the design and scalability is necessary to bring it up to a standard comparable to a standard wind turbine. With the identification of resonance lock in at $7ms^{-1}$ where shedding frequency appeared to match that of the rod's natural frequency, proves that precise tunning for optimal wind speeds, VIV devices can operate at higher efficiencies within specific niche environmental conditions, furthering its credibility as a feasible energy harvesting device.

The practical implementation of this technology would be most effective in fields that operate on Ultra-Low Power (ULP) networks. Given the largest average power output overserved being 0.954W, the technology is well suited to powering LoRaWAN gateways for backup communications in remote fields and applications of structural accelerometers on bridges for example, typically devices that require intermittent bursts of energy rather than a continuous high voltage supply increasing efficiency in the low power domain [18]. Transitioning this technology to a full-scale device would require the supplementation of battery technologies to rectify the lack of control over energy generation. This is a vital aspect, even though the device is a fit-and-forget mechanism, the device relies on prevailing wind and therefore cannot predict grid demands supplementing the required load when needed. Therefore, a battery system in coherence with a full-scale model would be required for suitable operation.

Whilst the VIV device might not surpass the conventional wind turbine as the leading method of capturing environmental energy, increasing the scope beyond dedicated energy harvesters the retrofitting of existing passive infrastructure with this technology could provide a great and sustainable energy generation practice though unconventional means. Specifically, the utilization of dead aerodynamic structures with

piezoelectric technologies would allow for passive generation. With the vast number of structures that are subject to high wind loading this could present an opportunity for localized power generation with telecommunication lines, transmission lines and streetlights. With 6.5 million streetlights in the UK [19] and over 4500Km of transmission lines within the UK [20], significant energy could be generated from these factors without the risk of added mechanical complexity that using a traditional rotating turbine design would cause. Furthermore, a key identifiable case to implement a VIV device would be on top of electricity pylons, aiding the reduction in transmission losses and providing power to pylon line monitoring systems.

Consequently, while the VIV aerofoil design may occupy a specific niche within the energy generation market when compared to traditional rotational turbines, the integration of this technology into the built environment could result in a purposeful step towards an energy secure and decarbonized energy grid providing a better and greener future for future generations.

Recommendations for Future Work

With time being a fundamental limiting factor during this project, further aspects of this project can be further explored in order to increase the devices efficiency at capturing energy within fluid flows.

Further, analysis into the development of the amplifier device regarding the diameter and distance from the central aerofoil section would provide a greater insight into the feasibility of the VIV device as a method of energy generation. This increased data set would allow for the optimisation of the prototype part used within this project creating larger and more consistent vortex generation in which the device can benefit from. With alternating the arc length geometry of the amplifier, potential increase to the shedding angle could be observed. This principle could allow for a greatly increased efficiency within the device as a whole, due to the pressure differential between the vortex shed and the ambient pressure can be utilised to a greater extent. In addition to this, alternating this geometry to for example a semi-oval shape, could be done to seek if differing geometry provide greater and more energy efficient shedding.

With wind in real life alternating its AoA the development of an omnidirectional variant of the VIV device would be the next logical step into creating a wind turbine alternative. To achieve this, further material analysis into a circular rod within the aerofoil would provide this multiaxial loading needed for this aspect. Complementing this, a redesigned mounting method and location would have to be developed to accommodate these added requirements. With the VIV device being omnidirectional the amplifier hardware would need to be able to angle itself into the prevailing wind direction to become operational. To achieve this the amplifier device would have to be mounted on low friction bearings so that when the basket catches the air, it self-orientates into the prevailing direction, increasing the oscillations doing so.

A general redesign with end plates and the removal of aerofoil geometry below the assembly location would drastically increase the performance of the device and should be further looked into. The inclusion of end plates would allow for the vortex sheet to be focused within a 2D plane instead of leaking outwards and reducing the efficiency. Research into mounting these end plates throughout the span of the aerofoil could also be performed seeking into whether it focuses and increases the magnitude of vortex shedding or breaks down the vortex sheet. With the lower geometry not being load baring; to increase the efficiency, aerodynamic shaping could occur on this device to produce stronger or direct the vortices outwards with a greater angle, whilst being cylindrical itself to preserve the vortex sheet.

Scalability of the design couldn't be tested during this project due to the size of the wind tunnel, further analysis into a device comparable to that of a medium scale wind turbine could be another logical step with real world testing. With the CFD preformed the is the prediction that increasing diameter and length of the wind device both linearly increases the force produced for vortex shedding in laminar flow with turbulent shedding regime.

Finally, as this project studied the feasibility of using VIV devices as a method of energy generation. Methods of extracting this energy will have to commence in order to produce the highest efficiency possible that are suited for oscillatory behaviour. This could prove to be particularly complex due to the alternating nature of the generation and thus producing a DC signal from this device would most likely require the use of a

rectifier circuit, reducing the overall efficiency of the device. Further research into electromagnetism could also be a method of energy generation like that of the Vortex Bladeless design to compare these methods together, creating a VIV device with a greater feasibility for future use as a carbon reducing energy generation method.

Critical Reflection

During the project I encountered multiple setbacks that prohibited my workflow and stunted my progression. Most notably, early on during the project I was experiencing multiple error messages when transferring the generated mesh into ANSYS Fluent. This validation error that kept occurring and prompted multiple attempts to re generated the mesh and iterate my way out of the error. This caused a week's delay within my projects timeline that had lasting effects as reruns of the simulations where required. In addition to this, during this phase of the project where I was learning how to suitably use Fluent, attempts to run the application locally were made on my personal computer. This caused issues during the project as after I saved the current workbench and closed it, the licencing prevented me from opening that workbench and carrying on the work I had just conducted. Therefore, the decision was made to use the virtual machine as well as the computer labs to carry out the simulations. However, with the use of the virtual machine and the computer labs, storage for the simulations was limited. This prevented my simulations from being ran as long as I initially wanted to, so I attempted to use cloud storage for the simulations I carried out. This resulted in another licencing error as the file location wasn't local where the licence is, the compromise was made to reduce the length of each simulation and reduce the autosaves from every time step to every 10.

During week 16 I was overviewing my current rod calculations at the time and had the realisation that the numerical results for the value of Cl from the simulations were wrong. This ended up being down to the reference geometry being set incorrectly, however, the unintended consequences were great as vast amounts of time had to allocated into allowing multiple of these simulations to run fully with 2000-time steps. This allowed my calculations to be amended and make them comparable to scientifically accepted values and increase the validity of the simulations.

In week 18, I started officially started testing a week after intended due to the metal rod being water cut from steel 1mm too thick. This was down to a communication issue between me and the technician at the time which got swiftly amended. However, during testing I developed a differential amplifier circuit which ended up being inoperable with the forces that the rod was experiencing. This was down to the electrical noise being produced within the circuit being too large even with attempts to reduce it with the addition of the shunt and bypass capacitors. In general, this circuit worked but was more efficient at detecting much larger forces which wasn't suitable for this project.

Overall, I enjoyed this project and learnt lots about this field of aerodynamics with how they intertwine with mechanical systems. The mistakes I've made whilst progressing though have allowed me to become a better engineer by creating methods to mitigate, bypass and fix the issues experienced.

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Appendix

Appendix – Chapter 2

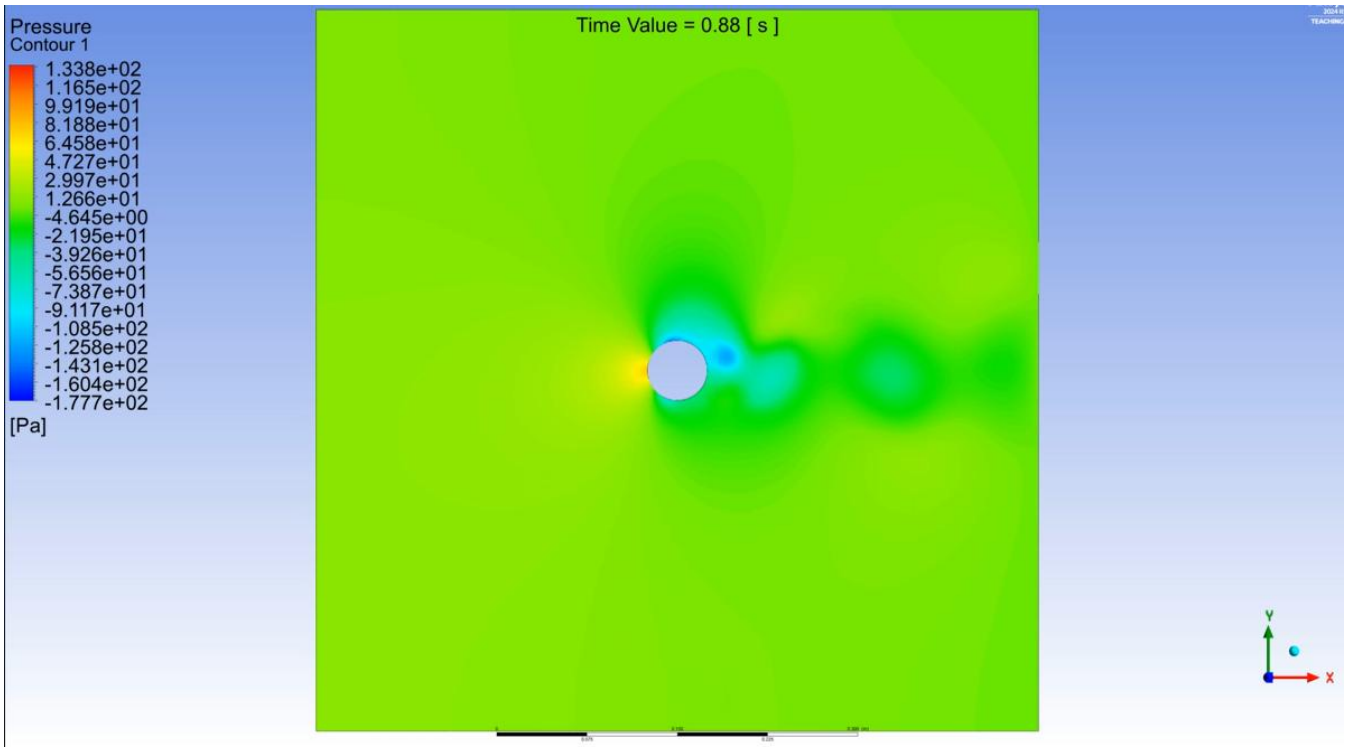


Figure A2.1 Pressure Contour Plot of 50mm Diameter at 10ms^{-1}

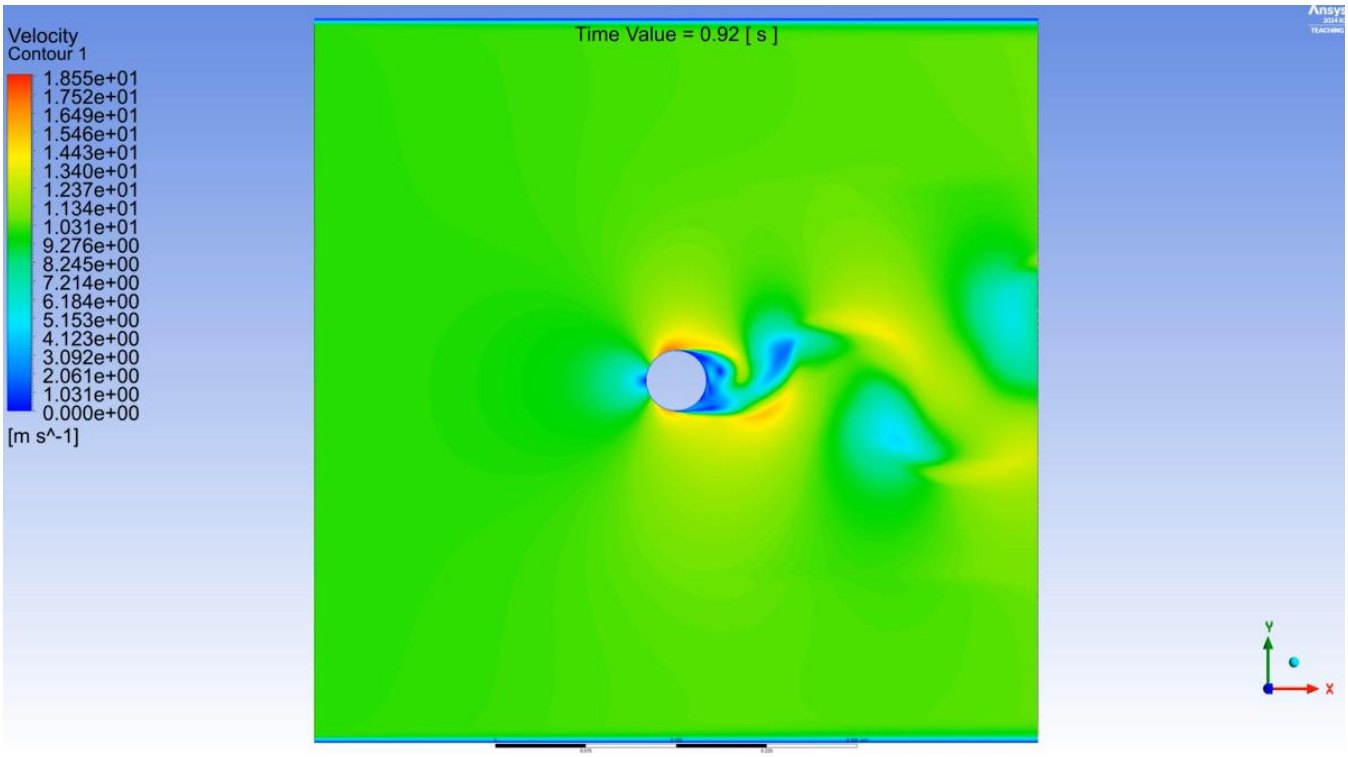


Figure A2.2 Velocity Contour Plot of 50mm Diameter at 10ms^{-1}

Appendix – Chapter 3

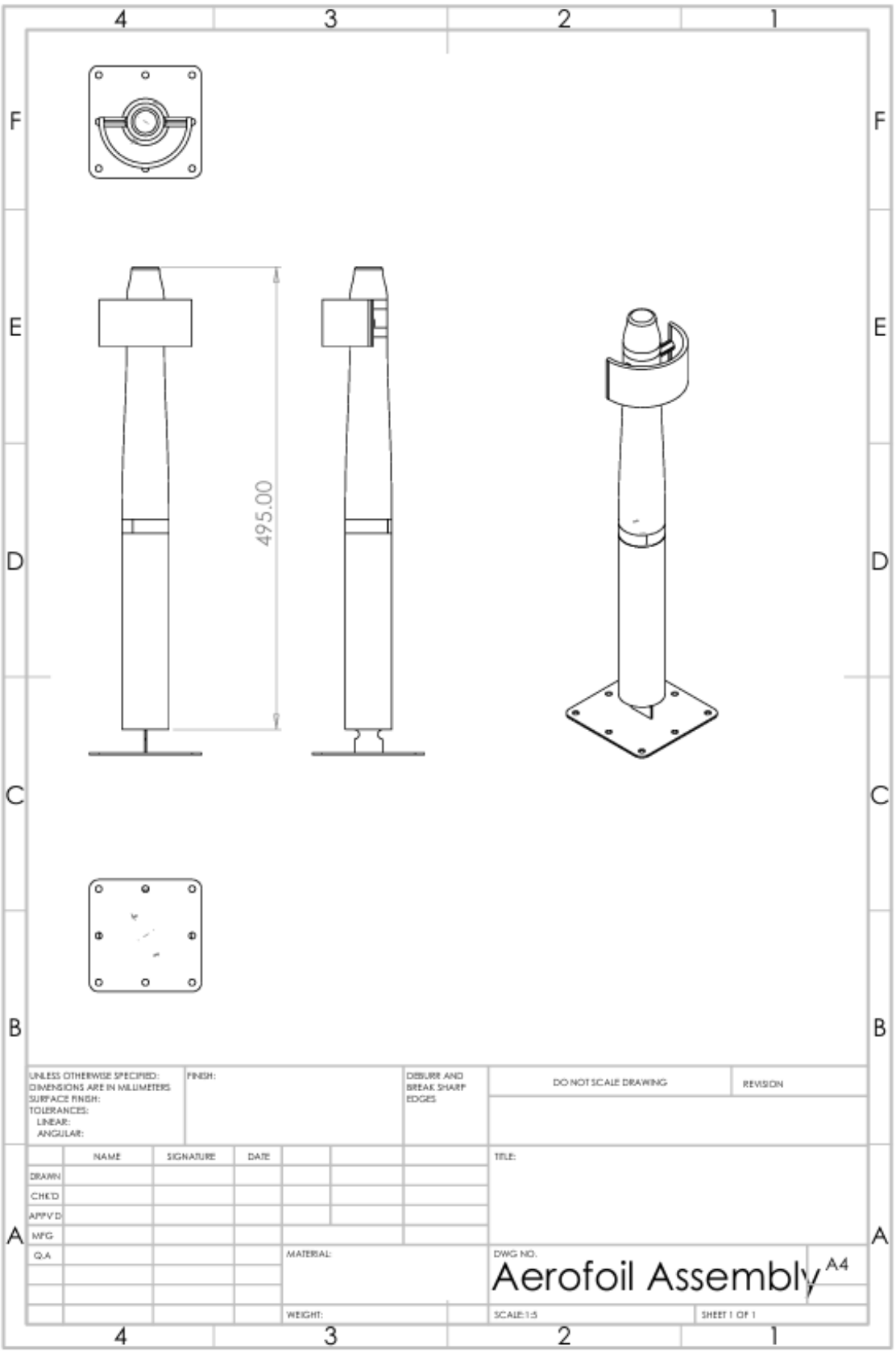


Figure A3.1 Aerofoil Amplifier Assembly

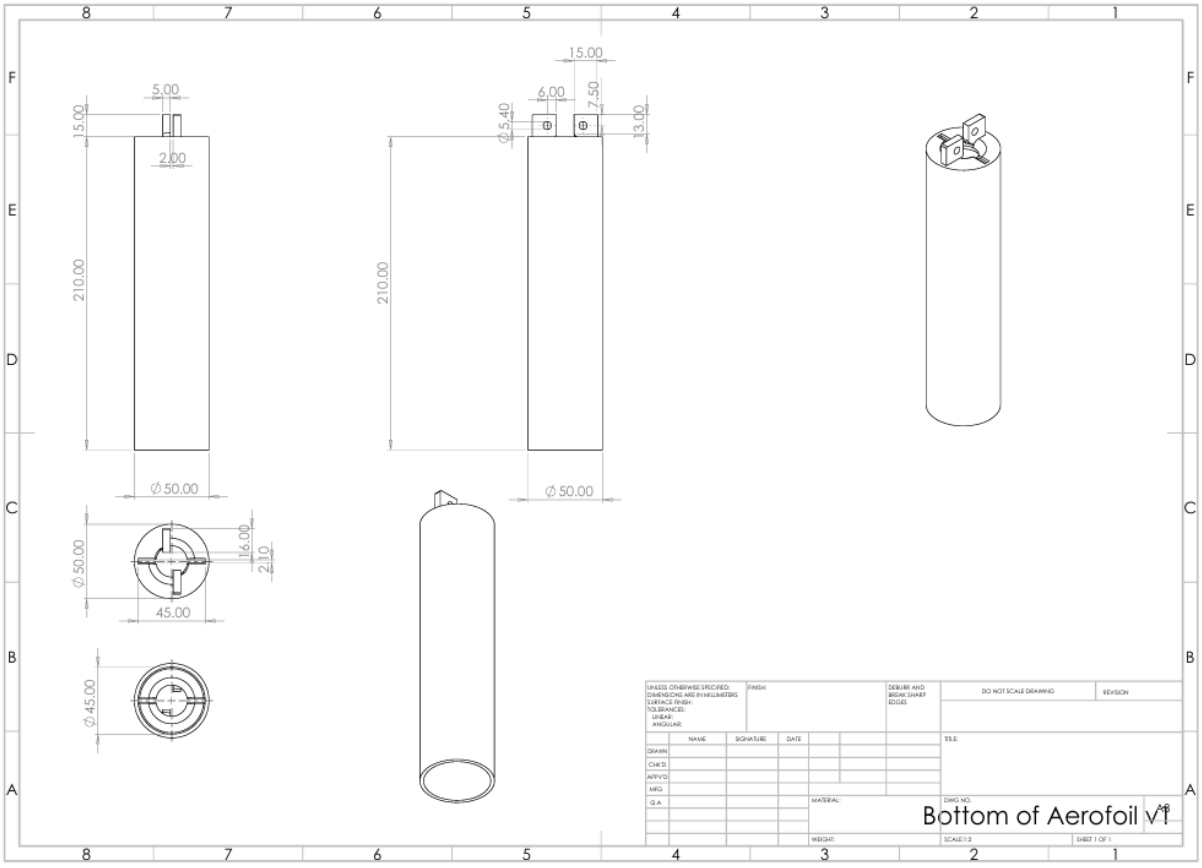


Figure A3.2 Bottom of Aerofoil

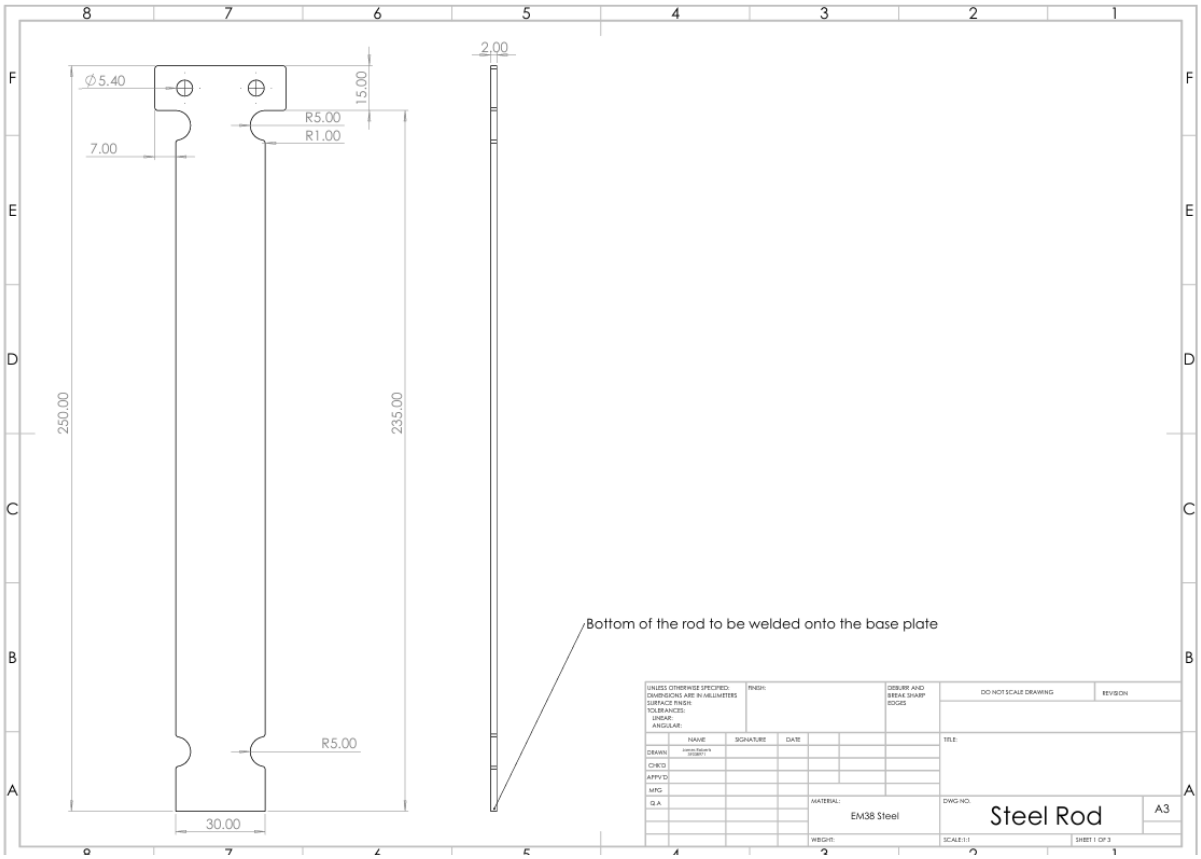


Figure A3.3 Steel Rod

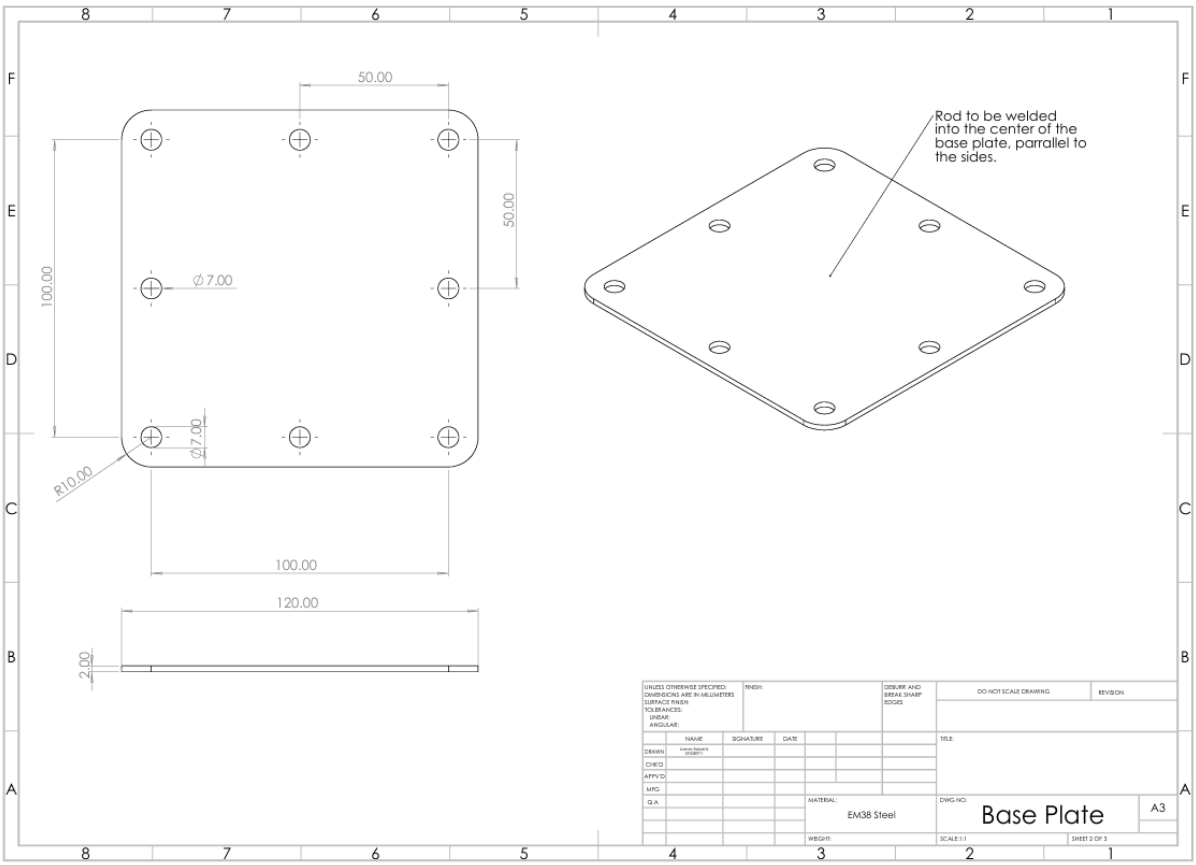


Figure A3.4 Steel Base Plate

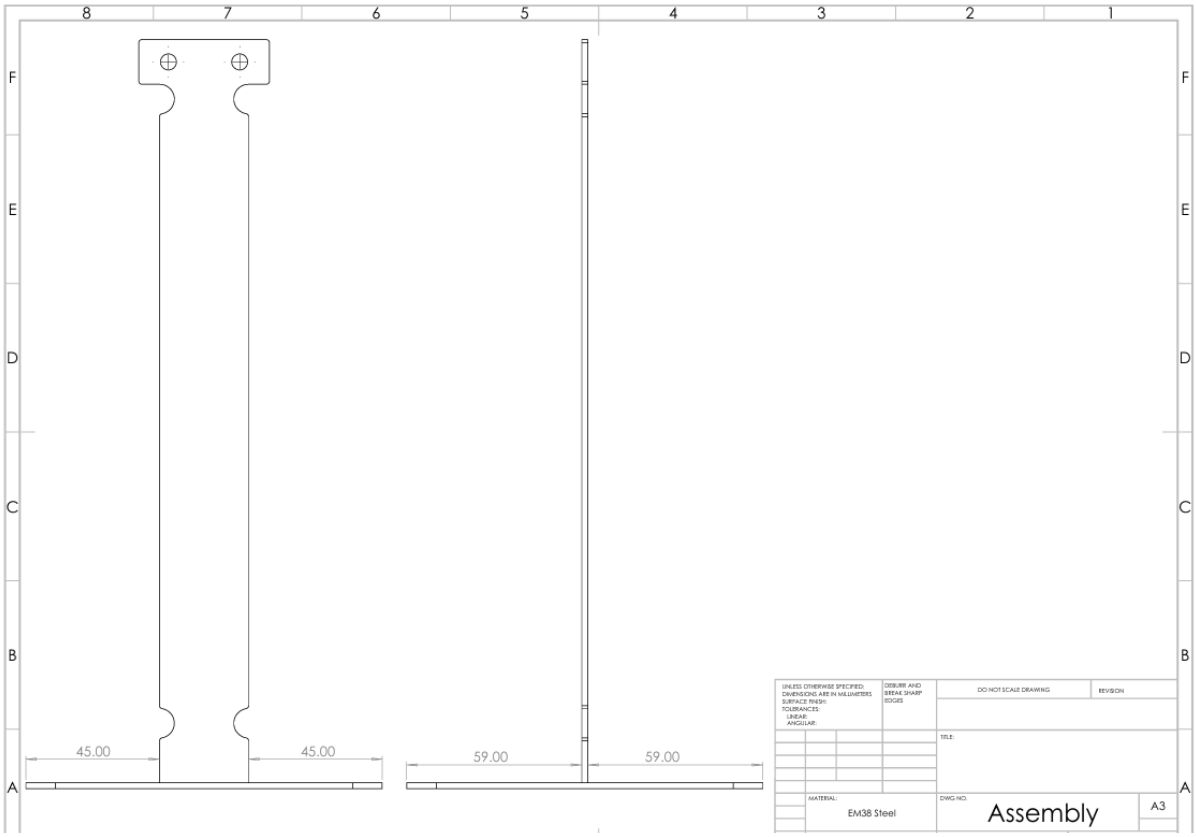


Figure A3.5 Rod Assembly

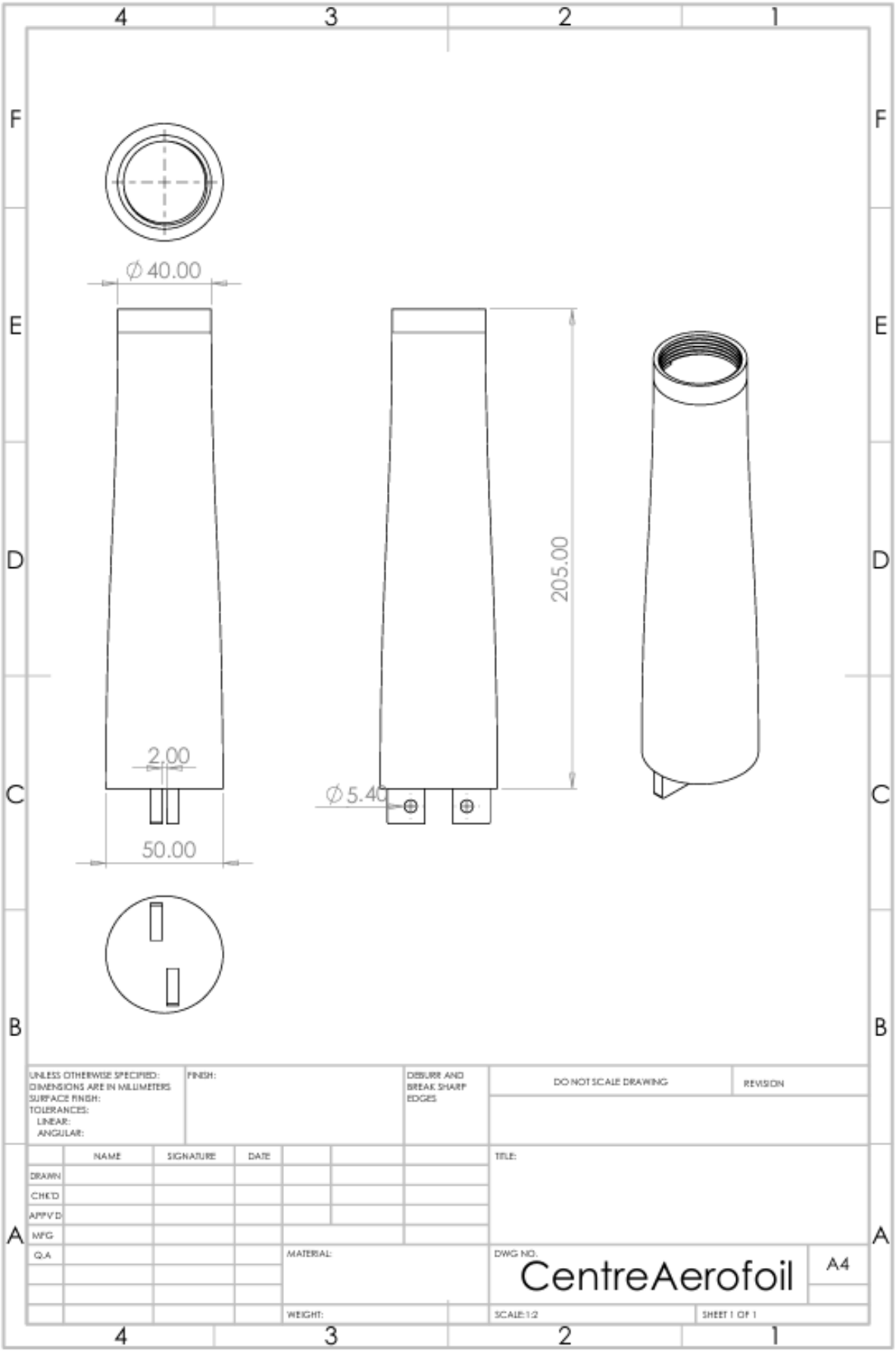


Figure A3.6 Centre Aerofoil

Appendix – Chapter 5

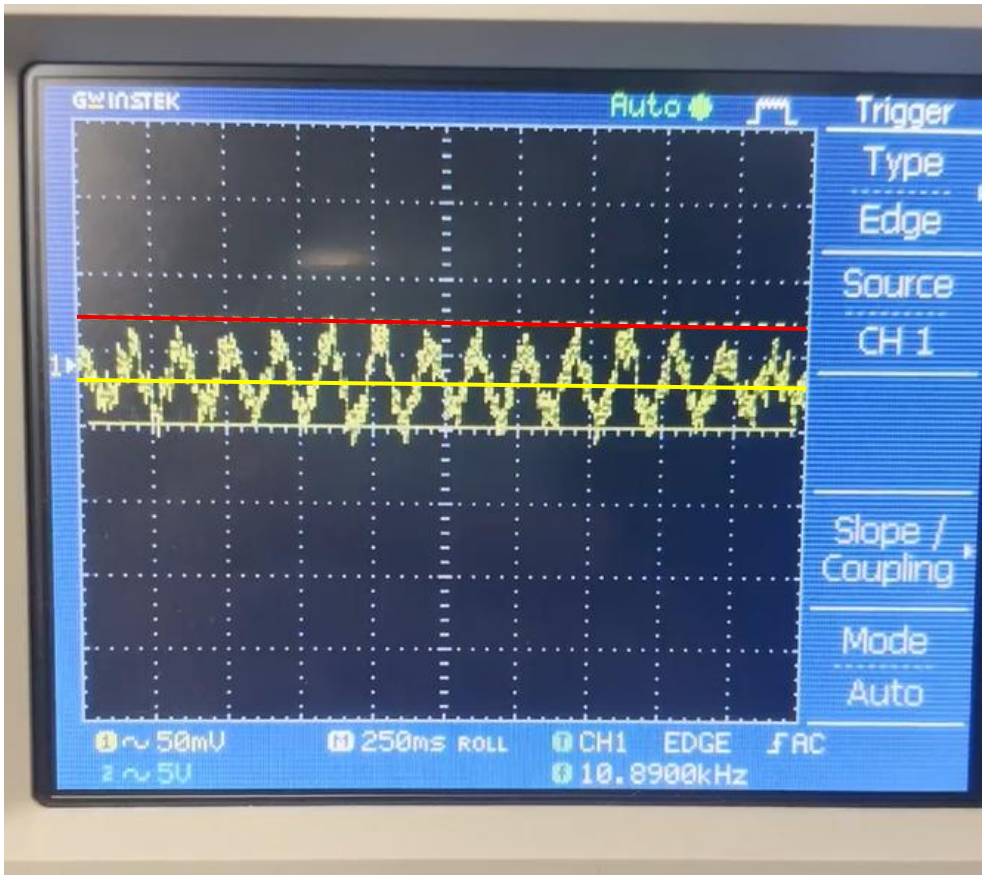


Figure A5.1 Control 10ms^{-1} Test

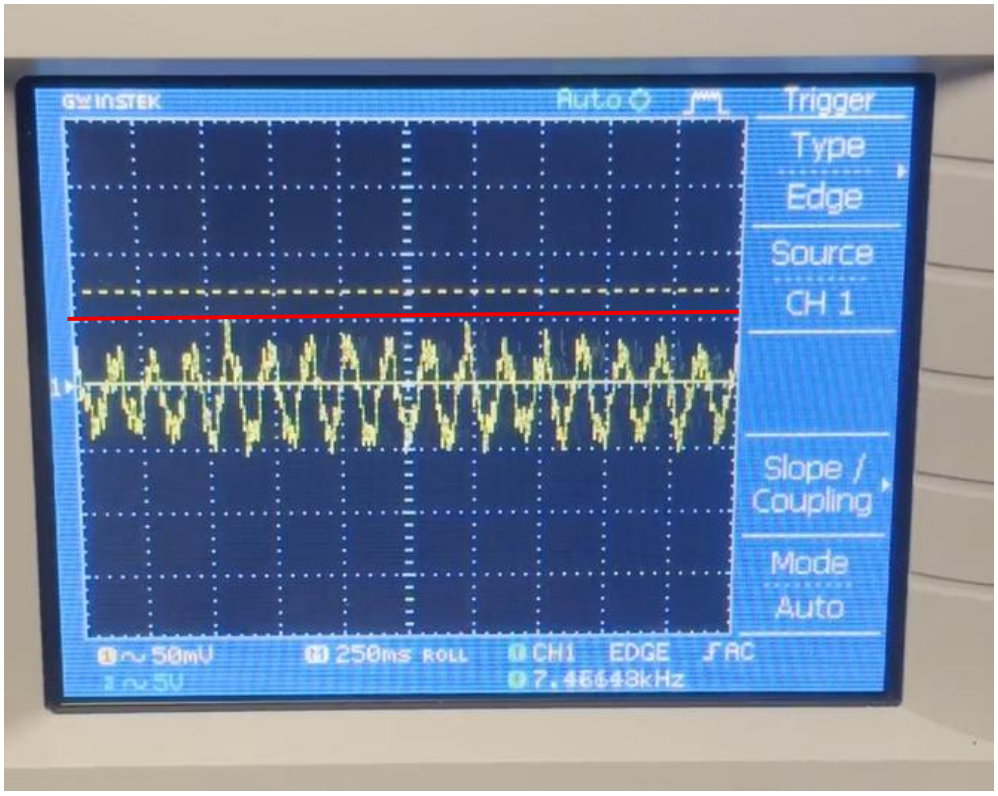


Figure A5.2 Control 15ms^{-1} Test

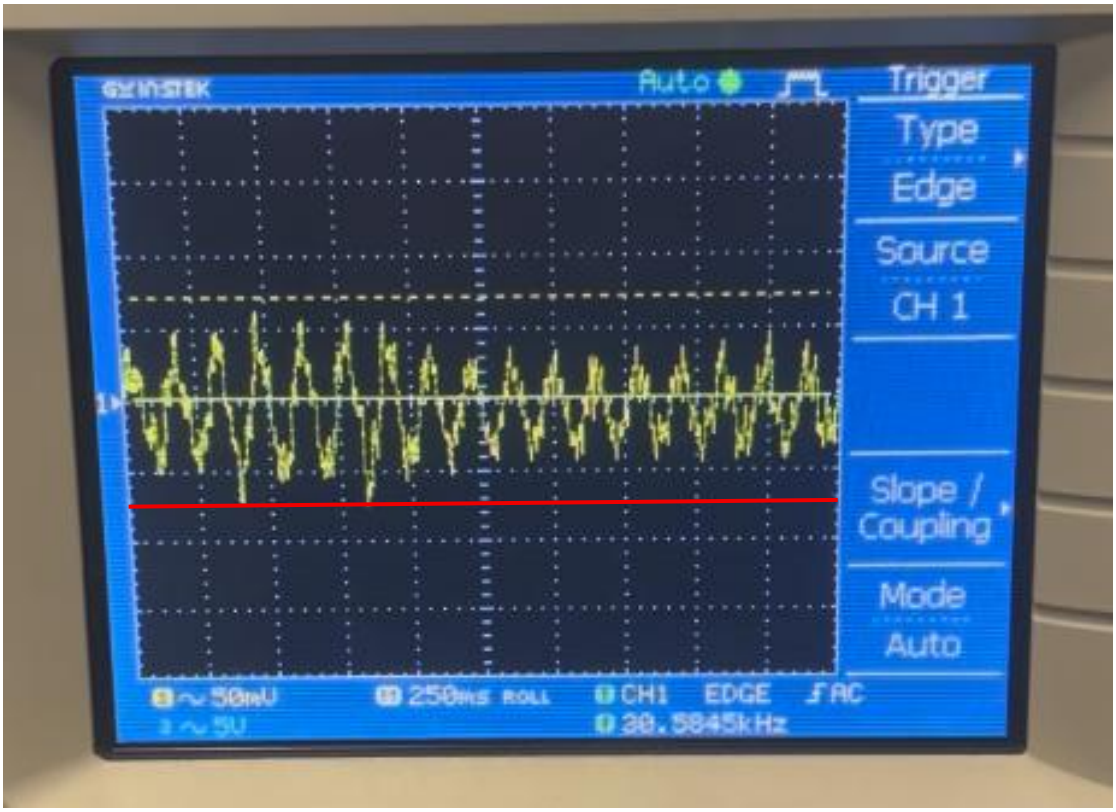


Figure A5.3 Dome Tip 10ms^{-1}

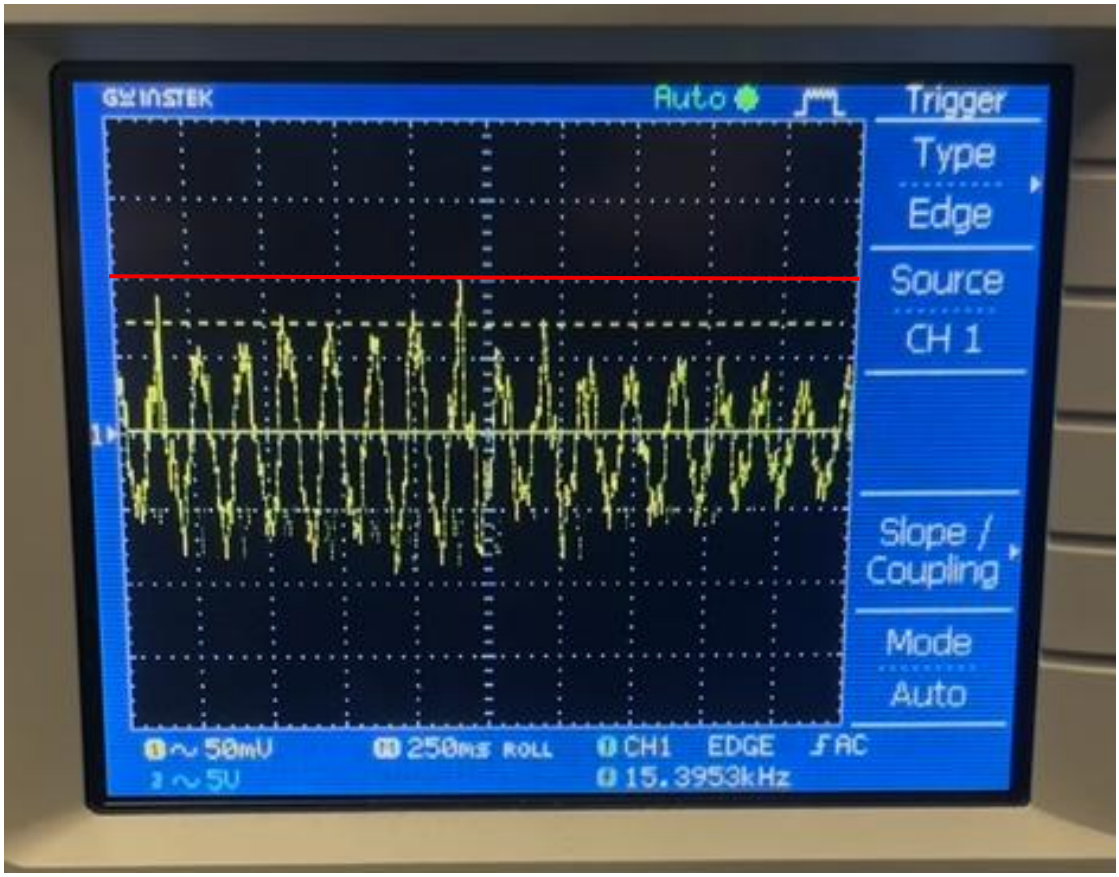


Figure A5.4 Dome Tip 15ms^{-1}

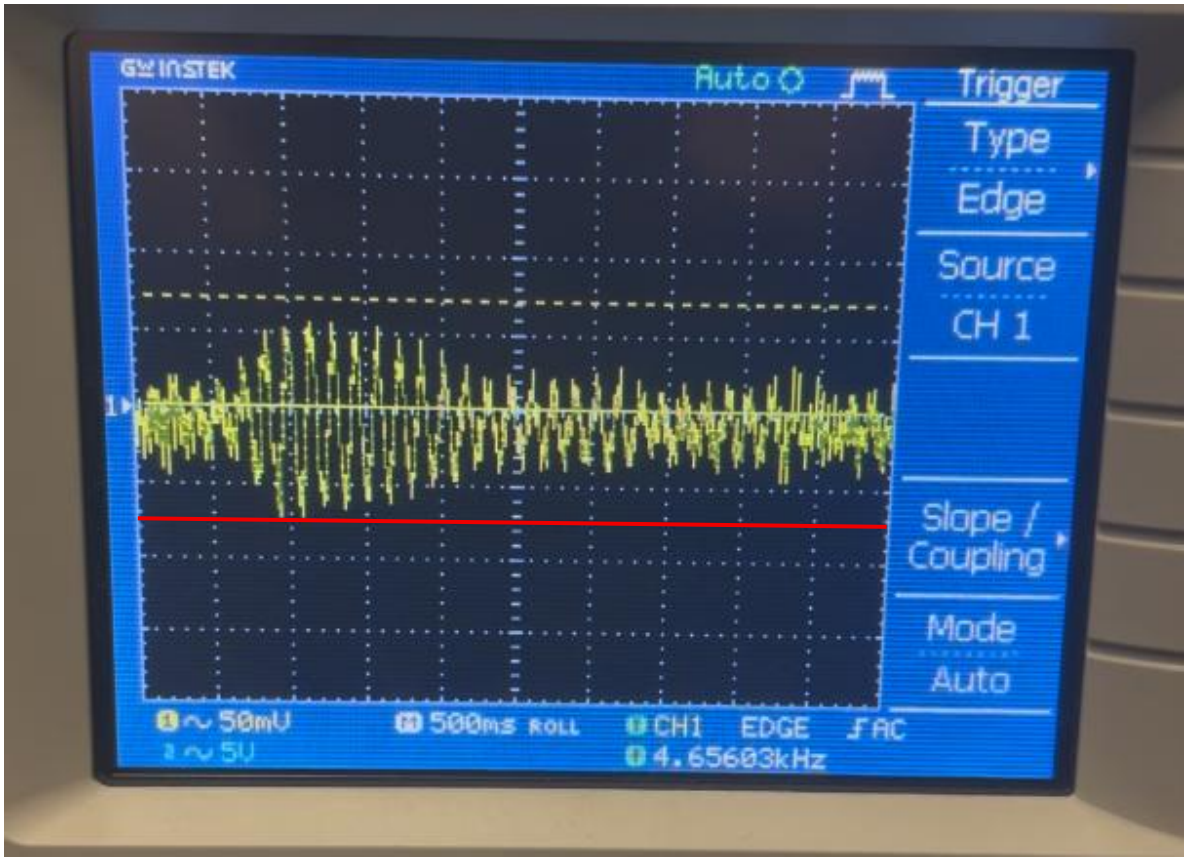


Figure A5.5 Tapered Tip 10ms^{-1}

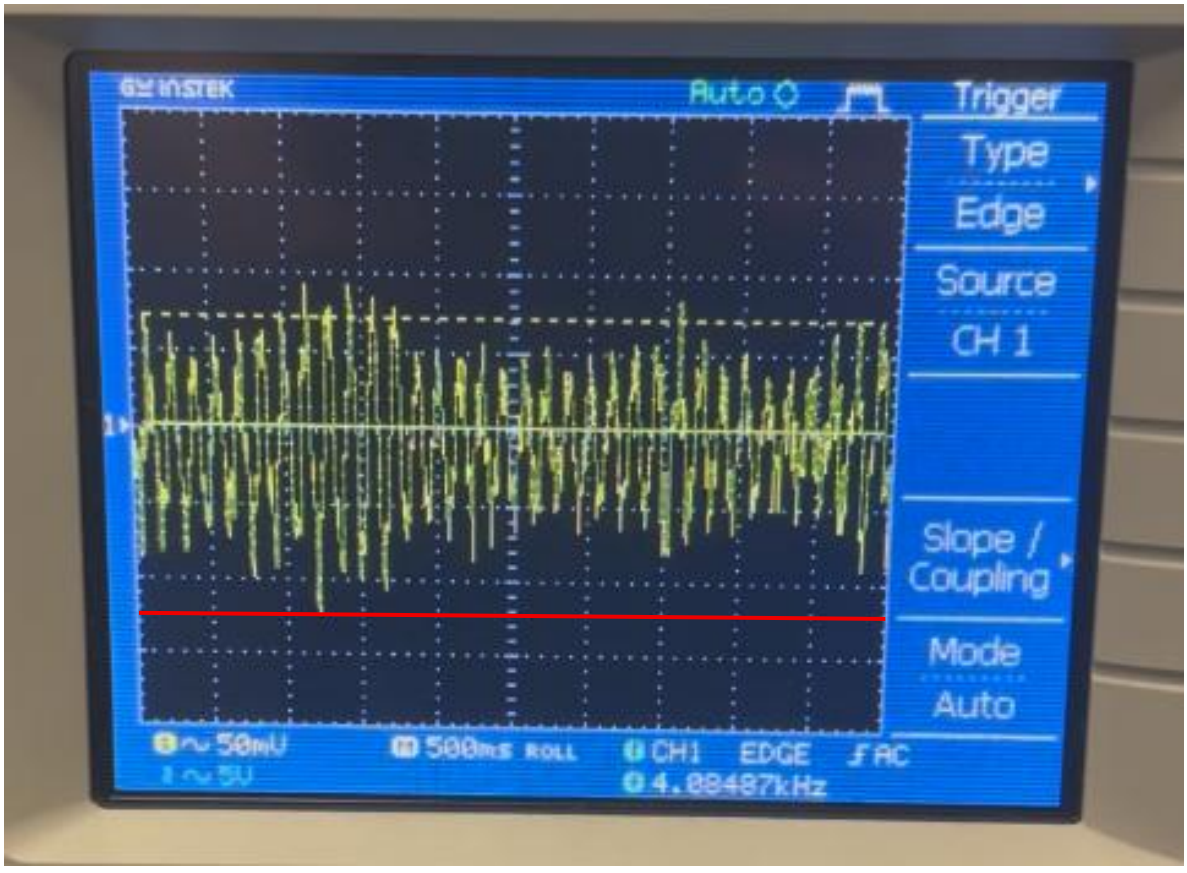


Figure A5.6 Tapered Tip 15ms^{-1}

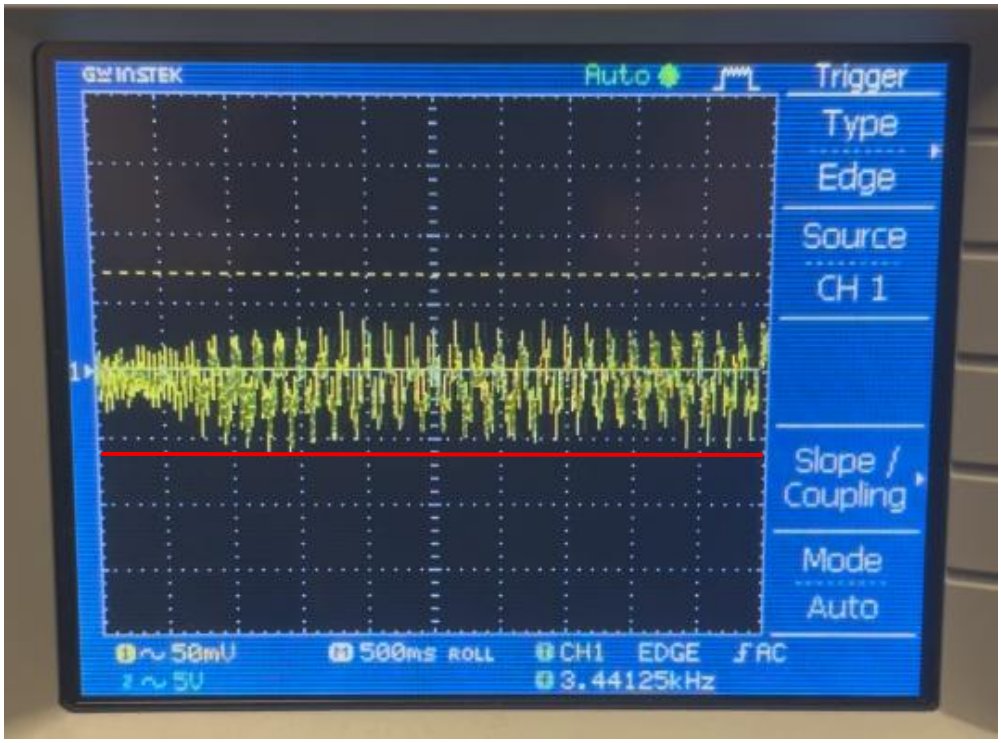


Figure A5.7 Expanding Tip 10ms^{-1}

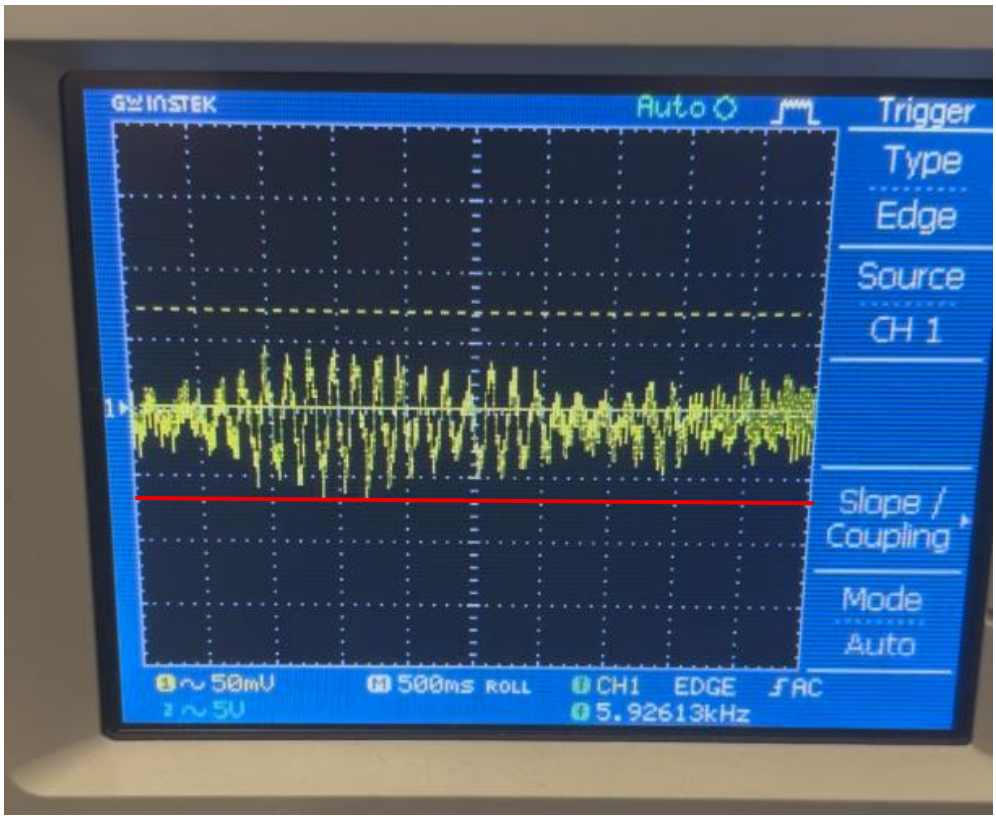


Figure A5.8 Expanding Tip 15ms^{-1}

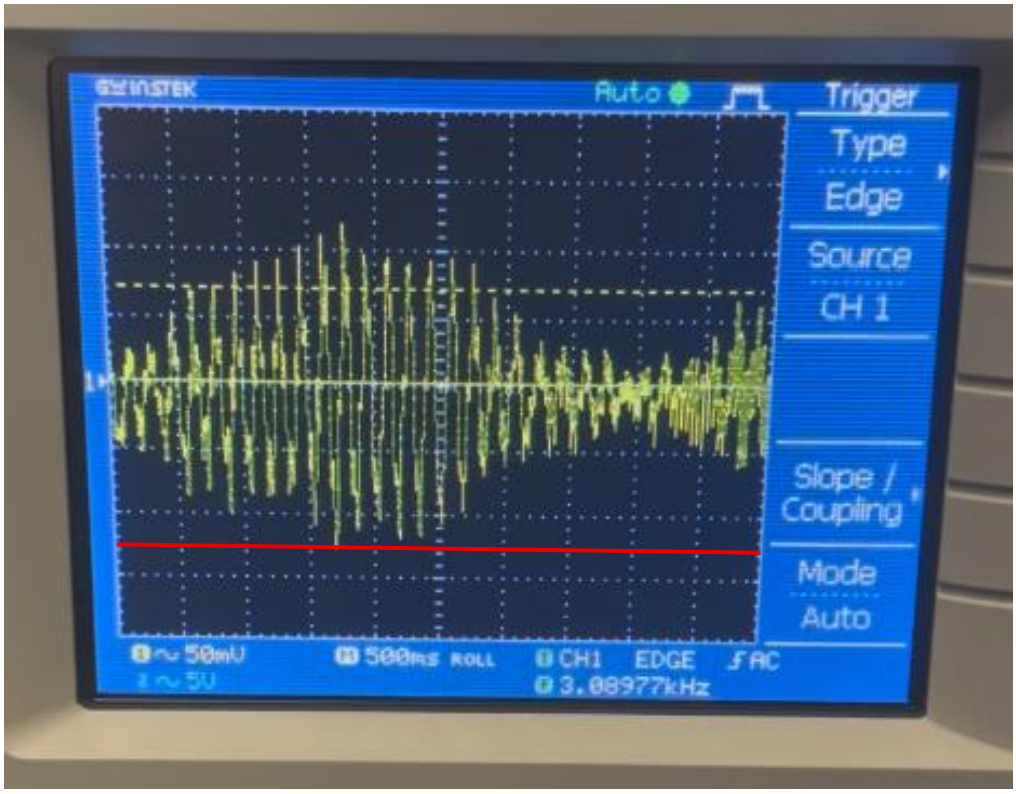


Figure A5.9 Expanding Tip 20ms^{-1}

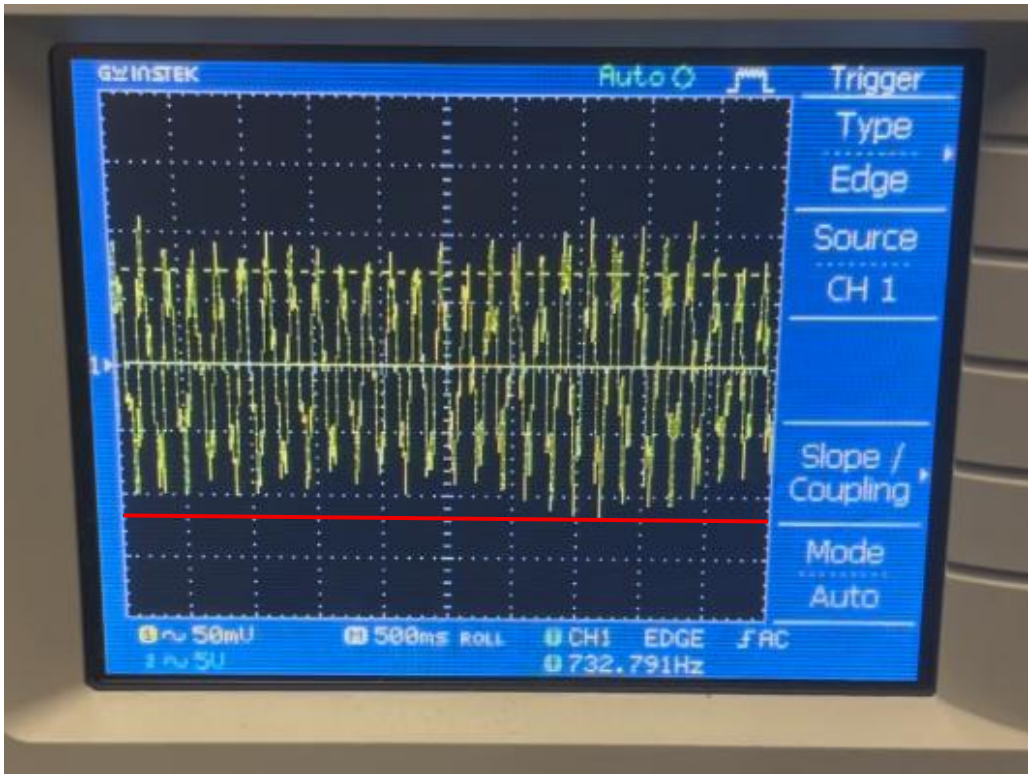


Figure A5.10 Amplifier Device 7ms^{-1}

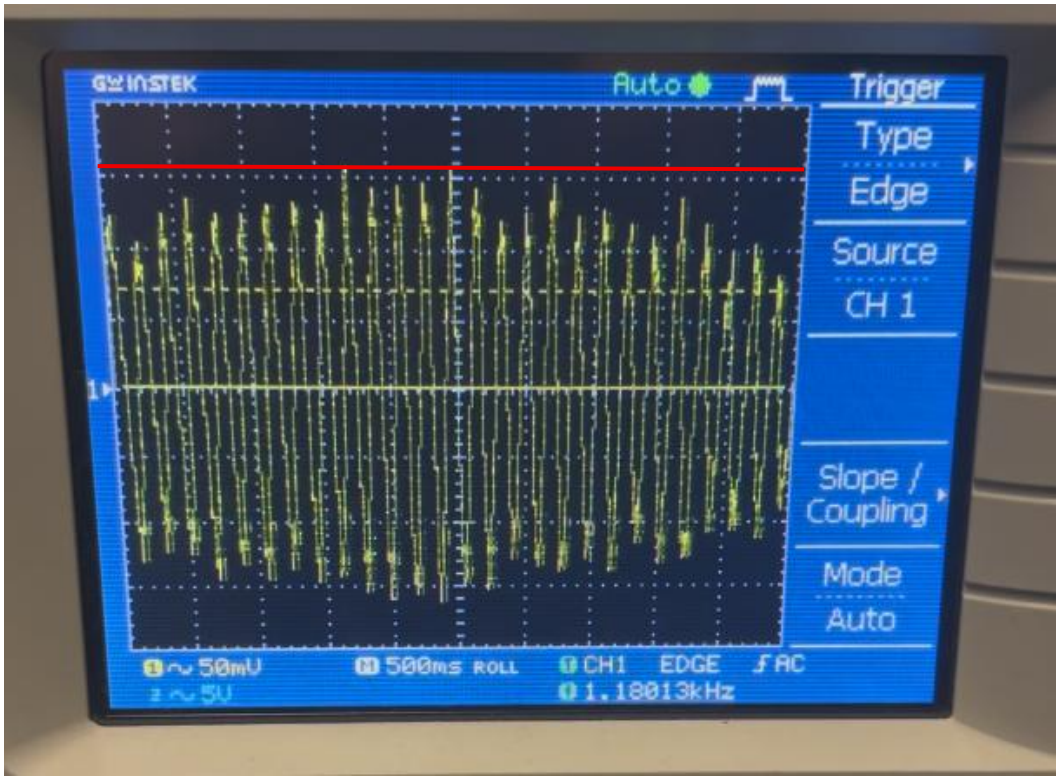


Figure A5.11 Amplifier Device 10ms^{-1}

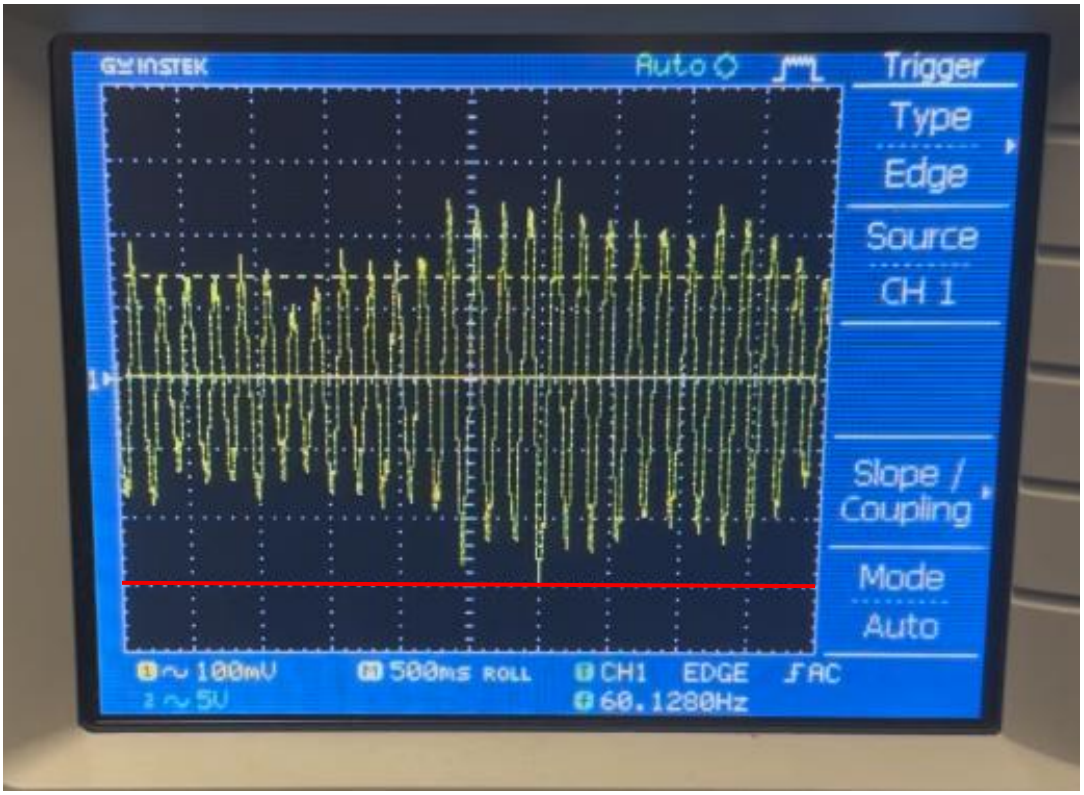


Figure A5.12 Amplifier Device 15ms^{-1}

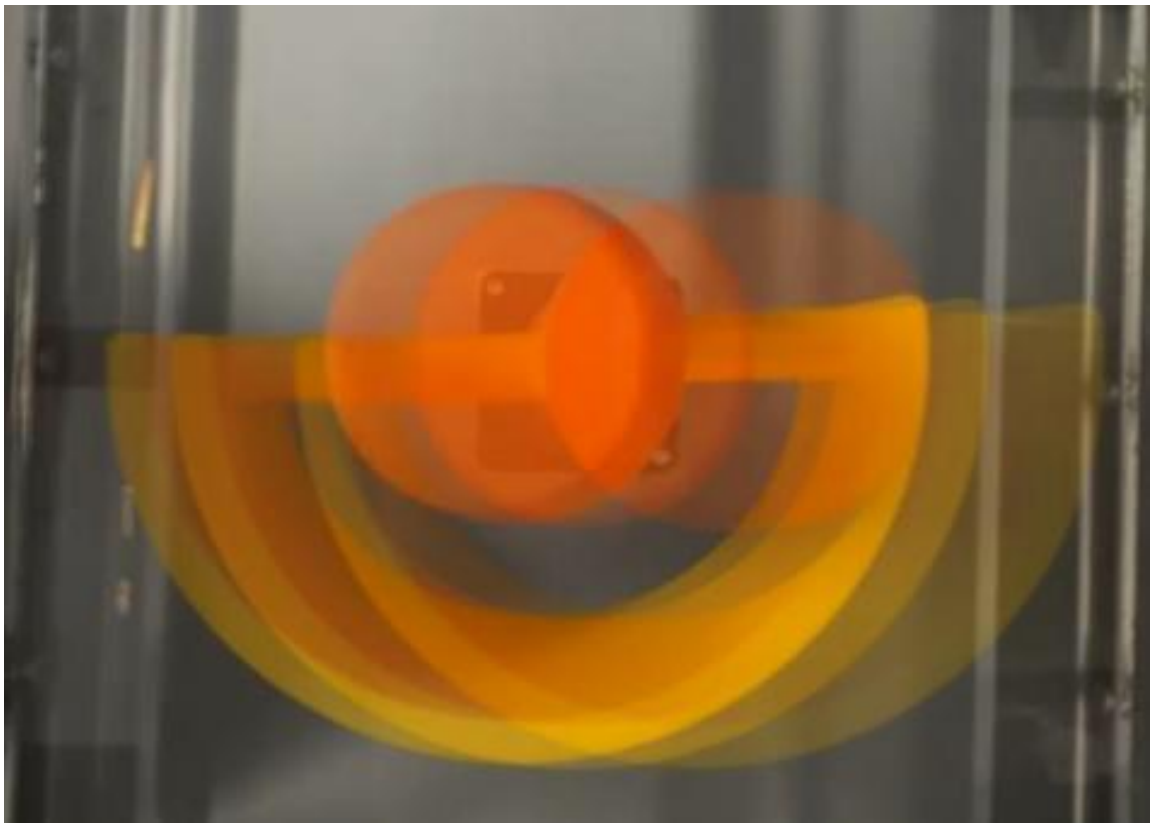


Figure A5.13 Maximum Amplifier Oscillation at 20ms^{-1}

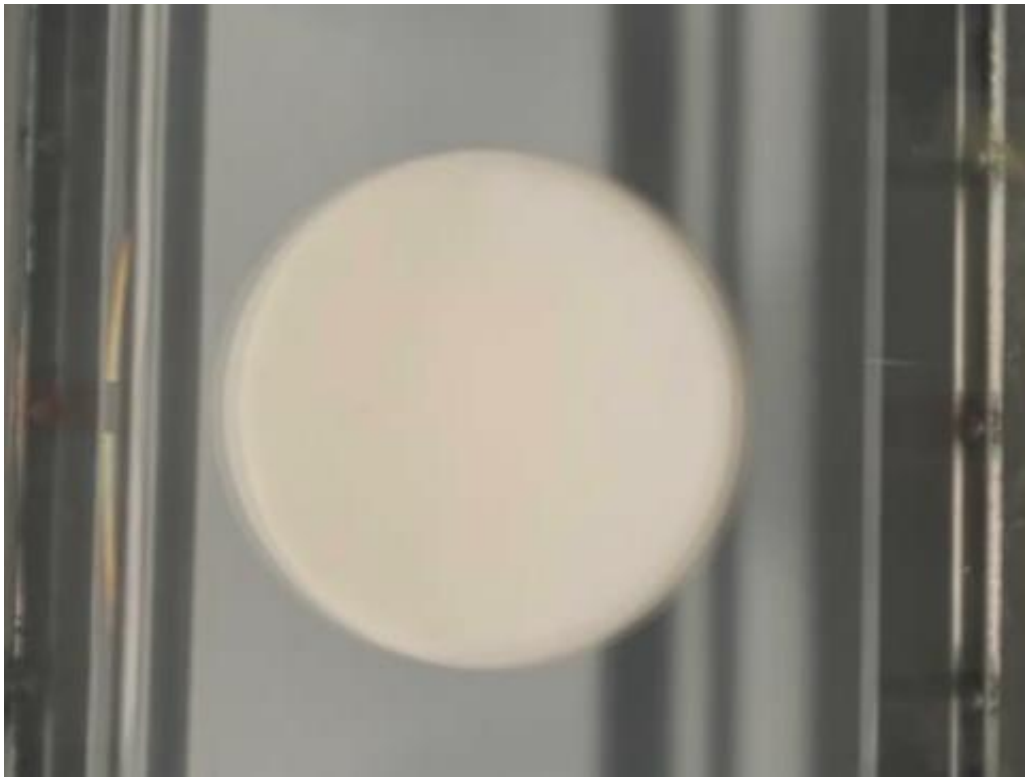


Figure A5.14 Maximum Expanding Tip Oscillation at 20ms^{-1}

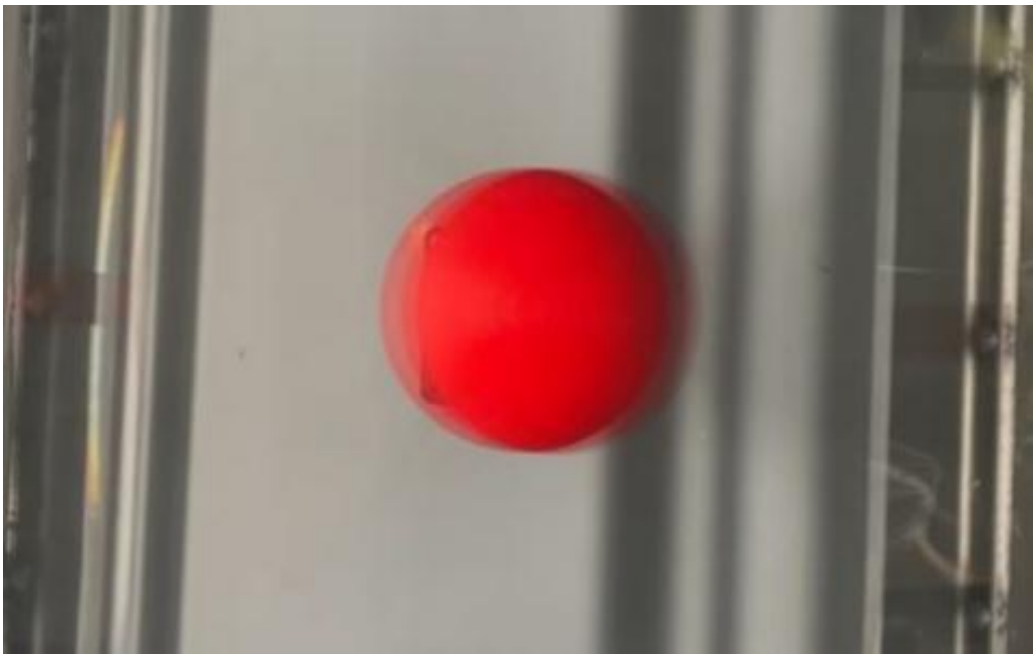


Figure A5.15 Maximum Dome Tip Oscillation at 20ms^{-1}



Figure A5.16 Maximum Tapered Tip Oscillation at 20ms^{-1}



Figure A5.17 P3 Strain Indicator and Recorder



Figure A5.18 Testing Setup for Oscilloscope

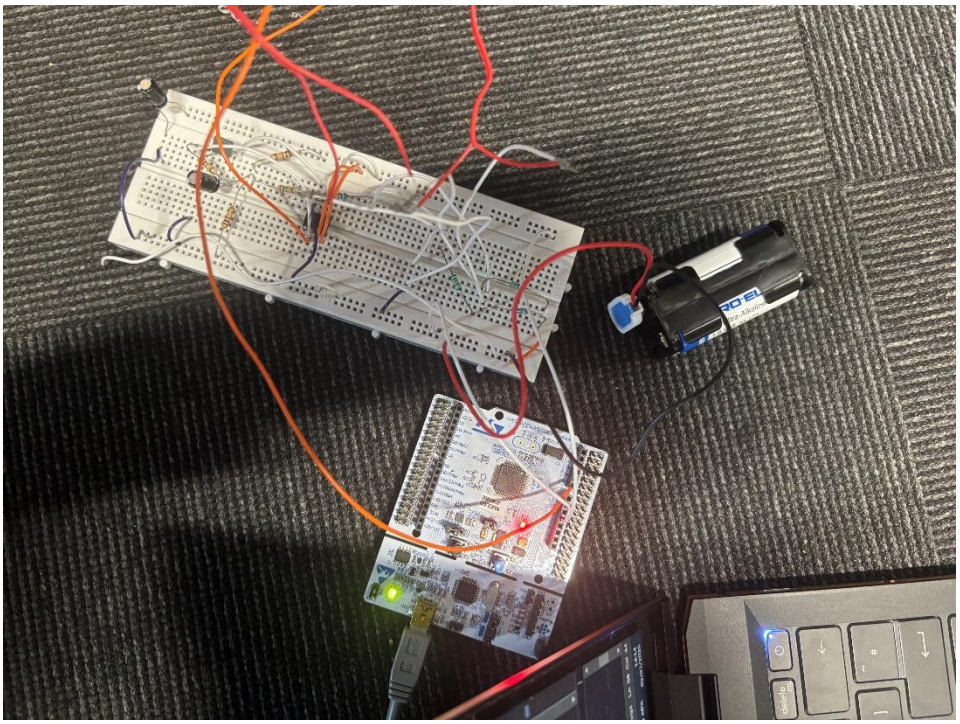


Figure A5.19 Failed Differential Amplifier Circuit for Data Logging

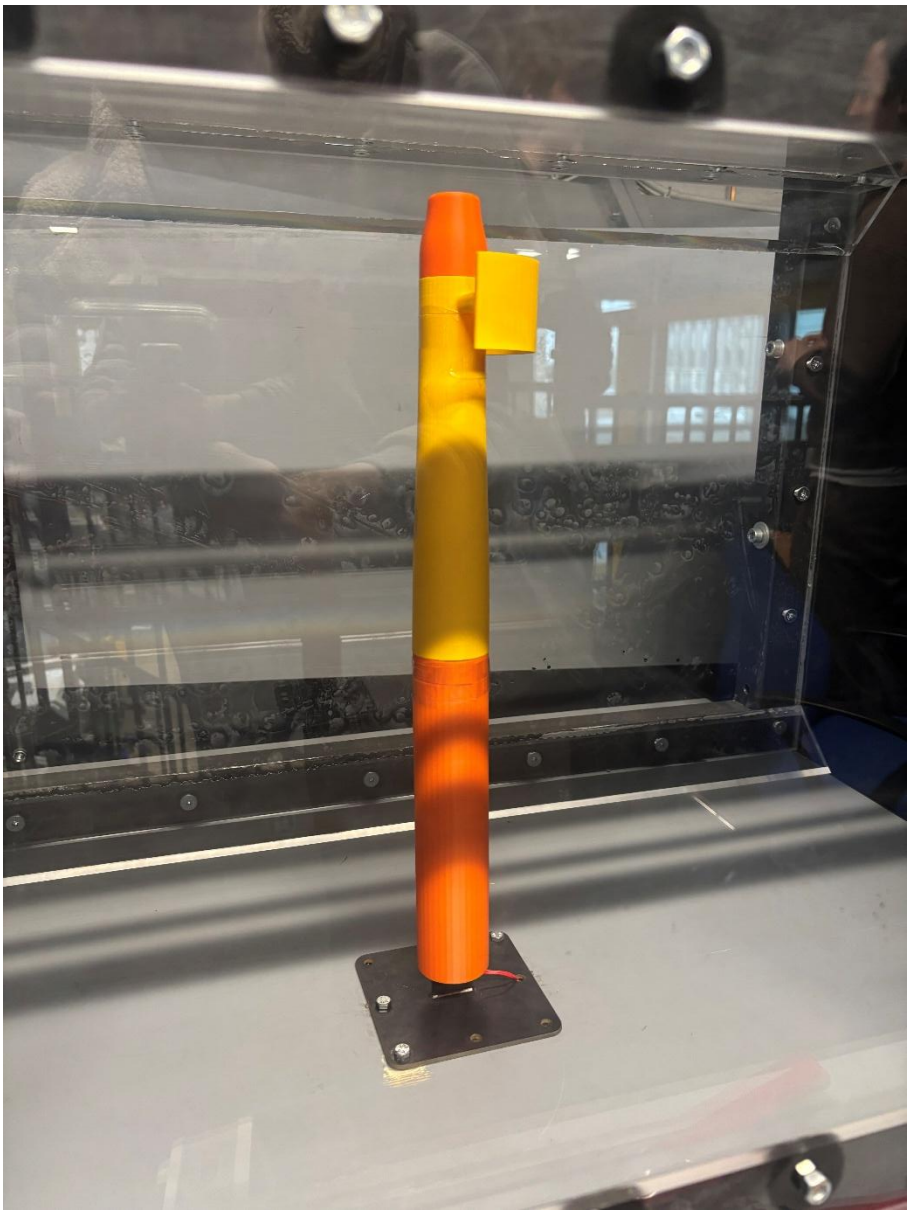


Figure A5.20 Full Amplifier Configuration within the Wind Tunnel

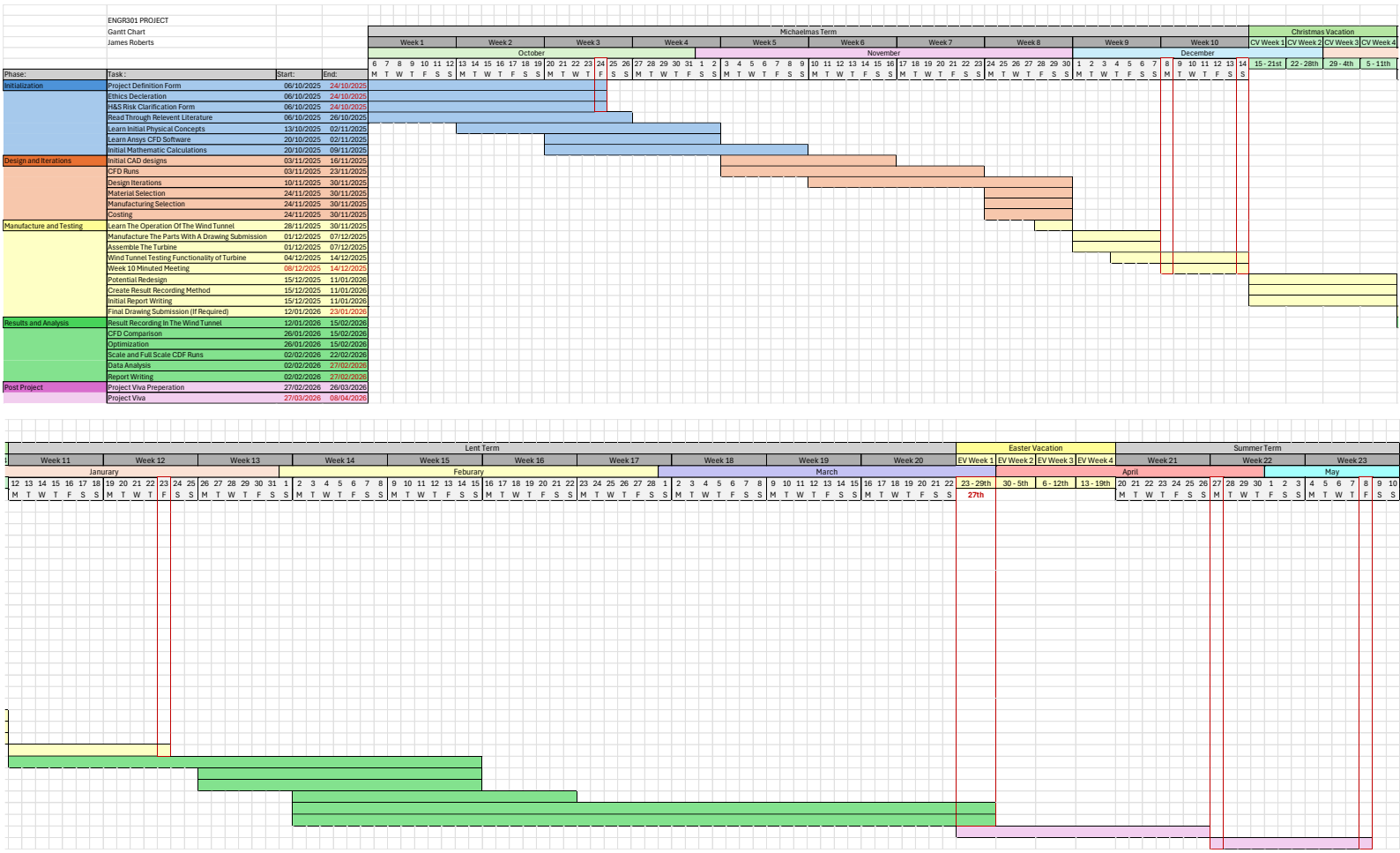


Figure A5.21 Project Gantt Chart